

PROBABILISTIC SYSTEMS ANALYSIS AND APPLIED PROBABILITY

MIHIR RAO

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LECTURE 1: PROBABILITY MODELS AND AXIOMS

Probability gives us a mathematical way to reason about uncertainty. A **PROBABILITY MODEL** specifies what can happen and assigns probabilities to events. Its two ingredients are a sample space and a probability law.

1.1 Sample spaces: what can happen?

Before assigning probabilities, decide what one complete outcome records. The detail must be enough to answer the questions of interest.

Definition 1.1 [SAMPLE SPACE]. The **SAMPLE SPACE**, denoted by Ω , is the set of all possible outcomes of an experiment. Its outcomes must be **MUTUALLY EXCLUSIVE**: only one outcome occurs in a run. They must also be **COLLECTIVELY EXHAUSTIVE**: every possible result is included.

Example: *Two rolls of a tetrahedral die.* Each roll produces one of 1, 2, 3, 4. Write X for the first result and Y for the second. One complete outcome is the ordered pair (X, Y) , so

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\}.$$

There are $4 \times 4 = 16$ outcomes. The order matters: $(1, 2)$ and $(2, 1)$ describe different results. We can display the pairs in a grid or describe the two rolls sequentially with a tree. Each full path through the tree ends at one of the same 16 outcomes.

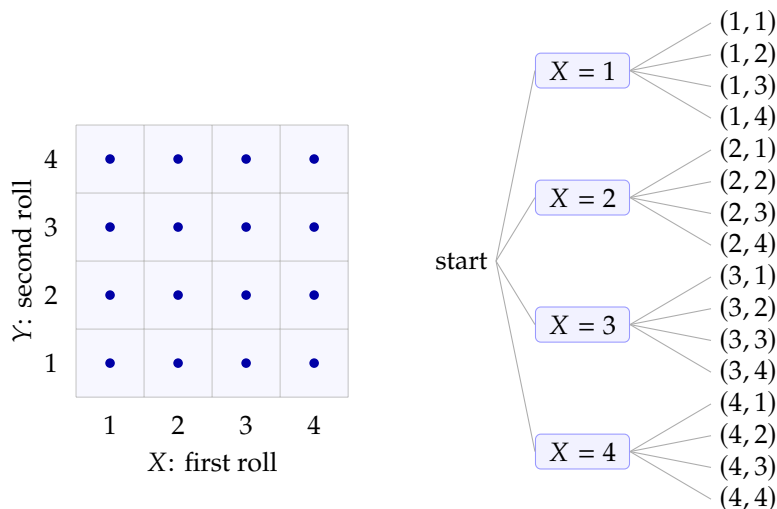


Figure 1: The grid and the tree describe the same sample space. A cell or a complete path records both rolls.

Example: *Two numbers in the unit interval.* If each result can be any real number from 0 to 1, the sample space is

$$\Omega = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\} = [0, 1]^2.$$

It is the entire unit square, including its boundary. The die space is finite; this space is continuous. Both still need a probability law.

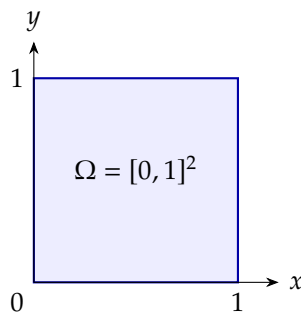


Figure 2: Each point in the unit square is one possible ordered pair.

1.2 Events and probability axioms

We often ask whether a result satisfies a condition, rather than whether one particular outcome occurs.

Definition 1.2 [EVENT AND PROBABILITY LAW]. An event A is a subset of Ω : it occurs when the outcome belongs to A . A PROBABILITY LAW assigns a number $P(A)$ to each event in its domain, subject to the probability axioms below.

For a finite or countably infinite sample space, every subset can be an event. On a continuous space, the domain of the law must be chosen with care; we use ordinary geometric regions here.

Definition 1.3 [PROBABILITY AXIOMS]. For events in the domain of P :

1. NONNEGATIVITY: $P(A) \geq 0$ for every event A .
2. NORMALIZATION: $P(\Omega) = 1$.
3. ADDITIVITY: for DISJOINT events A and B , meaning $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B).$$

The first two axioms say that probabilities cannot be negative and that some outcome in the sample space is certain to occur. The third lets us add probabilities when the events cannot occur together. For overlapping events, this sum would count the common outcomes twice.

Distinct individual outcomes give disjoint singleton events. Repeatedly applying additivity therefore gives

$$P(\{s_1, \dots, s_k\}) = P(\{s_1\}) + \dots + P(\{s_k\}).$$

We may abbreviate $P(\{s\})$ to $P(s)$. The two-event axiom handles finite unions; an infinite union will require a stronger form of additivity.

1.3 A finite probability law: two die rolls

Return to the 16 ordered die outcomes and assign probability $1/16$ to each. The probabilities are nonnegative and sum to $16(1/16) = 1$. An event containing k outcomes has probability $k/16$.

Example: *Either (1, 1) or (1, 2).* These are two distinct outcomes, so

$$P(\{(1, 1), (1, 2)\}) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}.$$

Example: *The first roll is 1.* The second roll can be 1, 2, 3, or 4. Thus

$$\{X = 1\} = \{(1, 1), (1, 2), (1, 3), (1, 4)\}, \quad P(X = 1) = \frac{4}{16} = \frac{1}{4}.$$

The notation $\{X = 1\}$ names the subset of outcomes satisfying the condition.

Example: *The sum is odd.* One roll must be odd and the other even. There are two odd values and two even values, so there are $2 \times 2 = 4$ pairs of each order. Hence

$$P(X + Y \text{ is odd}) = \frac{4 + 4}{16} = \frac{1}{2}.$$

Example: *The smaller roll is 2.* Both rolls must be at least 2, and at least one must equal 2. The event is

$$\{\min(X, Y) = 2\} = \{(2, 2), (2, 3), (2, 4), (3, 2), (4, 2)\}.$$

There are five outcomes, giving

$$P(\min(X, Y) = 2) = \frac{5}{16}.$$

Counting three pairs with $X = 2$ and three with $Y = 2$ would count $(2, 2)$ twice, so it must be included only once.

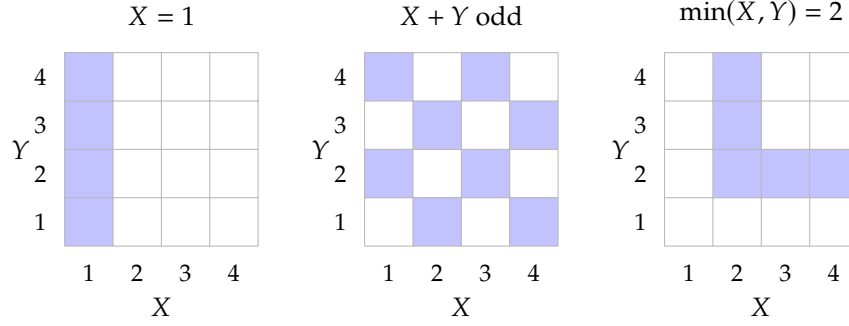


Figure 3: Highlighted cells show the last three die events. Each cell has probability $1/16$. In the last panel, the corner $(2, 2)$ is counted once.

1.4 The discrete uniform law

When a finite sample space has N equally likely outcomes, normalization forces each outcome to have probability $1/N$. This is the **DISCRETE UNIFORM LAW**. Additivity then gives

$$P(A) = \frac{|A|}{|\Omega|},$$

where $|A|$ is the number of outcomes in A and $|\Omega| = N$. Computing a probability becomes a counting problem. Fair coins, fair dice, and well-shuffled decks are modeled this way. The formula relies on the outcomes being equally likely.

1.5 The continuous uniform law

Now give the unit square a **CONTINUOUS UNIFORM LAW**: the probability of a region is its area. The square's total area is 1, so the law is normalized. Equal-area regions have equal probabilities, and areas add for disjoint regions.

Example: *The two numbers sum to at most $1/2$.* The event is

$$A = \{(x, y) \in [0, 1]^2 : x + y \leq \frac{1}{2}\}.$$

It is the triangle bounded by the two coordinate axes and the line $x + y = 1/2$. Both perpendicular sides have length $1/2$, so

$$P(X + Y \leq \frac{1}{2}) = \text{area}(A) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

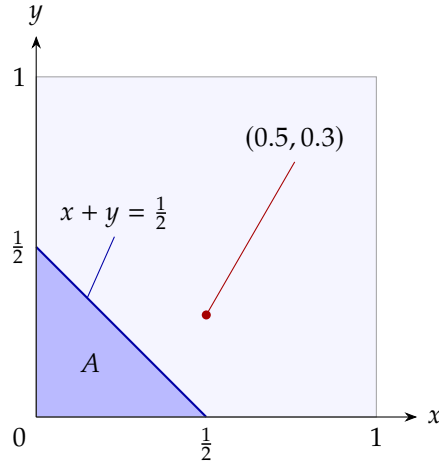


Figure 4: The shaded triangle has area $1/8$. The point $(0.5, 0.3)$ is a possible outcome, but its singleton has zero area.

Example: *The exact pair* $(0.5, 0.3)$. A single point has zero area, so

$$P((X, Y) = (0.5, 0.3)) = 0.$$

The point is still in the sample space. A probability-zero event need not be impossible: in this model every individual point has probability zero, while the whole square has probability one.

1.6 A countably infinite law and countable additivity

Consider the sample space $\Omega = \{1, 2, 3, \dots\}$ with

$$P(n) = 2^{-n}, \quad n = 1, 2, \dots$$

Each probability is positive, and the geometric series checks normalization:

$$\sum_{n=1}^{\infty} 2^{-n} = \frac{1/2}{1 - 1/2} = 1.$$

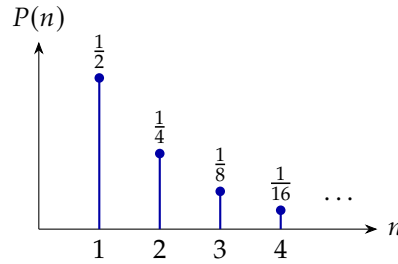


Figure 5: The first four outcome probabilities. Each is half the preceding one, and the pattern continues for every positive integer.

Example: *The outcome is even.* The event $\{2, 4, 6, \dots\}$ contains infinitely many disjoint singleton events. Adding their probabilities gives

$$\begin{aligned} P(\{2, 4, 6, \dots\}) &= \sum_{k=1}^{\infty} 2^{-2k} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \\ &= \frac{1/4}{1 - 1/4} = \frac{1}{3}. \end{aligned}$$

This calculation uses an infinite sum, which finite additivity alone does not justify.

Definition 1.4 [COUNTABLE ADDITIVITY]. If $A_1, A_2, \dots$ are pairwise disjoint events, the COUNTABLE ADDITIVITY axiom requires

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

This is the full additivity axiom for a probability law. It covers a sequence indexed by the positive integers and includes finite additivity as a special case. It justifies the even-outcome calculation above. It does not permit summing singleton probabilities over all the points of the unit square: that collection is uncountable.

Revision. A model needs both a sample space and a probability law. Events are sets of outcomes. Count outcomes under a finite uniform law, measure area under the uniform law on the unit square, and use countable additivity for a countable union of disjoint events.

Common mistakes: using the counting formula without equal likelihood, counting a shared outcome twice, or treating probability zero as impossibility.

Check yourself: Why do two tetrahedral-die rolls give 16 outcomes? Why does $\min(X, Y) = 2$ select five of them? Why can an infinite sum give the probability of an even integer, but cannot add up the probabilities of every point in the unit square?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 1: Probability Models and Axioms](#), PDF pp. 1–3. The calculations are completed and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 2: CONDITIONING AND BAYES' RULE

Learning that an event occurred changes which outcomes remain possible and how their probabilities should be compared. Conditioning makes this update precise. It also gives three useful tools: multiplying probabilities along a sequence, combining disjoint cases, and reversing a conditional probability.

2.1 Start with a probability model

The sample space Ω lists mutually exclusive, collectively exhaustive outcomes at the detail needed for the question. Events are subsets of Ω . Their probabilities are nonnegative, satisfy $P(\Omega) = 1$, and add over disjoint events. For a disjoint sequence $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

To solve a problem, specify the sample space, define its probability law, identify the event of interest, and then calculate. A calculation is only meaningful once these ingredients are clear.

2.2 Conditional probability: restrict and renormalize

Suppose we learn that B occurred. Outcomes outside B can be discarded. Among the outcomes in B , the event A occurs exactly on $A \cap B$. We compare the probability of this intersection with the probability of the new universe B .

Definition 2.1 [CONDITIONAL PROBABILITY]. For events A and B with $P(B) > 0$, the **CONDITIONAL PROBABILITY** of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This definition does not assign a value when $P(B) = 0$.

Restricting to B keeps only the outcomes compatible with the information. Dividing by $P(B)$ rescales the remaining probability to one: $P(B \mid B) = P(B)/P(B) = 1$.

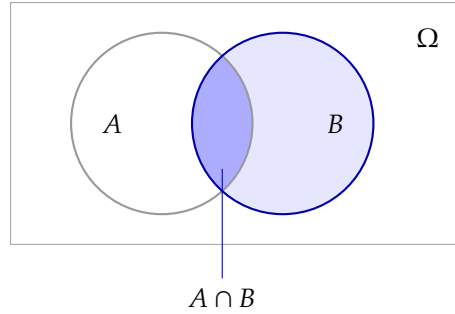


Figure 6: Conditioning keeps B and compares $A \cap B$ with all of B . This is a schematic set diagram; region sizes do not represent probabilities.

Example: *The minimum of two die rolls is 2.* Let X and Y be the two results of a tetrahedral die, with each of the 16 ordered pairs assigned probability $1/16$. Suppose we know

$$B = \{\min(X, Y) = 2\} = \{(2, 2), (2, 3), (2, 4), (3, 2), (4, 2)\}.$$

Then $P(B) = 5/16$. Each remaining pair has conditional probability $(1/16)/(5/16) = 1/5$.

Let $M = \max(X, Y)$. The event $M = 1$ has no outcome in common with B : if the minimum is 2, the maximum cannot be 1. Therefore

$$P(M = 1 \mid B) = \frac{0}{5/16} = 0.$$

For $M = 2$, the only compatible pair is $(2, 2)$, giving

$$P(M = 2 \mid B) = \frac{1/16}{5/16} = \frac{1}{5}.$$

The denominator counts the five outcomes allowed by the information, not all 16 outcomes in the original space.

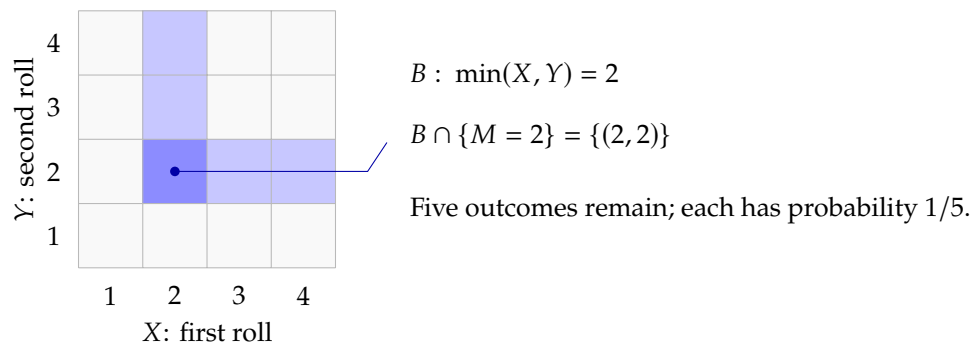


Figure 7: The five colored cells form B . The darker cell $(2, 2)$ is the only one where $M = 2$. Each colored cell has probability $1/5$ given B .

2.3 Building a model from conditional probabilities

A model can be specified in stages. First give the probability of one event, then give the probabilities of later events under each possible earlier result. Write A^c for the COMPLEMENT of A , the event that A does not occur.

Example: *An airplane and a radar reading.* Let A mean that an airplane is flying above, and let B mean that something registers on the radar screen. The model is

$$P(A) = 0.05, \quad P(A^c) = 0.95,$$

with the following conditional probabilities:

$$\begin{aligned} P(B | A) &= 0.99 & P(B^c | A) &= 0.01, \\ P(B | A^c) &= 0.10 & P(B^c | A^c) &= 0.90. \end{aligned}$$

Thus a registration is likely when a plane is present, but registrations also occur when no plane is present. Within each condition, the two radar probabilities sum to one.

Rearranging the conditional-probability definition gives

$$P(A \cap B) = P(A)P(B | A).$$

Multiply the probability of the first stage by the conditional probability of the second stage to obtain the probability of a complete path.

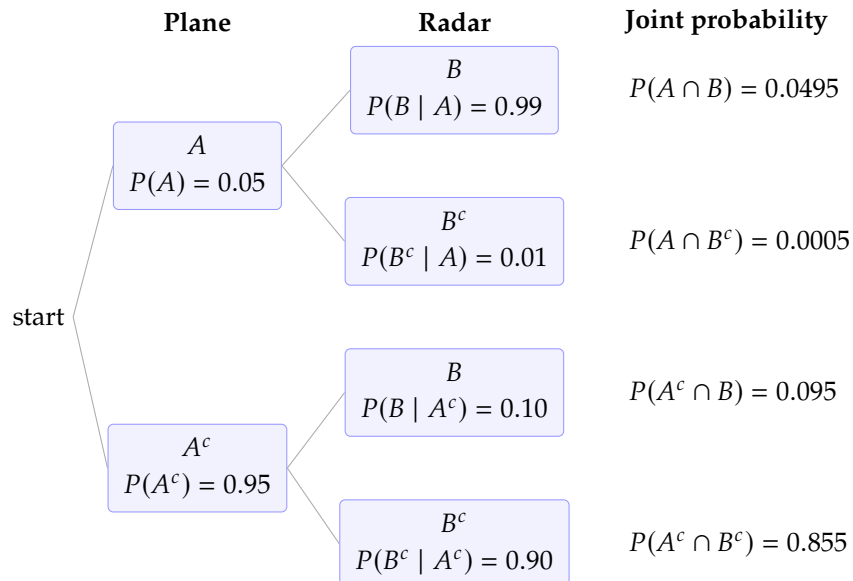


Figure 8: Each path specifies both plane presence and radar status. Multiply the two factors on a path to obtain the joint probability on the right.

For a plane and a registration,

$$P(A \cap B) = 0.05(0.99) = 0.0495.$$

A registration can come from either $A \cap B$ or $A^c \cap B$. These events are disjoint, so

$$\begin{aligned} P(B) &= P(A \cap B) + P(A^c \cap B) \\ &= 0.05(0.99) + 0.95(0.10) = 0.1445. \end{aligned}$$

Once a registration is observed, the probability of a plane is

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.0495}{0.1445} = \frac{99}{289} \approx 0.3426.$$

The two directions answer different questions: $P(B | A) = 0.99$ describes the radar given a plane; $P(A | B) \approx 0.3426$ describes plane presence given a registration. The latter must account for how often planes are present and how often the radar registers without one.

2.4 The multiplication rule

For three events, a complete path specifies A , then B given A , then C given both earlier events. Each factor must use all the information accumulated so far.

Proposition 2.2 [MULTIPLICATION RULE]. *If $P(A) > 0$ and $P(A \cap B) > 0$, then*

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B).$$

The positivity assumptions ensure that the conditional probabilities on the right are defined.

Proof. Apply the conditional-probability definition twice: first condition on the whole event $A \cap B$, then express $P(A \cap B)$ by conditioning on A .

$$\begin{aligned} P(A \cap B \cap C) &= P(A \cap B)P(C | A \cap B) \\ &= P(A)P(B | A)P(C | A \cap B). \end{aligned}$$

□

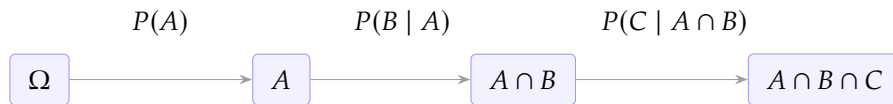


Figure 9: One path through a three-stage model. The last factor conditions on $A \cap B$, because both earlier events have been specified.

2.5 Total probability: combine disjoint cases

Sometimes the probability of B is easier to calculate after dividing the sample space into cases. Compute the contribution from each case, then add the contributions.

Definition 2.3 [PARTITION]. Events A_1, A_2, A_3 form a PARTITION of Ω if they are pairwise disjoint and cover the whole space:

$$A_i \cap A_j = \emptyset \quad (i \neq j), \quad A_1 \cup A_2 \cup A_3 = \Omega.$$

Every outcome belongs to exactly one of the three cases.

Theorem 2.4 [TOTAL PROBABILITY]. For a partition A_1, A_2, A_3 with $P(A_i) > 0$ for each i ,

$$P(B) = P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2) + P(A_3)P(B \mid A_3).$$

Proof. Each outcome of B belongs to exactly one part of the partition, so $B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)$ is a disjoint union. Additivity gives

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B).$$

For each piece, use $P(A_i \cap B) = P(A_i)P(B \mid A_i)$. Substituting these three products gives the formula. $\square$

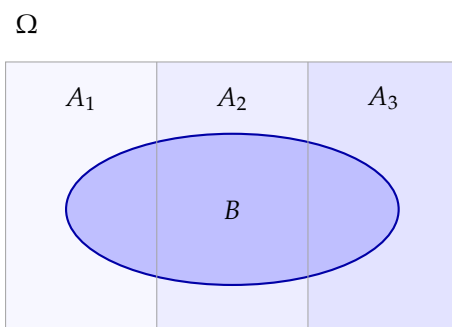


Figure 10: The partition cuts B into three disjoint pieces. Their probabilities add to $P(B)$. The geometry is schematic.

Each term weights a conditional probability by the probability of its case. In the radar example, the two cases are A and A^c , giving the calculation $P(B) = 0.05(0.99) + 0.95(0.10)$.

2.6 Bayes' rule: reverse the conditioning

Suppose we know how likely B is under each case A_i , but observe B and want to infer which case occurred. The PRIOR PROBABILITIES $P(A_i)$ describe the cases before using this observation. The updated probabilities are $P(A_i | B)$.

Theorem 2.5 [BAYES' RULE]. *For the partition above, if $P(A_i) > 0$ for each i and $P(B) > 0$, then*

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_{j=1}^3 P(A_j)P(B | A_j)}, \quad i = 1, 2, 3.$$

Proof. The conditional-probability definition gives

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}.$$

The multiplication rule supplies the numerator, $P(A_i \cap B) = P(A_i)P(B | A_i)$. The total probability theorem supplies the denominator by adding the contributions from all three cases. Substituting both expressions yields Bayes' rule. $\square$

In [figure 10](#), the numerator is the probability of the piece $A_i \cap B$ and the denominator is the probability of all of B . Thus Bayes' rule compares one case's contribution with the total probability of the observation. In the radar model, it updates $P(A) = 0.05$ to $P(A | B) \approx 0.3426$ after a registration.

Revision. Conditioning restricts the possible outcomes and divides by the probability of the conditioning event. Multiply conditional factors along a path; add disjoint paths for total probability. Bayes' rule divides one case's contribution by the total probability of the observed event.

Common mistakes: interchanging $P(A | B)$ and $P(B | A)$, using a denominator that ignores the information, or forgetting earlier events in the last factor of the multiplication rule.

Check yourself: Why is $P(M = 2 | \min(X, Y) = 2) = 1/5$? Why is the radar model's $P(B) = 0.1445$? Which two ingredients, besides $P(B | A)$, are needed to calculate $P(A | B)$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 2: Conditioning and Bayes' Rule](#), PDF pp. 1–2. The supplied calculations and derivations are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 3: INDEPENDENCE

Independent events do not change one another's probabilities. Independence depends on the probability law: conditioning can remove it or create it.

3.1 Review: probabilities along a coin-toss tree

For $P(B) > 0$, conditioning gives $P(A \mid B) = P(A \cap B)/P(B)$. The multiplication rule, total probability, and Bayes' rule then give

$$\begin{aligned} P(A \cap B) &= P(A)P(B \mid A) = P(B)P(A \mid B), \\ P(B) &= P(A)P(B \mid A) + P(A^c)P(B \mid A^c), \\ P(A \mid B) &= \frac{P(A)P(B \mid A)}{P(B)}. \end{aligned}$$

Each conditional factor requires a conditioning event of positive probability. Multiply along a path and add disjoint paths.

Example: *Three tosses of a biased coin.* At each stage, the probability of heads is p and of tails is $1 - p$, regardless of earlier results. The sample space consists of the eight strings $HHH, HHT, HTH, HTT, THH, THT, TTH, TTT$.

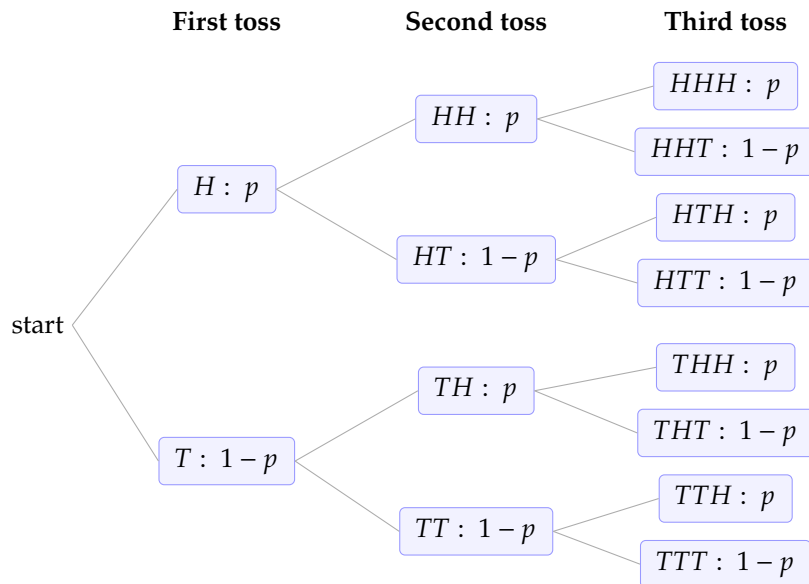


Figure 11: Each box gives the accumulated result and the probability of the newest toss. Multiply the three factors on a path. These factors remain p or $1 - p$ after every earlier history.

The path THT has probability

$$P(THT) = (1 - p)p(1 - p) = p(1 - p)^2.$$

Exactly one head occurs on the three disjoint paths HTT, THT, TTH . Each has the same probability, so

$$P(\text{exactly one head}) = 3p(1 - p)^2.$$

Given exactly one head, the first toss is heads only on HTT . For $0 < p < 1$,

$$P(\text{first toss is } H \mid \text{exactly one head}) = \frac{p(1 - p)^2}{3p(1 - p)^2} = \frac{1}{3}.$$

The bias cancels because all three remaining paths have equal probability. At $p = 0$ or $p = 1$, the conditioning event has probability zero, so this conditional probability is undefined.

3.2 Independence of two events

If learning that A occurred gives no information about B , then $P(B \mid A) = P(B)$, provided $P(A) > 0$. Substitution in the multiplication rule gives a definition that also works for probability-zero events.

Definition 3.1 [INDEPENDENCE]. Events A and B are INDEPENDENT if

$$P(A \cap B) = P(A)P(B).$$

The definition is symmetric in A and B . When $P(A) > 0$, it is equivalent to $P(B \mid A) = P(B)$; when $P(B) > 0$, it is equivalent to $P(A \mid B) = P(A)$.

Proof. If $P(A) > 0$, divide $P(A \cap B) = P(A)P(B)$ by $P(A)$ to obtain $P(B \mid A) = P(B)$. Conversely, multiplying that conditional equality by $P(A)$ gives the defining product. Interchanging A and B proves the other equivalence. $\square$

If $P(A) = 0$, then $A \cap B \subseteq A$ also has probability zero. Thus $P(A \cap B) = 0 = P(A)P(B)$ for every B , even though conditioning on A is undefined. Independence is a numerical property of the probability law; the shape of a set diagram alone does not establish it.

3.3 Conditioning can change independence

After learning that C occurred, use the conditional probability law $P(\cdot | C)$ throughout. Independence in this new universe is tested using conditional probabilities.

Definition 3.2 [CONDITIONAL INDEPENDENCE]. For $P(C) > 0$, events A and B are **CONDITIONALLY INDEPENDENT** given C if

$$P(A \cap B | C) = P(A | C)P(B | C).$$

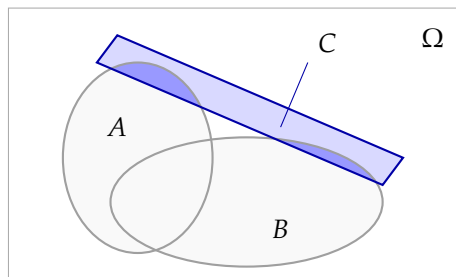


Figure 12: Within C , the pieces $A \cap C$ and $B \cap C$ are disjoint. The diagram is schematic; areas do not specify probabilities.

Suppose A and B are independent before conditioning, but C meets each of them while avoiding $A \cap B$. If both $P(A | C)$ and $P(B | C)$ are positive, then

$$P(A \cap B | C) = 0 \quad \text{while} \quad P(A | C)P(B | C) > 0.$$

They are therefore dependent given C . Unconditional independence does not guarantee conditional independence.

Example: *A coin chosen once, then tossed repeatedly.* Choose Coin A or Coin B with probability $1/2$ each. Coin A has head probability 0.9 ; Coin B has head probability 0.1 . Toss the chosen coin repeatedly, with independent tosses conditional on its identity. Let G be the event that Coin A was chosen, and let H_i mean that toss i is heads. Then

$$P(H_i | G) = 0.9, \quad P(H_i | G^c) = 0.1.$$

Knowing the identity fixes the head probability. Within either conditional model, earlier tosses give no further information about later tosses.

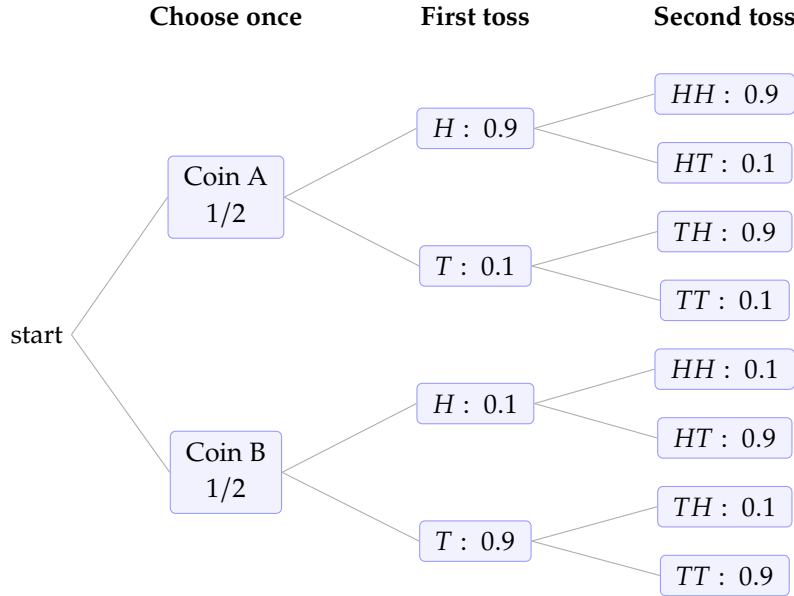


Figure 13: The first choice selects one coin for all tosses. The boxes give the probability of each new step conditional on the preceding history. The two tosses shown are independent given the chosen coin.

Without knowing the coin, total probability gives

$$P(H_{11}) = \frac{1}{2}(0.9) + \frac{1}{2}(0.1) = \frac{1}{2}.$$

Now suppose the first ten tosses were heads. Write $D = H_1 \cap \dots \cap H_{10}$. Conditional independence within each coin model gives $P(D \mid G) = 0.9^{10}$ and $P(D \mid G^c) = 0.1^{10}$. The observation is far more likely with Coin A. Bayes' rule gives

$$P(G \mid D) = \frac{0.9^{10}}{0.9^{10} + 0.1^{10}}.$$

For the next toss, divide the probability of eleven heads by that of ten:

$$\begin{aligned} P(H_{11} \mid D) &= \frac{\frac{1}{2}(0.9^{11}) + \frac{1}{2}(0.1^{11})}{\frac{1}{2}(0.9^{10}) + \frac{1}{2}(0.1^{10})} \\ &= \frac{0.9^{11} + 0.1^{11}}{0.9^{10} + 0.1^{10}} \approx 0.9. \end{aligned}$$

This differs from $P(H_{11}) = 1/2$. Earlier results reveal information about the shared coin and change the probability of later results. Thus conditional independence does not guarantee unconditional independence.

3.4 Independence of a collection of events

For a collection, the idea is stronger than checking two events at a time: information about any group of events must give no information about the remaining events.

Definition 3.3 [INDEPENDENCE OF A COLLECTION]. Events $A_1, \dots, A_n$ are **MUTUALLY INDEPENDENT** if, for every nonempty subset $S \subseteq \{1, \dots, n\}$,

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i).$$

Here $\prod$ means multiplication over the indicated indices. The condition must hold for every subcollection, not just all n events together.

For three events, this requires the three pairwise products and also

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3).$$

For a larger independent collection, even combinations of events and their complements involving disjoint groups of indices give no information about one another. For example, when $n \geq 6$ and $P(A_5 \cap A_6^c) > 0$,

$$P(A_1 \cap (A_2^c \cup A_3) \mid A_5 \cap A_6^c) = P(A_1 \cap (A_2^c \cup A_3)).$$

The condition concerns indices 5, 6, while the event of interest concerns indices 1, 2, 3.

3.5 Pairwise independence is not enough

Events are **PAIRWISE INDEPENDENT** if each pair of distinct events is independent. This checks only subcollections of size two. It need not establish mutual independence.

Example: *Two fair coin tosses and whether they match.* Consider two independent fair tosses, with the four equally likely outcomes HH, HT, TH, TT . Define

$$A = \{HH, HT\}, \quad B = \{HH, TH\}, \quad C = \{HH, TT\}.$$

Thus A means the first toss is heads, B the second is heads, and C that both results are the same.

		Second toss		
		H	T	
First toss	H	HH	HT	A: first toss is H
	T	TH	TT	B: second toss is H
				C: the tosses match

Figure 14: Every cell has probability $1/4$. The two colored cells form C . Knowing both A and B identifies HH , which is in C .

Each event contains two outcomes, so $P(A) = P(B) = P(C) = 1/2$. Each pairwise intersection is the singleton $\{HH\}$:

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}.$$

All three pairs are independent. But the triple intersection is also $\{HH\}$, giving

$$P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C).$$

Equivalently,

$$P(C \mid A \cap B) = \frac{1/4}{1/4} = 1 \neq \frac{1}{2} = P(C).$$

Neither toss alone changes the probability of matching; knowing both tosses determines whether they match. Pairwise independence misses this dependence on combined information.

3.6 The king's sibling: specify the experiment

Example: *A king comes from a family with two children. What is the probability that his sibling is female?* Assume the parents decided in advance to have exactly two children. Each child is a boy or girl with probability $1/2$, independently of the other. Assume that a boy takes precedence in succession, so a king exists exactly when at least one child is a boy.

Order the children by birth. The four outcomes BB, BG, GB, GG each have probability $1/4$, where B means boy and G girl. The information that there is a king is the event

$$K = \{BB, BG, GB\}, \quad P(K) = \frac{3}{4}.$$

His sibling is female in BG or GB . Writing F for this event within K gives

$$P(F \mid K) = \frac{P(F \cap K)}{P(K)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

		Younger child		
		B	G	
Older child	B	BB	BG	
	G	GB	GG excluded	K : at least one boy $F \cap K$: BG or GB $P(F K) = 2/3$

Figure 15: Under the fixed two-child model, conditioning on K removes GG . The other three outcomes are equally likely; two give a female sibling. The darker cells are $F \cap K$.

The answer depends on the experiment. If the parents instead have children until the first boy and then stop, having exactly two children forces the outcome GB . Under that rule, the king's sibling is female with probability one. The phrase "a family with two children" alone does not specify how the family came to have two children.

Revision. Independence is the product condition $P(A \cap B) = P(A)P(B)$. Conditional independence uses the same condition under $P(\cdot | C)$. Mutual independence checks every subcollection; pairwise independence checks only pairs.

Common mistakes: carrying independence into a conditional model without checking it, inferring mutual independence from pairwise products, or leaving the experiment's selection and stopping rules unspecified.

Check yourself: Why does exactly one head give a first-toss head probability of $1/3$, regardless of the bias? Why do ten heads change the next-toss probability for the unknown coin? Which product fails in the matching-toss example? What assumptions produce the sibling probability $2/3$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 3: Independence](#), PDF pp. 1–2, and the [accompanying transcript](#) (conditional independence and the sibling model, especially 41:29–46:28). The supplied calculations and derivations are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 4: COUNTING

Counting turns a probability question into two questions: how many outcomes are possible, and how many satisfy the event? The main task is to describe each outcome exactly once, taking care about order, repetition, and which groups are labeled.

4.1 The discrete uniform law

If a finite sample space Ω has equally likely outcomes, each outcome has probability $1/|\Omega|$. For an event $A \subseteq \Omega$,

$$P(A) = \frac{|A|}{|\Omega|}.$$

Here $|A|$ is the number of outcomes in A . Count the numerator and denominator using the same kind of outcome. The formula requires equal likelihood; counting alone does not establish that assumption.

4.2 The basic counting principle

Construct an outcome through successive choices. If every partial outcome has the same number of extensions at a given stage, the total can be found by multiplying the numbers of choices.

Proposition 4.1 [BASIC COUNTING PRINCIPLE]. *Suppose an outcome is constructed in r stages. At stage i , there are n_i choices for every permitted history of earlier stages. If each complete sequence of choices describes exactly one outcome, and each outcome has exactly one such description, the number of outcomes is*

$$n_1 n_2 \cdots n_r.$$

Proof. After the first stage there are n_1 partial outcomes. Each has n_2 extensions, giving $n_1 n_2$ partial outcomes after stage two. Repeating this reasoning at each stage gives the product. The unique-description assumption prevents counting the same outcome more than once. $\square$

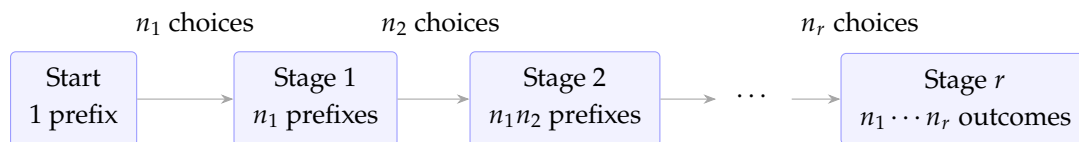


Figure 16: Each stage extends every existing prefix by the indicated number of choices. The boxes count prefixes, not probabilities.

Example: License plates with three letters followed by four digits. Use 26 letters and the ten digits 0, . . . , 9 in these fixed positions. With repetition allowed, the number of plates is

$$26^3 10^4 = 175,760,000.$$

If letters cannot repeat and digits cannot repeat, the choices decrease as positions are filled:

$$(26 \cdot 25 \cdot 24)(10 \cdot 9 \cdot 8 \cdot 7) = 78,624,000.$$

	Three letters			Four digits			
Repetition allowed	26	26	26	10	10	10	10
No repetition	26	25	24	10	9	8	7

Figure 17: Numbers in the slots count the available choices. Multiply across a row. Without repetition, used symbols are unavailable later.

A PERMUTATION of n distinct elements is an ordering of all of them. There are n choices for the first position, $n - 1$ for the second, and so on, giving

$$n(n - 1) \cdots 2 \cdot 1 = n!.$$

The notation $n!$ is the FACTORIAL of n ; by convention, $0! = 1$.

To construct a subset of $\{1, \dots, n\}$, decide separately for each element whether to include it. There are two choices at each of n stages, so the number of subsets, including the empty set, is

$$2^n.$$

Example: Six die rolls, all different. Roll a fair six-sided die independently six times. An outcome is an ordered sequence of six results. All 6^6 sequences are equally likely. For the results to be different, the numbers of choices are 6, 5, 4, 3, 2, 1, giving $6!$ favorable sequences. Thus

$$P(\text{all six different}) = \frac{6!}{6^6} = \frac{720}{46,656} = \frac{5}{324} \approx 0.01543.$$

The denominator allows repeated results; the numerator forbids them.

4.3 From ordered choices to combinations

Choosing k distinct elements in order from an n -element set is a k -PERMUTATION. For integers $0 \leq k \leq n$, its count is

$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

For $k = 0$, there is one empty sequence. An unordered selection must identify all the different orderings of the same chosen elements.

Definition 4.2 [COMBINATIONS]. The BINOMIAL COEFFICIENT $\binom{n}{k}$, read “ n choose k ,” is the number of k -element subsets of a given n -element set. These unordered selections are called COMBINATIONS.

Proposition 4.3 [COUNTING COMBINATIONS]. For integers $0 \leq k \leq n$,

$$\boxed{\binom{n}{k} = \frac{n!}{k!(n-k)!}}.$$

Proof. Count ordered selections of k distinct elements in two ways. Choosing the elements one at a time gives $n!/(n-k)!$ selections. Alternatively, choose a k -element subset in $\binom{n}{k}$ ways, then order its elements in $k!$ ways. Both procedures describe the same ordered selections, so

$$\binom{n}{k} k! = \frac{n!}{(n-k)!}.$$

Dividing by $k!$ gives the formula. □

The factor $k!$ removes repeated counts of the same subset. Also, every subset has one of the sizes $0, 1, \dots, n$. Counting subsets by size or by the include/exclude decisions above gives

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

4.4 Binomial probabilities

Consider n independent tosses of a coin with head probability p . A specified sequence with k heads and $n - k$ tails has probability $p^k(1-p)^{n-k}$. For the sequence $HTTHHH$, for example,

$$P(HTTHHH) = p(1-p)(1-p)ppp = p^4(1-p)^2.$$

The positions of the heads identify the sequence. Choosing k of the n positions therefore counts the sequences with exactly k heads.

Proposition 4.4 [BINOMIAL PROBABILITY]. For $0 < p < 1$ and $k = 0, 1, \dots, n$,

$$P(\text{exactly } k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Proof. Exactly k heads is a disjoint union of the $\binom{n}{k}$ sequences whose head positions form a k -element subset. Independence gives each sequence probability $p^k (1-p)^{n-k}$. Add their equal probabilities to obtain the formula. $\square$

If $p = 0$, the all-tail sequence has probability one; if $p = 1$, the all-head sequence has probability one. These are the boundary cases of the same model. For a biased coin, sequences with different head counts need not be equally likely. Equal likelihood is being used only among sequences with a fixed head count.

4.5 Conditioning on the number of heads

Example: *Three heads in ten tosses.* Toss a coin independently ten times, with $0 < p < 1$. Let B be the event that exactly three tosses are heads, and let A be the event that the first two tosses are heads. Find $P(A | B)$.

Every sequence in B has probability $p^3 (1-p)^7$. After conditioning on B , these sequences remain equally likely, even when the coin is biased. There are

$$|B| = \binom{10}{3} = 120$$

possible sets of head positions. To be in $A \cap B$, positions 1 and 2 must be heads, leaving exactly one head to place among positions 3 through 10. Hence

$$|A \cap B| = \binom{8}{1} = 8.$$

In B : choose three of all ten positions

In $A \cap B$: the first two heads are fixed

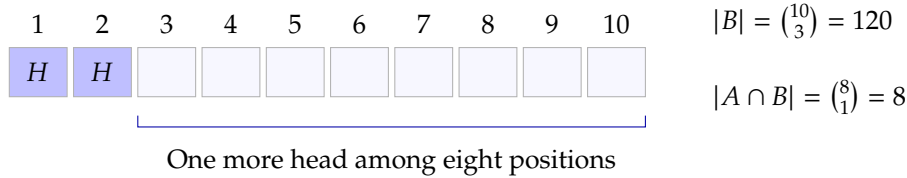


Figure 18: Fixing the first two heads leaves eight possible positions for the last head. Without those fixed heads, any three of the ten positions may be the head positions.

The conditional uniform law gives

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{\binom{8}{1}}{\binom{10}{3}} = \frac{8}{120} = \frac{1}{15}.$$

Equivalently, the common factor $p^3(1-p)^7$ cancels from the conditional-probability ratio. The denominator counts only sequences compatible with B .

4.6 Partitions: dealing a deck to four players

Example: *Each player receives an ace.* Deal a uniformly shuffled 52-card deck to four labeled players, with 13 cards per player. What is the probability that each receives one ace? An outcome records each player's hand, with order within a hand ignored. The four hands are disjoint and cover the deck, so they form a partition.

To count all outcomes, order the 52 distinct cards and divide the list into four consecutive blocks of 13 cards. Each fixed deal appears $(13!)^4$ times, because cards within any hand can be reordered without changing the deal. Thus

$$|\Omega| = \frac{52!}{(13!)^4}.$$

Every partition has the same number of full-deck orderings, so these outcomes are equally likely. Do not divide by $4!$: the players are labeled, and exchanging their hands changes the outcome.

For a favorable deal, first give the four distinct aces to the four players, one each. There are $4! = 4 \cdot 3 \cdot 2 \cdot 1$ assignments. Then distribute the 48 non-aces, giving 12 to each player. The same ordering-and-grouping argument gives $48!/(12!)^4$ assignments of the non-aces. Therefore

$$|A| = 4! \frac{48!}{(12!)^4},$$

where A is the event that each player has an ace.

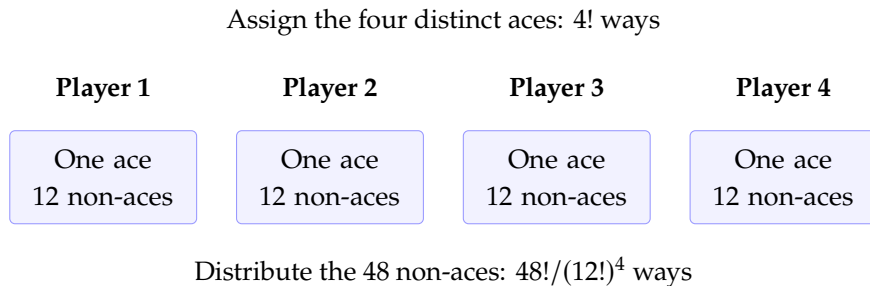


Figure 19: A favorable deal puts one distinct ace and 12 non-aces in each labeled hand. Choose the ace assignment, then the distribution of the remaining cards.

Taking the ratio of favorable to total outcomes and canceling factorials gives

$$\begin{aligned} P(A) &= \frac{4! 48! / (12!)^4}{52! / (13!)^4} \\ &= \frac{4! 13^4}{52 \cdot 51 \cdot 50 \cdot 49} = \frac{2197}{20825} \approx 0.10550. \end{aligned}$$

The numerator and denominator both count unordered hands assigned to labeled players.

Revision. Multiply choices across stages when each stage has a fixed number of extensions. Ordered selections of k distinct items have count $n!/(n-k)!$; unordered selections have count $\binom{n}{k}$. For independent coin tosses, exactly k heads has probability $\binom{n}{k} p^k (1-p)^{n-k}$.

Common mistakes: using a uniform counting ratio without equal likelihood, mixing ordered and unordered outcomes, allowing repetition in only one of the two counts, or dividing by permutations of labeled players.

Check yourself: Why does counting combinations require division by $k!$? Why does the bias cancel after conditioning on three heads? Why is the deck count divided by $(13!)^4$, but not by $4!$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 4: Counting](#), PDF pp. 1–2. The supplied calculations and counting arguments are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 5: DISCRETE RANDOM VARIABLES, PMFs, AND EXPECTATIONS

A probability model describes outcomes. A random variable records a number from each outcome. Its probability mass function collects the probability attached to each number; expectation describes the center of this distribution, and variance describes its spread.

5.1 Random variables: numbers assigned to outcomes

Definition 5.1 [RANDOM VARIABLE]. A RANDOM VARIABLE is a function $X : \Omega \rightarrow \mathbb{R}$. For each outcome ω , the number $X(\omega)$ is the value recorded by X .

The rule X is fixed; the outcome determines which value it takes. Use a capital letter X for the random variable and a lower-case letter x for a possible numerical value. Several random variables can record different numbers from the same outcome.

A random variable is DISCRETE when its possible values form a finite or countably infinite set. Random variables can also take continuous values, but the probability mass functions below describe the discrete case.

5.2 Probability mass functions

To find the probability of a value x , collect the outcomes that produce that value. The event $\{X = x\}$ is the set $\{\omega \in \Omega : X(\omega) = x\}$.

Definition 5.2 [PROBABILITY MASS FUNCTION]. The PROBABILITY MASS FUNCTION, or PMF, of a discrete random variable X is

$$p_X(x) = P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\}).$$

It describes the PROBABILITY DISTRIBUTION of X .

Each value has nonnegative probability, and the probabilities of all possible values add to one:

$$p_X(x) \geq 0, \quad \sum_x p_X(x) = 1.$$

The sum runs over the possible values of X , rather than over individual outcomes. The events $\{X = x\}$ are disjoint and cover the sample space, which explains normalization.

Example: *Tosses until the first head.* Toss a coin independently, with head probability $p > 0$. Let X be the number of tosses up to and including the first head. For $X = k$, the first $k - 1$ tosses must be tails and the k th must be heads. For $0 < p < 1$,

$$p_X(k) = P(\underbrace{TT \cdots T}_{k-1 \text{ tails}} H) = (1 - p)^{k-1} p, \quad k = 1, 2, \dots$$

This is the GEOMETRIC PMF. Its probabilities sum to $p \sum_{j=0}^{\infty} (1 - p)^j = p/[1 - (1 - p)] = 1$. Equivalently, the probability of seeing no head in the first m tosses is $(1 - p)^m$, which tends to zero. A head therefore occurs eventually with probability one. If $p = 1$, then $X = 1$ with probability one.

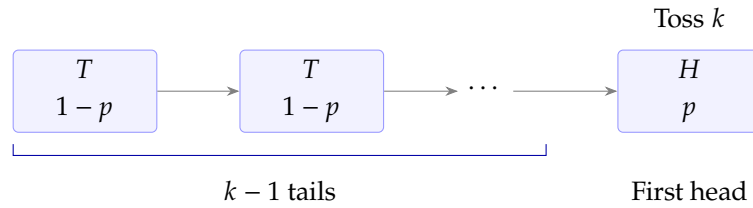


Figure 20: For the event $X = k$, every earlier toss is a tail and the last toss is a head. Independence lets us multiply the probabilities of these required results.

5.3 Computing a PMF by grouping outcomes

For each possible value x , collect all outcomes on which $X = x$ and add their probabilities. Repeat for the other values. Different outcomes can contribute to the same PMF value.

Example: *The minimum of two die rolls.* Roll a fair tetrahedral die twice, independently. Let F and S be the first and second results, and set $X = \min(F, S)$. The 16 ordered pairs (f, s) are equally likely, each with probability $1/16$.

For $X = 2$, both results must be at least 2 and at least one must equal 2. The five pairs are

$$(2, 2), (2, 3), (2, 4), (3, 2), (4, 2),$$

so $p_X(2) = 5/16$. Count $(2, 2)$ once, even though both rolls equal 2. Grouping the other pairs by their minimum gives the complete PMF:

x	1	2	3	4
$p_X(x)$	7/16	5/16	3/16	1/16

These four probabilities sum to one. The values of X are not equally likely, even though the underlying die outcomes are.

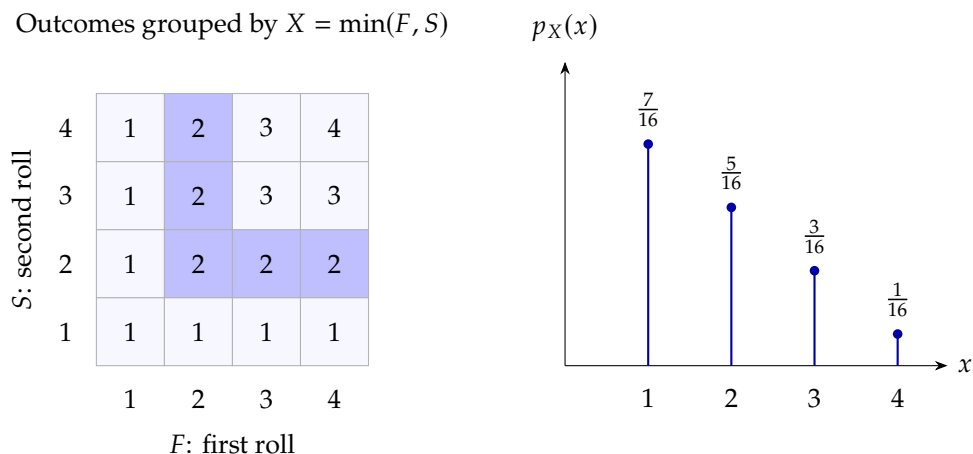


Figure 21: Each grid cell shows $X = \min(F, S)$ and has probability $1/16$. The highlighted cells give $X = 2$. Grouping cells with the same number produces the PMF on the right.

5.4 The binomial PMF

Let X count the heads in n independent coin tosses, with head probability p . A sequence with k heads and $n - k$ tails has probability $p^k(1 - p)^{n-k}$. There are $\binom{n}{k}$ such sequences, one for each choice of the k head positions. Thus the BINOMIAL PMF is

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n, \quad 0 < p < 1.$$

The PMF is zero at other values. At $p = 0$, $X = 0$ with probability one; at $p = 1$, $X = n$ with probability one.

Example: *Two heads in four tosses.* For $n = 4$, the event $X = 2$ consists of six disjoint sequences:

$$HHTT, HTHT, HTTH, THHT, THTH, TTHH.$$

Each has probability $p^2(1 - p)^2$. Adding them gives

$$p_X(2) = 6p^2(1 - p)^2 = \binom{4}{2} p^2(1 - p)^2.$$

The coefficient counts sequences, and the remaining factor gives the probability of one such sequence.

5.5 Expectation: a probability-weighted average

Values with more probability receive more weight in the average.

Definition 5.3 [EXPECTATION]. For a discrete random variable X , its EXPECTATION, OR MEAN, is

$$\mathbb{E}[X] = \sum_x x p_X(x),$$

provided $\sum_x |x| p_X(x) < \infty$.

For finitely many possible values, the sum is automatically finite. For infinite sets of values, the expectation identities below assume that the relevant sums are finite. Think of $\mathbb{E}[X]$ as the center of gravity when mass $p_X(x)$ is placed at position x . It also represents the average approached over many independent repetitions of the same experiment; this interpretation requires a limit result.

Example: *Uniform on $0, 1, \dots, n$.* Let n be a nonnegative integer. Each value has probability $1/(n+1)$, so

$$\mathbb{E}[X] = \frac{0 + 1 + \dots + n}{n+1}.$$

To evaluate the numerator, write $S = 0 + 1 + \dots + n$ and also $S = n + (n-1) + \dots + 0$. Adding the two expressions gives $2S = n(n+1)$. Therefore

$$\mathbb{E}[X] = \frac{n(n+1)/2}{n+1} = \left\lfloor \frac{n}{2} \right\rfloor.$$

Equal probabilities at positions equally far from $n/2$ also explain why this is the center.

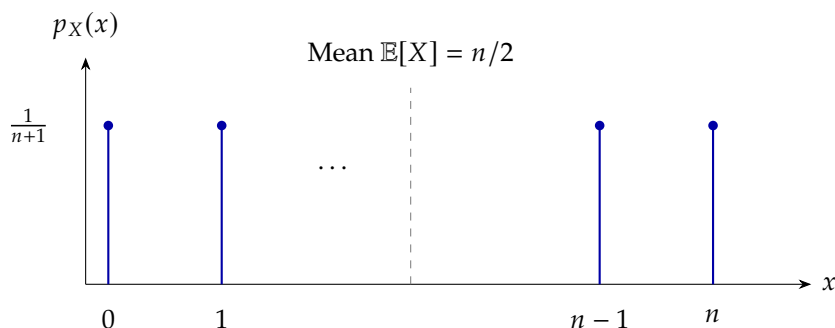


Figure 22: The uniform PMF assigns equal mass to all $n+1$ positions. Its center is $n/2$. The spacing is schematic; omitted positions are indicated by dots.

5.6 Expectations of functions and affine expressions

Suppose $Y = g(X)$. One way to calculate $\mathbb{E}[Y]$ is to find the PMF of Y and sum $y p_Y(y)$. We can instead use the PMF of X directly.

Proposition 5.4 [EXPECTATION OF A FUNCTION]. *If $\sum_x |g(x)|p_X(x) < \infty$, then*

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x).$$

Proof. Group the values of X according to the common value they produce under g . For a possible value y of Y ,

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x).$$

Substitute this into the definition of expectation:

$$\begin{aligned} \mathbb{E}[Y] &= \sum_y y \sum_{x:g(x)=y} p_X(x) \\ &= \sum_y \sum_{x:g(x)=y} g(x)p_X(x) = \sum_x g(x)p_X(x). \end{aligned}$$

Every value of X belongs to one group. Absolute convergence justifies regrouping an infinite sum. $\square$

In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. Taking a function of an average need not give the average of that function. An **AFFINE FUNCTION** has the form $g(x) = \alpha x + \beta$, with constants α, β . For these functions, the two operations agree.

Proposition 5.5 [CONSTANTS AND AFFINE FUNCTIONS]. *For constants α, β and a random variable with finite mean,*

$$\mathbb{E}[\alpha] = \alpha, \quad \mathbb{E}[\alpha X] = \alpha \mathbb{E}[X], \quad \boxed{\mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta}.$$

Proof. The PMF sums to one, so a constant contributes $\sum_x \alpha p_X(x) = \alpha$. Applying the expectation-of-a-function formula to $g(x) = \alpha x + \beta$ gives

$$\sum_x (\alpha x + \beta)p_X(x) = \alpha \sum_x x p_X(x) + \beta \sum_x p_X(x) = \alpha \mathbb{E}[X] + \beta.$$

Taking $\beta = 0$ gives the scaling identity. $\square$

5.7 Variance: squared distance from the mean

The mean locates the center, but does not describe how far values lie from it. Squaring their deviations from the mean prevents positive and negative deviations from canceling.

Definition 5.6 [SECOND MOMENT AND VARIANCE]. The SECOND MOMENT of X is $\mathbb{E}[X^2] = \sum_x x^2 p_X(x)$. When this is finite, write $\mu = \mathbb{E}[X]$. The VARIANCE of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x).$$

The expectation-of-a-function formula applies with $g(x) = (x - \mu)^2$. It also gives a useful way to compute variance.

Proposition 5.7 [VARIANCE IDENTITIES]. For finite second moment,

$$\boxed{\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2}, \quad \text{Var}(X) \geq 0.$$

For constants α, β ,

$$\boxed{\text{Var}(\alpha X + \beta) = \alpha^2 \text{Var}(X)}.$$

Proof. Expand the square inside the defining sum and use $\sum_x x p_X(x) = \mu$ and $\sum_x p_X(x) = 1$:

$$\begin{aligned} \text{Var}(X) &= \sum_x (x^2 - 2\mu x + \mu^2) p_X(x) \\ &= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2. \end{aligned}$$

Nonnegativity follows from the defining sum of nonnegative squared deviations, not from treating the last difference as automatically nonnegative.

For $Y = \alpha X + \beta$, its mean is $\mathbb{E}[Y] = \alpha\mu + \beta$. Consequently

$$Y - \mathbb{E}[Y] = \alpha(X - \mu), \quad \text{Var}(Y) = \mathbb{E}[\alpha^2(X - \mu)^2] = \alpha^2 \text{Var}(X).$$

□

A shift β moves the values and their mean by the same amount, leaving deviations unchanged. Scaling by α multiplies each deviation by α , so squared deviations scale by α^2 .

Revision. A random variable assigns a number to each outcome. Its PMF groups outcomes by that number and adds their probabilities. Expectation is the probability-weighted average; a function's expectation can be computed directly from the original PMF. Variance is the average squared deviation from the mean, equivalently $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Common mistakes: confusing an outcome with a value of X , assuming equally likely outcomes give equally likely values, replacing $\mathbb{E}[g(X)]$ by $g(\mathbb{E}[X])$ for an arbitrary function, or scaling variance by α instead of α^2 .

Check yourself: Which five die outcomes give $X = 2$? Why does the geometric PMF contain $k - 1$ tail factors? Why does adding a constant change the mean but leave variance unchanged?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 5: Discrete Random Variables; Probability Mass Functions; Expectations](#), PDF pp. 1–2. The supplied calculations and expectation and variance arguments are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 6: DISCRETE RANDOM VARIABLE EXAMPLES AND JOINT PMFs

Expectations describe averages, but a function of an average need not equal the average of that function. Conditioning changes the probabilities used in an average. A joint PMF records how two random variables behave together.

6.1 Review: means, deviations, and standard deviation

A random variable X assigns a real number to each outcome. For a discrete X , its PMF is $p_X(x) = P(X = x)$. With the relevant sums finite,

$$\mathbb{E}[X] = \sum_x x p_X(x), \quad \mathbb{E}[g(X)] = \sum_x g(x) p_X(x), \quad \mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta.$$

Writing $\mu = \mathbb{E}[X]$, the deviations from the mean average to zero:

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mu = 0.$$

Their squares need not average to zero. They give the variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x) = \mathbb{E}[X^2] - \mu^2.$$

Definition 6.1 [STANDARD DEVIATION]. The STANDARD DEVIATION of X is

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Variance has squared units; standard deviation has the same units as X .

6.2 Random speed and average travel time

Travel a distance of 200 miles at a speed V that is constant during the trip but random before it begins. The speed is either 1 or 200 miles per hour, each with probability $1/2$.

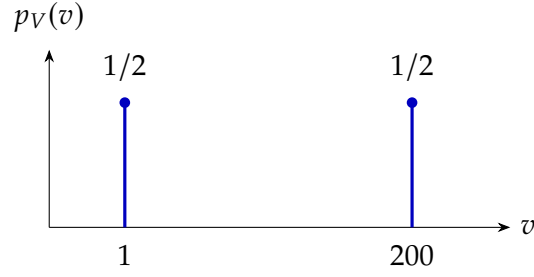


Figure 23: Both speeds have probability $1/2$. The horizontal spacing is schematic; the two values are 1 and 200 miles per hour.

The mean speed is

$$\mathbb{E}[V] = \frac{1}{2}(1) + \frac{1}{2}(200) = \frac{201}{2} = 100.5 \text{ miles per hour.}$$

Both possible speeds are $199/2$ from this mean, so

$$\text{Var}(V) = \frac{1}{2} \left(-\frac{199}{2} \right)^2 + \frac{1}{2} \left(\frac{199}{2} \right)^2 = \frac{39601}{4}, \quad \sigma_V = \frac{199}{2} = 99.5 \text{ miles per hour.}$$

Travel time, in hours, is the function $T = t(V) = 200/V$:

V	1	200
$T = 200/V$	200	1
Probability	$1/2$	$1/2$

Average the times using their probabilities:

$$\mathbb{E}[T] = \sum_v \frac{200}{v} p_V(v) = \frac{1}{2}(200) + \frac{1}{2}(1) = \frac{201}{2} = 100.5 \text{ hours.}$$

Using the mean speed instead would give

$$\frac{200}{\mathbb{E}[V]} = \frac{400}{201} \text{ hours,} \quad \mathbb{E}[200/V] \neq \frac{200}{\mathbb{E}[V]}.$$

The slow trip contributes 200 hours to the possible times, so it strongly affects their average. Taking a reciprocal does not commute with averaging.

Also, $TV = 200$ on every outcome, hence

$$\mathbb{E}[TV] = 200 \neq \mathbb{E}[T]\mathbb{E}[V] = \frac{40401}{4}.$$

The speed and time come from the same trip. Their product must be averaged using the values that occur together.

6.3 Conditional PMFs and conditional expectations

Once we know that an event A occurred, use conditional probabilities as the new weights. Assume $P(A) > 0$.

Definition 6.2 [CONDITIONAL PMF AND EXPECTATION]. The CONDITIONAL PMF of X given A is

$$p_{X|A}(x) = P(X = x | A) = \frac{P(\{X = x\} \cap A)}{P(A)}.$$

Its probabilities sum to one. The CONDITIONAL EXPECTATION is

$$\mathbb{E}[X | A] = \sum_x x p_{X|A}(x),$$

when the corresponding absolute-value sum is finite.

Example: *A uniform variable after conditioning.* Let X be uniform on $\{1, 2, 3, 4\}$, so $p_X(x) = 1/4$ at these four values. Take $A = \{X \geq 2\}$. Then $P(A) = 3/4$. Conditioning removes the value 1 and divides each remaining probability by $3/4$:

$$p_{X|A}(x) = \begin{cases} 1/3, & x \in \{2, 3, 4\}, \\ 0, & \text{otherwise.} \end{cases} \quad \mathbb{E}[X | A] = \frac{2 + 3 + 4}{3} = 3.$$

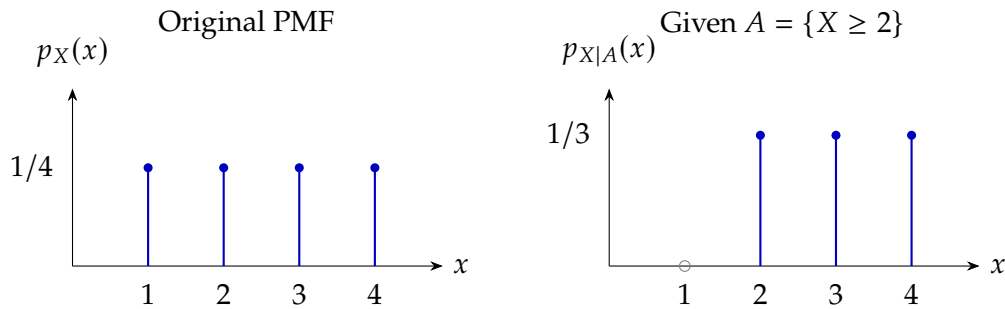


Figure 24: Conditioning on $X \geq 2$ removes the probability at 1. The retained probabilities are rescaled to sum to one.

6.4 The geometric PMF and memorylessness

Toss a coin independently until the first head, with head probability $0 < p < 1$. Let X count all tosses, including the head, and write $q = 1 - p$. The first $k - 1$ tosses must be tails, giving

$$p_X(k) = q^{k-1}p, \quad k = 1, 2, \dots, \quad \mathbb{E}[X] = \sum_{k=1}^{\infty} k q^{k-1} p.$$

Suppose we already know that $X > 2$. The first two tosses were tails, with probability q^2 . For $k \geq 3$,

$$p_{X|X>2}(k) = \frac{q^{k-1}p}{q^2} = q^{k-3}p.$$

$$P(R = j \mid X > 2) = P(X = j + 2 \mid X > 2) = q^{j-1}p.$$

Figure 25: The geometric PMFs are schematic. Conditioning on two tails renormalizes the tail of the original PMF. Subtracting the two elapsed tosses restores its original shape and starting value.

Split the sample space into disjoint events $A_1, \dots, A_n$ whose union is Ω . First average within each event, then weight these averages by how likely the events are. Events of probability zero can be omitted.

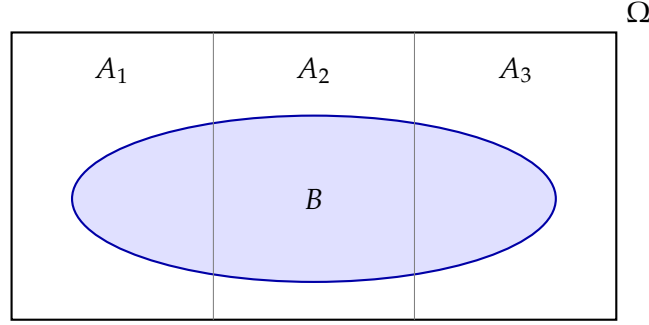


Figure 26: The events A_1, A_2, A_3 partition Ω . The portions of B in these events are disjoint and together make up B .

For any event B , total probability gives

$$P(B) = \sum_{i=1}^n P(A_i)P(B | A_i).$$

Taking $B = \{X = x\}$ yields

$$p_X(x) = \sum_{i=1}^n P(A_i)p_{X|A_i}(x).$$

Theorem 6.3 [TOTAL EXPECTATION]. *For a discrete random variable with finite absolute expectation,*

$$\mathbb{E}[X] = \sum_{i=1}^n P(A_i)\mathbb{E}[X | A_i].$$

Proof. Substitute the total-probability expression for the PMF into the expectation, then collect terms from each event:

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x \sum_{i=1}^n P(A_i)p_{X|A_i}(x) \\ &= \sum_{i=1}^n P(A_i) \sum_x x p_{X|A_i}(x) = \sum_{i=1}^n P(A_i)\mathbb{E}[X | A_i]. \end{aligned}$$

Absolute convergence justifies exchanging the sums. □

Example: *The geometric mean.* Return to the tosses until the first head. Write $m = \mathbb{E}[X]$ and partition by $A_1 = \{X = 1\}$ and $A_2 = \{X > 1\}$. Their probabilities are p and q . On A_1 , $X = 1$. On A_2 , one tail has elapsed, and the remaining tosses start the same experiment. Thus $\mathbb{E}[X | X > 1] = 1 + m$. Total expectation gives

$$m = p \cdot 1 + q(1 + m) = 1 + qm.$$

Since $1 - q = p$, solving gives

$$\boxed{\mathbb{E}[X] = \frac{1}{p}}.$$

If $p = 1$, the first toss is a head with certainty and $\mathbb{E}[X] = 1$, which agrees with this formula.

6.6 Joint PMFs, marginals, and conditioning

Two random variables can record different features of the same outcome. To describe them together, assign a probability to each pair of values.

Definition 6.4 [JOINT PMF]. The JOINT PMF of discrete random variables X, Y is

$$p_{X,Y}(x, y) = P(\{X = x\} \cap \{Y = y\}).$$

It is nonnegative and satisfies

$$\sum_x \sum_y p_{X,Y}(x, y) = 1.$$

The pair events are disjoint and cover the sample space, which explains the double sum.

y	$p_{X,Y}(x, y)$				$p_Y(y)$
4	$\frac{1}{20}$	$\frac{2}{20}$	$\frac{2}{20}$	0	$\frac{5}{20}$
3	$\frac{2}{20}$	$\frac{4}{20}$	$\frac{1}{20}$	$\frac{2}{20}$	$\frac{9}{20}$
2	0	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{1}{20}$	$\frac{5}{20}$
1	0	$\frac{1}{20}$	0	0	$\frac{1}{20}$
	x				
	1	2	3	4	
$p_X(x)$	$\frac{3}{20}$	$\frac{8}{20}$	$\frac{6}{20}$	$\frac{3}{20}$	

Figure 27: Each cell gives the probability of one pair (x, y) . Column sums give $p_X(x)$ and row sums give $p_Y(y)$. The highlighted row contains the pairs retained when conditioning on $Y = 3$.

To find the probability of $X = x$, allow every possible value of Y :

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

These are the **MARGINAL** PMFs. In the displayed example, the column totals are $(3, 8, 6, 3)/20$, which sum to one. For instance, $p_X(3) = (0 + 3 + 1 + 2)/20 = 6/20$.

If $p_Y(y) > 0$, conditional probability gives

$$p_{X|Y}(x | y) = P(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Here the denominator is the whole row's probability, so

$$\sum_x p_{X|Y}(x | y) = \frac{\sum_x p_{X,Y}(x, y)}{p_Y(y)} = 1.$$

For $Y = 3$, the row has probability $9/20$. Its conditional PMF is

x	1	2	3	4
$p_{X Y}(x 3)$	2/9	4/9	1/9	2/9

Marginalizing adds probabilities over the other variable. Conditioning retains one row and rescales it.

Revision. Conditioning changes the weights in a PMF and in an expectation. Total expectation combines conditional means using event probabilities. The geometric remaining waiting time is memoryless, and its mean is $1/p$. A joint PMF gives pair probabilities; marginals sum across the other variable, while a conditional PMF divides a retained row by its probability.

Common mistakes: replacing $\mathbb{E}[200/V]$ by $200/\mathbb{E}[V]$, leaving retained probabilities unnormalized, forgetting to subtract elapsed tosses in the memoryless statement, or using a conditional PMF when the conditioning event has probability zero.

Check yourself: Why is the conditional mean of the uniform example 3? Why does the geometric recursion contain $1 + m$? How do the joint table's highlighted entries become $2/9, 4/9, 1/9, 2/9$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 6: Discrete Random Variable Examples; Joint PMFs](#), PDF pp. 1–2. The supplied calculations and arguments are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 7: MULTIPLE DISCRETE RANDOM VARIABLES: EXPECTATIONS, CONDITIONING, AND INDEPENDENCE

A joint PMF records which values occur together. Independence lets us replace joint probabilities by products of marginal probabilities. Expectations can then be used to count successes or matches without first finding the full distribution of their total.

7.1 Joint PMFs and the multiplication rule

For discrete random variables X, Y ,

$$p_X(x) = P(X = x), \quad p_{X,Y}(x, y) = P(X = x, Y = y).$$

Sum over the other variable to obtain a marginal PMF:

$$p_X(x) = \sum_y p_{X,Y}(x, y).$$

When $p_Y(y) > 0$, the conditional PMF is

$$p_{X|Y}(x | y) = P(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Rearranging the corresponding formula with X as the condition gives

$$p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y | x), \quad p_X(x) > 0.$$

The first factor describes X ; the second uses the information that $X = x$ to describe Y .

For three variables, apply the multiplication rule once more:

$$p_{X,Y,Z}(x, y, z) = p_X(x)p_{Y|X}(y | x)p_{Z|X,Y}(z | x, y).$$

The conditional factors require $p_X(x) > 0$ and $p_{X,Y}(x, y) > 0$. If a conditioning event has probability zero, the joint probability is zero, but the conditional expression is undefined.

7.2 Independence and the effect of conditioning

Under independence, knowing the values of some variables does not change the probabilities for the others.

Definition 7.1 [INDEPENDENT RANDOM VARIABLES]. Discrete random variables X, Y are INDEPENDENT RANDOM VARIABLES if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y.$$

Three variables X, Y, Z are independent if

$$p_{X,Y,Z}(x, y, z) = p_X(x)p_Y(y)p_Z(z) \quad \text{for every } x, y, z.$$

For two independent variables and $p_X(x) > 0$, division gives $p_{Y|X}(y | x) = p_Y(y)$. The product condition itself also applies at values of probability zero. It must hold for *every* pair, rather than just one selected pair.

Example: *A joint table before and after conditioning.* Consider the joint PMF on $\{1, 2, 3, 4\}^2$ shown below; all other pairs have probability zero. Its marginals give $p_X(1) = 3/20$ and $p_Y(1) = 1/20$. Yet $p_{X,Y}(1, 1) = 0$, whereas $p_X(1)p_Y(1) = 3/400 > 0$. Thus X, Y are not independent.

Now condition on $A = \{X \leq 2, Y \geq 3\}$. The retained probabilities sum to $(1 + 2 + 2 + 4)/20 = 9/20$. Dividing by $P(A)$ gives

	$x = 1$	$x = 2$
$y = 4$	$1/9$	$2/9$
$y = 3$	$2/9$	$4/9$

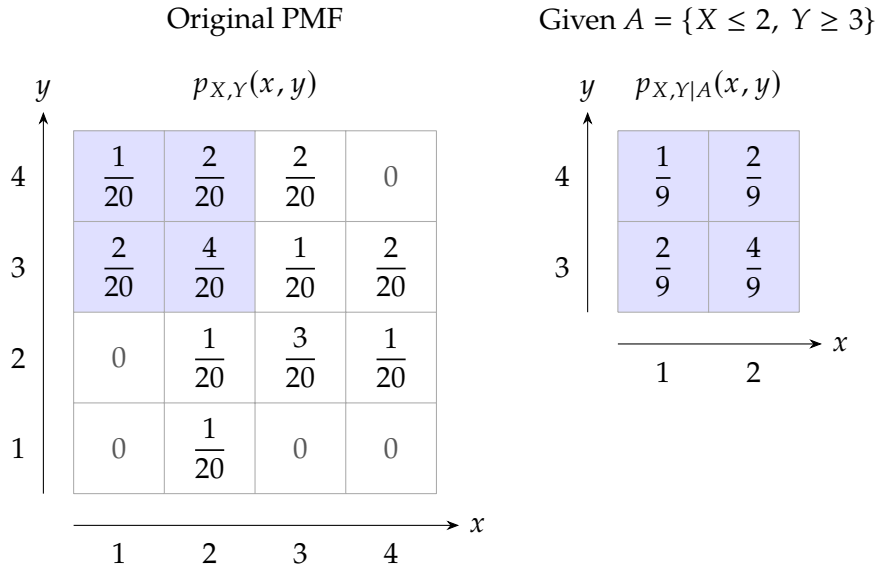


Figure 28: The highlighted cells are the event A . Conditioning keeps these four probabilities and divides them by $9/20$. The resulting two-by-two table factors into its conditional marginals.

The conditional marginals are

$$\begin{aligned} p_{X|A}(1) &= 1/3, & p_{X|A}(2) &= 2/3, \\ p_{Y|A}(3) &= 2/3, & p_{Y|A}(4) &= 1/3. \end{aligned}$$

Their four products are precisely the four entries above; all other entries and the corresponding products are zero. Therefore

$$p_{X,Y|A}(x, y) = p_{X|A}(x)p_{Y|A}(y) \quad \text{for every } x, y.$$

The variables are independent under the conditional probability law, even though they are dependent under the original law.

7.3 Expectations of functions, sums, and products

To average a function of two variables, weight its value at each pair by the joint probability of that pair. Assume the relevant expectation sums converge absolutely.

Proposition 7.2 [EXPECTATION OF A FUNCTION OF TWO VARIABLES]. *For $W = g(X, Y)$,*

$$\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y)p_{X,Y}(x, y).$$

Proof. Collect the pairs that give the same value w of W :

$$p_W(w) = \sum_{(x,y):g(x,y)=w} p_{X,Y}(x, y).$$

Substitute into $\mathbb{E}[W] = \sum_w wp_W(w)$. Each pair belongs to one group, and within that group $w = g(x, y)$. Regrouping gives the stated double sum. $\square$

In general, $\mathbb{E}[g(X, Y)]$ need not equal $g(\mathbb{E}[X], \mathbb{E}[Y])$. Affine functions satisfy $\mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta$. Sums of variables have a further useful property.

Proposition 7.3 [LINEARITY OF EXPECTATION]. *For variables with finite absolute expectations, no independence assumption is needed for*

$$\mathbb{E}[X + Y + Z] = \mathbb{E}[X] + \mathbb{E}[Y] + \mathbb{E}[Z].$$

Proof. For two variables, split the joint-PMF sum and then use the marginals:

$$\begin{aligned}\mathbb{E}[X + Y] &= \sum_x \sum_y (x + y) p_{X,Y}(x, y) \\ &= \sum_x x \sum_y p_{X,Y}(x, y) + \sum_y y \sum_x p_{X,Y}(x, y) \\ &= \mathbb{E}[X] + \mathbb{E}[Y].\end{aligned}$$

Apply this identity first to $X + Y$ and Z , then to X and Y . □

Products require more care: values that occur together matter. Independence makes the double sum factor.

Proposition 7.4 [PRODUCTS UNDER INDEPENDENCE]. *If X, Y are independent and $g(X), h(Y)$ have finite absolute expectations, then*

$$\boxed{\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]}.$$

In particular, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ when X, Y have finite absolute expectations.

Proof. Replace the joint PMF by the product of its marginals:

$$\begin{aligned}\mathbb{E}[g(X)h(Y)] &= \sum_x \sum_y g(x)h(y)p_X(x)p_Y(y) \\ &= \left(\sum_x g(x)p_X(x) \right) \left(\sum_y h(y)p_Y(y) \right).\end{aligned}$$

The corresponding absolute-value sum is the product of two finite sums, so the factorization is justified. □

7.4 Variances of sums

A constant shift leaves deviations from the mean unchanged; scaling by a scales their squares by a^2 . Thus

$$\text{Var}(X + a) = \text{Var}(X), \quad \text{Var}(aX) = a^2 \text{Var}(X).$$

For a sum, expand its squared deviation. This reveals where independence enters.

Proposition 7.5 [VARIANCE OF AN INDEPENDENT SUM]. *If X, Y are independent with finite second moments, then*

$$\boxed{\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)}.$$

Proof. Write $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Linearity gives the mean of $X + Y$, so expanding its squared deviation yields

$$\begin{aligned}\text{Var}(X + Y) &= \mathbb{E}[(X - \mu_X) + (Y - \mu_Y)]^2 \\ &= \text{Var}(X) + \text{Var}(Y) + 2\mathbb{E}[(X - \mu_X)(Y - \mu_Y)].\end{aligned}$$

Independence factors the last expectation into $\mathbb{E}[X - \mu_X]\mathbb{E}[Y - \mu_Y] = 0$. This removes the cross term. $\square$

The independence assumption cannot simply be dropped:

- If $X = Y$, then $X + Y = 2X$, so $\text{Var}(X + Y) = 4 \text{Var}(X)$.
- If $X = -Y$, then $X + Y = 0$, so $\text{Var}(X + Y) = 0$.
- If X, Y are independent and $Z = X - 3Y$, scaling Y and applying the independent-sum rule gives $\text{Var}(Z) = \text{Var}(X) + 9 \text{Var}(Y)$.

These cases distinguish adding means, which always works for finite absolute expectations, from adding variances.

7.5 Binomial mean and variance using indicators

Let X count the successes in n independent trials, each with success probability p . For $0 < p < 1$, a direct PMF calculation gives

$$\mathbb{E}[X] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}.$$

We can compute the mean through the individual trials. Define the INDICATOR RANDOM VARIABLE

$$X_i = \begin{cases} 1, & \text{trial } i \text{ succeeds,} \\ 0, & \text{otherwise.} \end{cases} \quad X = \sum_{i=1}^n X_i.$$

An indicator turns an event into a number. Its expectation is the event's probability:

$$\mathbb{E}[X_i] = 1 \cdot p + 0 \cdot (1-p) = p.$$

Apply linearity repeatedly:

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = np.$$

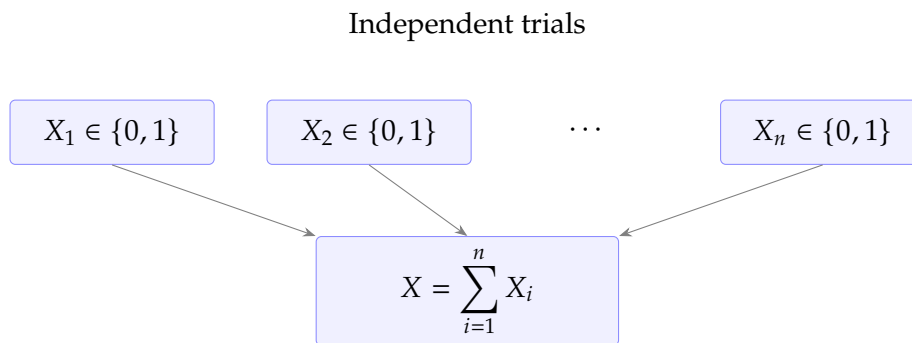


Figure 29: Each trial contributes 0 or 1 to the success count. Linearity adds the indicator means; independence is used when adding their variances.

Since $X_i^2 = X_i$, its second moment is also p . Therefore

$$\text{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = p - p^2 = p(1 - p).$$

The independent trials make the indicators independent. In the squared deviation of their sum, every cross term has expectation zero by the same product argument used for two variables. Hence

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p).$$

At $p = 0$ or $p = 1$, the success count is constant, and these mean and variance formulas still hold. They also give zero mean and variance when $n = 0$.

7.6 The hat problem: an expected number of matches

There are $n \geq 1$ people, each with a distinct hat. Put the hats in a box and return them at random, without replacement. Every assignment of the n hats to the n people is equally likely. Let X count the people who receive their own hats.

Define indicators for the individual matches:

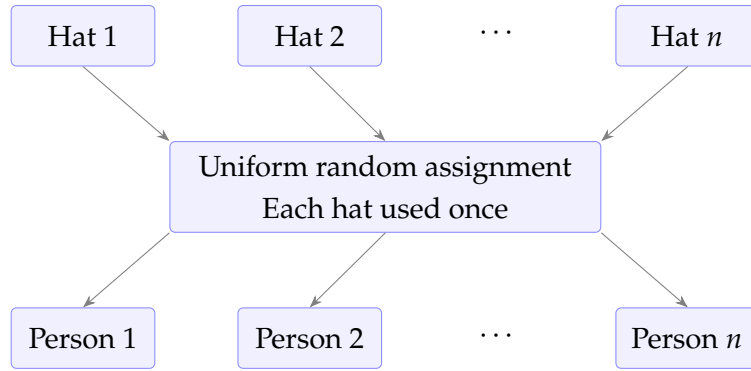
$$X_i = \begin{cases} 1, & \text{person } i \text{ receives their own hat,} \\ 0, & \text{otherwise.} \end{cases} \quad X = X_1 + \cdots + X_n.$$

Each person is equally likely to receive any of the n hats, so

$$P(X_i = 1) = \frac{1}{n}, \quad \mathbb{E}[X_i] = \frac{1}{n}.$$

For $n \geq 2$, these indicators are not independent. Given that person 1 receives their own hat, person 2 has probability $1/(n - 1)$ of receiving theirs, rather than $1/n$. Nevertheless, linearity gives

$$\mathbb{E}[X] = \sum_{i=1}^n \frac{1}{n} = 1.$$



$X_i = 1$ exactly when person i receives hat i

Figure 30: All hats are used once in a uniformly random assignment. The match indicators share this assignment, so they are dependent for $n \geq 2$. Their means can still be added.

7.7 The hat problem: variance and pair probabilities

To find the variance, compute the second moment of the total. Assume $n \geq 2$. Since $\mathbb{E}[X] = 1$,

$$\text{Var}(X) = \mathbb{E}[X^2] - 1.$$

Expand the square before taking expectations:

$$X^2 = \sum_{i=1}^n X_i^2 + \sum_{i \neq j} X_i X_j.$$

The second sum uses *ordered* pairs: both (i, j) and (j, i) appear. There are $n(n - 1)$ terms.

An indicator satisfies $X_i^2 = X_i$, so $\mathbb{E}[X_i^2] = 1/n$. For two distinct people, $X_i X_j$ is itself the indicator that both receive their own hats. Conditioning on the first match gives

$$\begin{aligned} \mathbb{E}[X_i X_j] &= P(X_i = 1, X_j = 1) \\ &= P(X_i = 1)P(X_j = 1 \mid X_i = 1) \\ &= \frac{1}{n} \frac{1}{n-1} = \frac{1}{n(n-1)}. \end{aligned}$$

After fixing one correct hat, the remaining $n - 1$ hats are still uniformly assigned to the remaining people. In particular, this joint probability is not $1/n^2$.

Now sum the individual and pair contributions:

$$\mathbb{E}[X^2] = n \frac{1}{n} + n(n-1) \frac{1}{n(n-1)} = 2.$$

Consequently,

$$\boxed{\text{Var}(X) = 2 - 1 = 1 \quad (n \geq 2)}.$$

When $n = 1$, the sole person always receives their own hat: $X = 1$ and $\text{Var}(X) = 0$. The pair calculation does not apply in this case.

Revision. Independence factors the entire joint PMF. Conditioning can change whether that factorization holds. Means add without independence; products factor and variances add under the stated independence assumptions. Indicator variables express a count as a sum of event occurrences. For the binomial count, the mean is np and variance is $np(1-p)$. In the hat problem, the mean is 1 and the variance is 1 for $n \geq 2$.

Common mistakes: checking independence at only one favorable pair, assuming conditioning preserves it, requiring independence for linearity, adding dependent variances, or counting unordered pairs in the ordered-pair expansion of X^2 .

Check yourself: Why does the conditioned joint table factor? Which step in the variance proof needs independence? Why is a pair of correct hats more likely than $1/n^2$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 7: Multiple Discrete Random Variables: Expectations, Conditioning, Independence](#), PDF pp. 1–2. The supplied arguments and calculations are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 8: CONTINUOUS RANDOM VARIABLES

For a continuous random variable, probability is spread over intervals. A density describes how concentrated it is near each value; integrating the density gives probability. A cumulative distribution function records the probability accumulated up to a chosen value.

8.1 Probability as area under a density

Definition 8.1 [CONTINUOUS RANDOM VARIABLE AND DENSITY]. A random variable X is CONTINUOUS if its probabilities can be described by a PROBABILITY DENSITY FUNCTION, or PDF, f_X , with

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1,$$

such that, for $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

The total area is one. The area over a particular interval is the probability that the value of X belongs to that interval. More generally, for a set B over which the integral is defined,

$$P(X \in B) = \int_B f_X(x) dx.$$

For a union of disjoint intervals, add their areas.

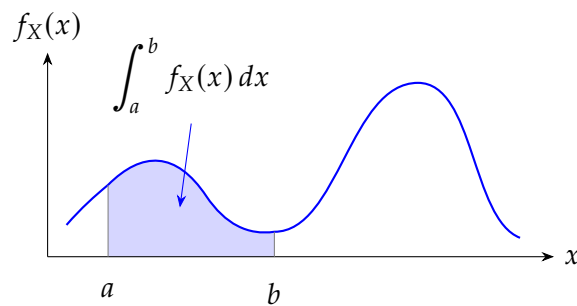


Figure 31: The shaded area represents $P(a \leq X \leq b)$. The curve is schematic: its height describes density, while its area describes probability.

A PDF value is not a point probability. An interval of zero length has zero area, so $P(X = x) = 0$ for every x . Including or excluding an interval's endpoints therefore gives the same probability. A density may exceed one; its integral, rather than its height, must equal one.

For $\delta > 0$,

$$P(x \leq X \leq x + \delta) = \int_x^{x+\delta} f_X(s) ds \approx f_X(x)\delta.$$

The approximation replaces a short strip of area by a rectangle. It is accurate when the density varies little across the interval, for example near a point where it is continuous as δ becomes small.

8.2 Means, variances, and the continuous uniform law

The expectation formulas use the same weighting idea as for a PMF, with an integral in place of a sum:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

These formulas assume that the corresponding absolute-value integrals are finite. With $\mu = \mathbb{E}[X]$ and finite second moment,

$$\text{Var}(X) = \sigma_X^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f_X(x) dx.$$

Thus variance still averages squared distances from the mean.

Example: *Uniform on $[a, b]$.* Suppose $a < b$ and the density is constant on this interval and zero outside it. This is the CONTINUOUS UNIFORM DISTRIBUTION. Normalization gives the height: if it is c , then $c(b - a) = 1$. Hence

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

The mean is the midpoint:

$$\mathbb{E}[X] = \frac{1}{b-a} \int_a^b x dx = \frac{b^2 - a^2}{2(b-a)} = \boxed{\frac{a+b}{2}}.$$

To compute variance, write $m = (a + b)/2$ and $L = b - a$. The substitution $u = x - m$ centers the interval:

$$\begin{aligned} \text{Var}(X) &= \frac{1}{L} \int_{-L/2}^{L/2} u^2 du = \frac{1}{L} \left[\frac{u^3}{3} \right]_{-L/2}^{L/2} \\ &= \frac{L^2}{12} = \boxed{\frac{(b-a)^2}{12}}. \end{aligned}$$

8.3 Cumulative distribution functions

A CDF describes the probability to the left of a value, and works for both continuous and discrete variables.

Definition 8.2 [CUMULATIVE DISTRIBUTION FUNCTION]. The CUMULATIVE DISTRIBUTION FUNCTION, or CDF, of X is

$$F_X(x) = P(X \leq x).$$

For a continuous variable with density f_X ,

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

As x increases, the event $\{X \leq x\}$ grows, so the CDF cannot decrease. Its limits at $-\infty$ and ∞ are zero and one. For $a < b$,

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

This subtracts the probability already accumulated by a . For a continuous variable, endpoints do not matter. Where the density is continuous, differentiation recovers it: $F'_X(x) = f_X(x)$.

For the uniform distribution on $[a, b]$, the accumulated area gives

$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

The CDF is flat before the interval, increases linearly through it, and is flat again once all the probability has been included.

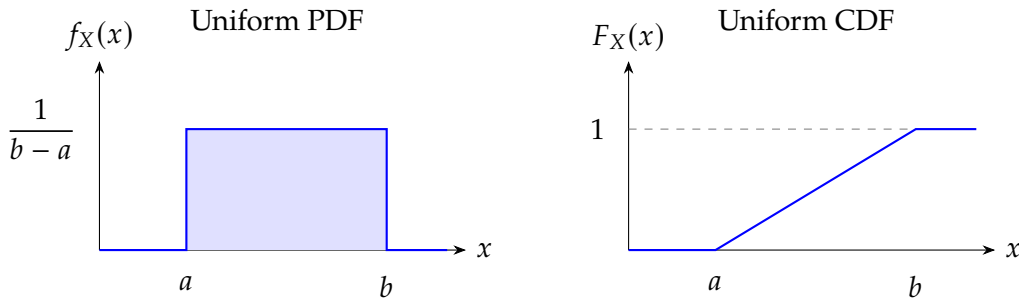


Figure 32: A constant uniform density produces a linear increase in the CDF. The density has total area one; the CDF reaches one at b . Horizontal spacing is schematic.

For a discrete variable, accumulate masses instead of areas:

$$F_X(x) = \sum_{k \leq x} p_X(k),$$

where the sum runs over its possible values. Each positive mass produces a jump at that value.

Example: *Masses at 1, 2, and 4.* Let $p_X(1) = 1/6$, $p_X(2) = 3/6$, and $p_X(4) = 2/6$. These masses sum to one. The CDF is

$$F_X(x) = \begin{cases} 0, & x < 1, \\ 1/6, & 1 \leq x < 2, \\ 4/6, & 2 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

At a mass location, F_X includes that mass because its defining event uses $X \leq x$.

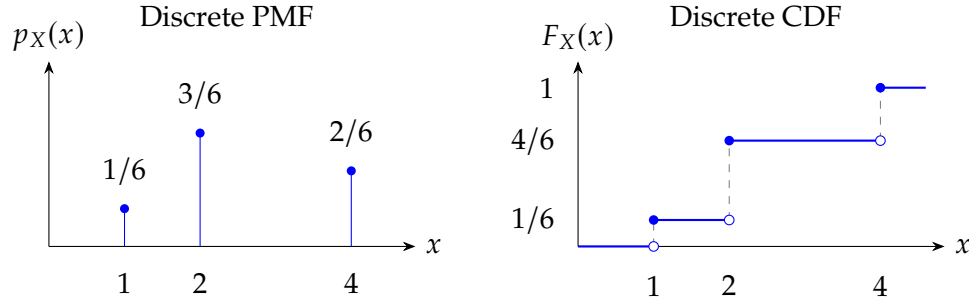


Figure 33: The jump sizes are $1/6$, $3/6$, and $2/6$. Filled circles show the CDF's value at each jump; open circles show the value approached from the left.

8.4 Mixed distributions: area and point mass

A distribution can have both a continuous part and point masses. Such a **MIXED DISTRIBUTION** is described by adding the two kinds of probability. Its CDF has continuous increases as well as jumps.

Consider probability $1/2$ spread uniformly over $[0, 1]$, with the other $1/2$ concentrated at $x = 1/2$. The continuous part has density height $1/2$ on $[0, 1]$, so its area is $1/2$. For a set B ,

$$P(X \in B) = \frac{1}{2} \int_{B \cap [0,1]} 1 \, dx + \frac{1}{2} \mathbf{1}_{\{1/2 \in B\}},$$

where the indicator is one if B contains $1/2$, and zero otherwise. The point mass is not an ordinary density height.

For $0 \leq x < 1/2$, the CDF accumulates only the continuous area $x/2$. At $1/2$, it also includes the point mass. Thus

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x/2, & 0 \leq x < 1/2, \\ x/2 + 1/2, & 1/2 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

Immediately before $1/2$, the accumulated probability approaches $1/4$. At $1/2$, it is $3/4$. The jump of $1/2$ is exactly the point mass. The remaining continuous area brings the CDF to one.

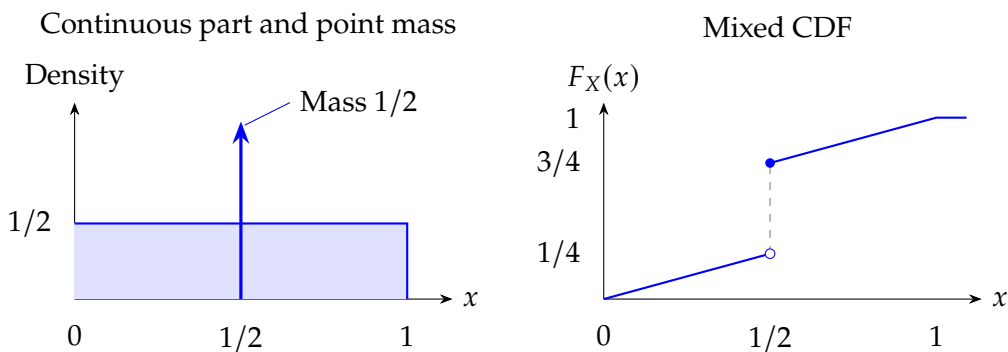


Figure 34: The arrow marks a point mass of $1/2$; its drawn height has no density meaning. The continuous part contributes area $1/2$. The CDF combines a linear increase with a jump of $1/2$.

8.5 Gaussian densities and their moments

The normal family has a symmetric, bell-shaped density. Its parameters specify the center and spread.

Definition 8.3 [STANDARD NORMAL DISTRIBUTION]. A STANDARD NORMAL RANDOM VARIABLE Z , written $Z \sim N(0, 1)$, has density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in \mathbb{R}.$$

Its CDF is denoted by

$$\Phi(z) = \int_{-\infty}^z \phi(t) dt.$$

The normalization constant makes the total area one. Since ϕ is even, $z\phi(z)$ is odd and its integral is zero. Therefore $\mathbb{E}[Z] = 0$. For the second moment, use $\phi'(z) = -z\phi(z)$ and integration by parts:

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \phi(z) dz = \left[-z\phi(z) \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z) dz = 1.$$

The boundary term vanishes because $ze^{-z^2/2}$ tends to zero at both ends. Hence $\text{Var}(Z) = 1$.

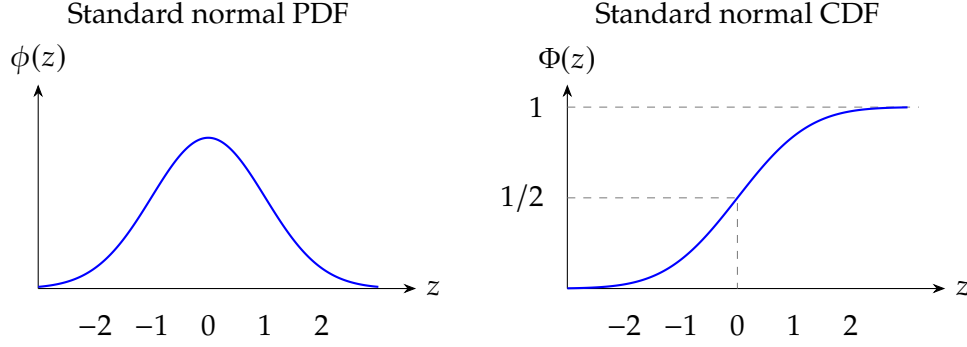


Figure 35: The standard normal density is symmetric about zero. Its CDF is smooth and increasing, with $\Phi(0) = 1/2$. Both curves are drawn from their defining formulas.

Definition 8.4 [GENERAL NORMAL DISTRIBUTION]. A NORMAL, or GAUSSIAN, random variable $X \sim N(\mu, \sigma^2)$, with $\sigma > 0$, has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

The substitution $z = (x - \mu)/\sigma$ changes $f_X(x) dx$ into $\phi(z) dz$. Thus

$$\mathbb{E}[X] = \mu + \sigma\mathbb{E}[Z] = \mu, \quad \text{Var}(X) = \sigma^2\mathbb{E}[Z^2] = \sigma^2.$$

The second parameter of $N(\mu, \sigma^2)$ is the variance; σ is the standard deviation.

For $Y = aX + b$, the expectation and variance rules give

$$\mathbb{E}[Y] = a\mu + b, \quad \text{Var}(Y) = a^2\sigma^2.$$

When $a \neq 0$, the transformed variable is again normal:

$$\boxed{Y \sim N(a\mu + b, a^2\sigma^2)}.$$

Indeed, after centering and dividing by its positive standard deviation $|a|\sigma$,

$$\frac{Y - (a\mu + b)}{|a|\sigma} = \frac{a}{|a|} \frac{X - \mu}{\sigma}.$$

This is either Z or $-Z$. The symmetric standard normal density gives them the same distribution. If $a = 0$, then $Y = b$ is constant with variance zero, so it has no nondegenerate normal density.

8.6 Normal probabilities by standardization

The normal CDF has no elementary antiderivative. Numerical values of Φ let us compute probabilities for every normal distribution. For $X \sim N(\mu, \sigma^2)$, STANDARDIZATION means

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Subtracting the mean centers the variable; dividing by its standard deviation expresses distance from the mean in units of spread. Because $\sigma > 0$, the direction of an inequality is preserved:

$$P(X \leq x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Example: A normal variable with mean 2 and variance 16. If $X \sim N(2, 16)$, then its standard deviation is 4. Hence

$$P(X \leq 3) = P\left(\frac{X - 2}{4} \leq \frac{3 - 2}{4}\right) = \Phi(0.25) \approx \boxed{0.5987}.$$

In the table, use row 0.2 and column 0.05: together they give the argument 0.25. The entry is accumulated probability to the left, not a density height.

Table 1: Standard normal CDF $\Phi(z)$, rounded to four decimal places. Add the row label and column label to obtain z . The bold entry is $\Phi(0.25)$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

8.7 How the descriptions fit together

A PMF assigns mass to values; a PDF assigns density whose integral gives probability. Both lead to a CDF, a mean, and a variance. The notation also distinguishes a single-variable description from a joint or conditional one:

Description	Discrete	Continuous
Single variable	$p_X(x)$	$f_X(x)$
Joint	$p_{X,Y}(x, y)$	$f_{X,Y}(x, y)$
Conditional	$p_{X Y}(x y)$	$f_{X Y}(x y)$

The CDF notation $F_X(x) = P(X \leq x)$ and the meanings of $\mathbb{E}[X]$ and $\text{Var}(X)$ are shared by both columns.

Revision. Integrate a density to get probability, and weight it by x or $(x - \mu)^2$ to get mean or variance. A CDF accumulates probability: densities give continuous increases, while point masses give jumps. The uniform law on $[a, b]$ has mean $(a + b)/2$ and variance $(b - a)^2/12$. A normal variable has mean μ and variance σ^2 ; subtract μ and divide by σ to use Φ .

Common mistakes: treating a PDF height as probability, forgetting a mixed distribution's point mass, reading a CDF jump at its lower endpoint, or dividing by variance instead of standard deviation when standardizing.

Check yourself: Why do endpoints not affect a continuous interval probability? Why is the mixed CDF's value at $1/2$ equal to $3/4$? Which row and column give $\Phi(0.25)$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 8: Continuous Random Variables](#), PDF pp. 1–2. The supplied definitions and calculations are completed; the diagrams are redrawn and the displayed normal-table values retained. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 9: MULTIPLE CONTINUOUS RANDOM VARIABLES

A joint density describes two values together. Integrating over one variable gives the density of the other. Fixing one value and normalizing the resulting slice gives a conditional density.

9.1 Review: densities and expectations

The discrete and continuous descriptions use parallel notation:

Description	Discrete	Continuous
Single variable	$p_X(x)$	$f_X(x)$
Joint	$p_{X,Y}(x, y)$	$f_{X,Y}(x, y)$
Given an event A	$p_{X A}(x)$	$f_{X A}(x)$
Given a value y	$p_{X Y}(x y)$	$f_{X Y}(x y)$

Both descriptions share the CDF $F_X(x) = P(X \leq x)$ and the meanings of expectation and variance.

For a continuous random variable X , probabilities are areas under a nonnegative density whose total integral is one:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

The event collects all outcomes whose recorded value lies in $[a, b]$. Its probability becomes area over that interval in the value space.

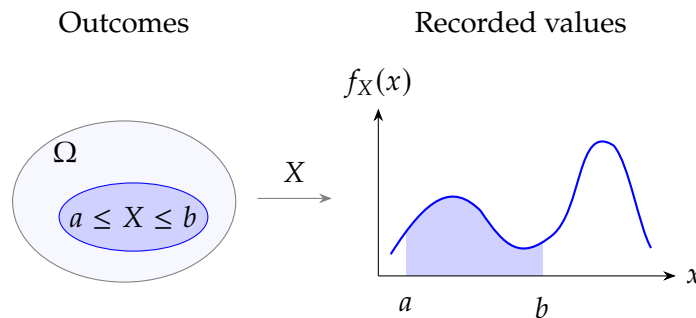


Figure 36: The event in the sample space consists of outcomes with $a \leq X \leq b$. Its probability is represented by the shaded area under the PDF. Both drawings are schematic.

For a short interval on which the density varies little,

$$P(x \leq X \leq x + \delta) \approx f_X(x)\delta.$$

Weighting the density by a function gives its expectation:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx,$$

provided the corresponding absolute-value integral is finite. Taking $g(x) = x$ gives the mean; taking $g(x) = (x - \mathbb{E}[X])^2$ gives the variance when the second moment is finite.

9.2 Joint densities, marginals, and independence

For two variables, probability is spread over the plane of pairs (x, y) . A double integral collects the probability in a region.

Definition 9.1 [JOINT PROBABILITY DENSITY]. A JOINT PROBABILITY DENSITY FUNCTION $f_{X,Y}$ is nonnegative and satisfies

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy dx = 1.$$

For a region S over which the integral is defined,

$$P((X, Y) \in S) = \iint_S f_{X,Y}(x, y) dx dy.$$

The density height is not the probability of the pair. For a small square of side δ , where the density varies little,

$$P(x \leq X \leq x + \delta, y \leq Y \leq y + \delta) \approx f_{X,Y}(x, y) \delta^2.$$

The factor δ^2 is the square's area. Expectations similarly weight each pair by its joint density:

$$\mathbb{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dy dx,$$

assuming the absolute-value double integral is finite.

To describe X alone, allow every possible value of Y . The event $\{a \leq X \leq b\}$ is a vertical strip in the pair plane:

$$P(a \leq X \leq b) = \int_a^b \left(\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right) dx.$$

The quantity in parentheses therefore serves as the density of X . These single-variable densities are the MARGINAL DENSITIES:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

Marginalizing adds all contributions from the variable being removed. It does not select one value of that variable.

Definition 9.2 [INDEPENDENT CONTINUOUS VARIABLES]. Variables X, Y with a joint density are INDEPENDENT if their joint density factors as

$$f_{X,Y}(x, y) = f_X(x)f_Y(y).$$

The equality may be taken throughout the plane; changing density values on boundaries of zero area does not change probabilities.

For independent variables, a rectangle probability separates into two interval probabilities:

$$\begin{aligned} P(a \leq X \leq b, c \leq Y \leq d) &= \int_a^b \int_c^d f_X(x)f_Y(y) dy dx \\ &= \left(\int_a^b f_X(x) dx \right) \left(\int_c^d f_Y(y) dy \right). \end{aligned}$$

Thus information restricting one variable to an interval does not change the other's interval probabilities, when the conditioning event has positive probability.

9.3 Buffon's needle

Example: *A needle crossing parallel lines.* Parallel lines are spaced a distance $d > 0$ apart. A needle has length ℓ , with $0 < \ell < d$. Find the probability that it intersects a line.

Use two variables to describe its placement. Let X be the distance from its midpoint to the nearest line, so $0 \leq X \leq d/2$. Let Θ be its acute angle to the lines, with $0 \leq \Theta \leq \pi/2$. Model these variables as uniform and independent. Their densities and joint density are

$$f_X(x) = \frac{2}{d}, \quad f_\Theta(\theta) = \frac{2}{\pi}, \quad f_{X,\Theta}(x, \theta) = \frac{4}{\pi d}$$

on the rectangle $0 \leq x \leq d/2, 0 \leq \theta \leq \pi/2$, and zero outside it.

Half the needle has length $\ell/2$. Its component perpendicular to the lines is $(\ell/2) \sin \Theta$. The needle reaches the nearest line exactly when

$$X \leq \frac{\ell}{2} \sin \Theta.$$

Since $\ell < d$, this threshold is always below $d/2$. If the needle reaches any line, it reaches a nearest one.

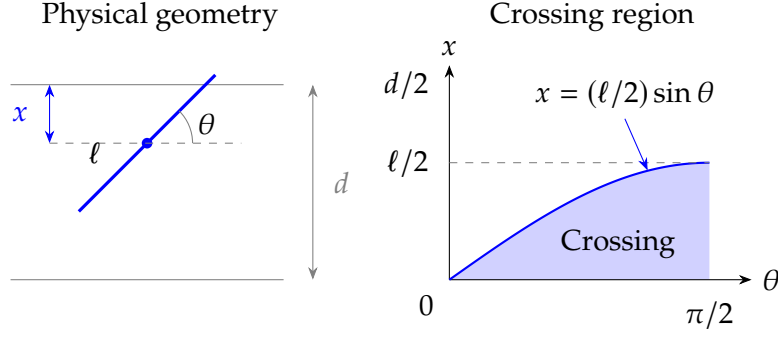


Figure 37: The distance x is measured perpendicular to a nearest line; θ is measured from a parallel direction. In the value plane, crossing occurs below $x = (\ell/2) \sin \theta$. The geometry is schematic.

For each angle, integrate over the distances that permit crossing, then integrate over all angles:

$$\begin{aligned}
 P(\text{crossing}) &= \frac{4}{\pi d} \int_0^{\pi/2} \int_0^{(\ell/2) \sin \theta} 1 \, dx \, d\theta \\
 &= \frac{4}{\pi d} \int_0^{\pi/2} \frac{\ell}{2} \sin \theta \, d\theta \\
 &= \frac{2\ell}{\pi d} [-\cos \theta]_0^{\pi/2} = \boxed{\frac{2\ell}{\pi d}}.
 \end{aligned}$$

The joint density is constant, so this integral multiplies the crossing region's area by its density.

9.4 Conditional densities: normalize a slice

For a continuous variable, $P(Y = y) = 0$. The ordinary event ratio cannot define conditioning on $Y = y$. Instead, use the density near that value.

To see the ratio, restrict Y to a short interval $[y, y + \varepsilon]$. Near a point where the densities are continuous and $f_Y(y) > 0$,

$$\begin{aligned}
 P(x \leq X \leq x + \delta \mid y \leq Y \leq y + \varepsilon) \\
 \approx \frac{f_{X,Y}(x, y) \delta \varepsilon}{f_Y(y) \varepsilon} = \frac{f_{X,Y}(x, y)}{f_Y(y)} \delta.
 \end{aligned}$$

This suggests a density obtained by dividing a slice by its total area.

Definition 9.3 [CONDITIONAL PROBABILITY DENSITY]. For a value y with $f_Y(y) > 0$, the **CONDITIONAL PROBABILITY DENSITY FUNCTION** of X given $Y = y$ is

$$\boxed{f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}}.$$

For fixed y , the numerator is a slice through the joint density. Its integral over x is $f_Y(y)$, so division normalizes it:

$$\int_{-\infty}^{\infty} f_{X|Y}(x | y) dx = \frac{1}{f_Y(y)} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = 1.$$

The conditional interval probability is the area under this density:

$$P(a \leq X \leq b | Y = y) = \int_a^b f_{X|Y}(x | y) dx.$$

Interchanging the roles of the variables gives $f_{Y|X}(y | x) = f_{X,Y}(x, y)/f_X(x)$ when $f_X(x) > 0$.

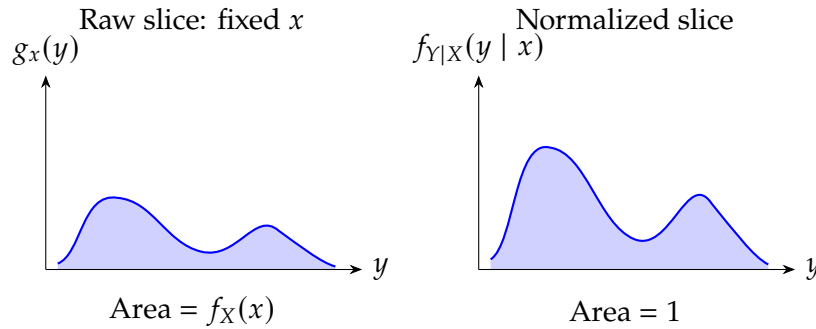


Figure 38: For fixed x , the slice $g_x(y) = f_{X,Y}(x, y)$ has area $f_X(x)$. Dividing by this area gives the conditional density of Y , with area one. Curves and vertical scales are schematic.

For independent variables and $f_Y(y) > 0$,

$$f_{X|Y}(x | y) = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x).$$

The conditional density is unchanged by the observed value. For dependent variables, different slices can give different conditional densities.

9.5 Stick breaking: construct the joint density

Example: *Two successive breaks.* Start with a stick of length $\ell > 0$. Choose a first break point X uniformly along $[0, \ell]$. Keep the part from 0 to X , and choose a second break point Y uniformly along $[0, X]$, conditional on the first break. Both coordinates are measured from the same left endpoint.

The first density is $f_X(x) = 1/\ell$ for $0 < x < \ell$. Given $X = x > 0$, the second break is uniform on an interval of length x , so

$$f_{Y|X}(y | x) = \begin{cases} 1/x, & 0 < y < x, \\ 0, & \text{otherwise.} \end{cases}$$

The possibility $X = 0$ has probability zero. Density values at individual endpoints do not affect these distributions.

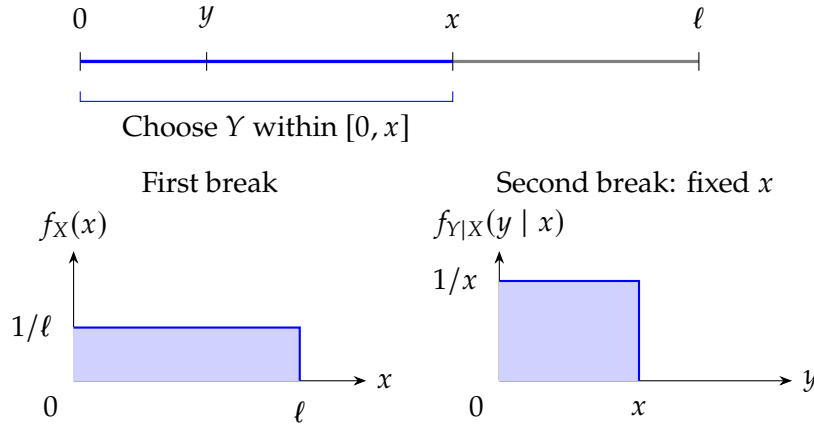


Figure 39: The second break is chosen within the retained length x . The conditional density has height $1/x$, so its area is one for each fixed $x > 0$. Horizontal spacing is schematic.

Rearrange the conditional-density definition to obtain the continuous multiplication rule:

$$f_{X,Y}(x, y) = f_X(x)f_{Y|X}(y | x).$$

Consequently,

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{\ell x}, & 0 < y < x < \ell, \\ 0, & \text{otherwise.} \end{cases}$$

The **SUPPORT**, the region where the density can be positive, is the triangle below $y = x$ and to the left of $x = \ell$. The density is not constant on this triangle: smaller retained lengths give larger conditional density heights. Normalization checks the construction:

$$\int_0^\ell \int_0^x \frac{1}{\ell x} dy dx = \int_0^\ell \frac{1}{\ell} dx = 1.$$

For a fixed first break, the conditional mean averages y over $[0, x]$:

$$\mathbb{E}[Y | X = x] = \int_0^x y \frac{1}{x} dy = \frac{1}{x} \left[\frac{y^2}{2} \right]_0^x = \left[\frac{x}{2} \right], \quad x > 0.$$

9.6 Stick breaking: marginal density and mean

To find the distribution of Y alone, hold y fixed and integrate over all possible first breaks. A second break at y requires $x \geq y$, while the stick ends at ℓ .

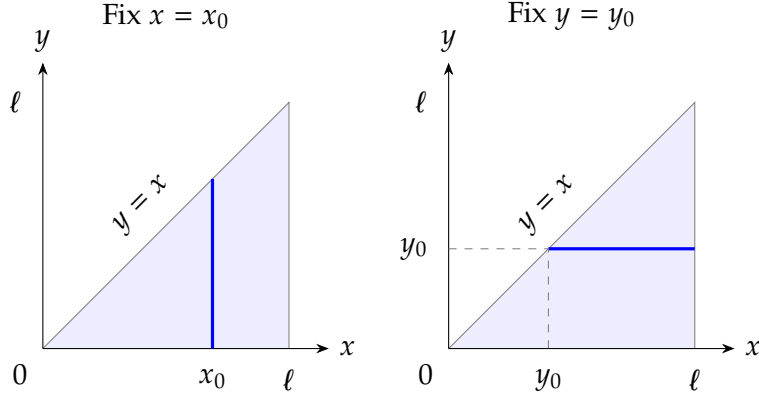


Figure 40: A vertical slice at x_0 allows $0 < y < x_0$. A horizontal slice at y_0 allows $y_0 < x < \ell$. These bounds determine the conditional and marginal integrals. The triangle shows support, not a uniform joint density.

For $0 < y < \ell$,

$$\begin{aligned} f_Y(y) &= \int_y^\ell \frac{1}{\ell x} dx = \frac{1}{\ell} [\log x]_y^\ell \\ &= \boxed{\frac{1}{\ell} \log \frac{\ell}{y}}. \end{aligned}$$

Here $\log$ denotes the natural logarithm. Outside $[0, \ell]$, the density is zero. It grows without bound as y approaches zero from the right, but its area is finite:

$$\int_0^\ell \frac{1}{\ell} \log \frac{\ell}{y} dy = \frac{1}{\ell} \left[y \log \frac{\ell}{y} + y \right]_0^\ell = 1,$$

using $y \log(\ell/y) \rightarrow 0$ as $y \downarrow 0$. This is a continuous distribution; the large density near zero does not create a point mass there.

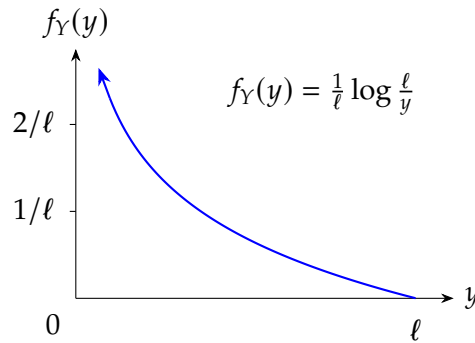


Figure 41: The second break's density decreases toward zero at ℓ and grows without bound near zero. The plotted portion follows $\ell f_Y(y) = \log(\ell/y)$; the upward arrow indicates the omitted continuation near zero.

Its mean is obtained by weighting this density by y :

$$\mathbb{E}[Y] = \frac{1}{\ell} \int_0^\ell y \log \frac{\ell}{y} dy.$$

Integration by parts uses $u = \log(\ell/y)$ and $dv = y dy$. Then $du = -dy/y$ and $v = y^2/2$, giving

$$\begin{aligned} \mathbb{E}[Y] &= \frac{1}{\ell} \left[\frac{y^2}{2} \log \frac{\ell}{y} + \frac{y^2}{4} \right]_0^\ell \\ &= \frac{1}{\ell} \frac{\ell^2}{4} = \boxed{\frac{\ell}{4}}. \end{aligned}$$

The lower endpoint is a limit, with $y^2 \log(\ell/y) \rightarrow 0$ as $y \downarrow 0$. Although the second break is uniform given the first, it is not uniform over the original stick. Its available interval depends on X , so X and Y are dependent.

Revision. Integrate a joint density over a region to get a pair probability. Integrate out one variable to get a marginal density. A conditional density is a slice divided by its marginal area; independence makes it equal to the original marginal density. Buffon's needle crosses when $X \leq (\ell/2) \sin \Theta$, with probability $2\ell/(\pi d)$. For successive stick breaks, the joint density is $1/(\ell x)$ on $0 < y < x < \ell$, the conditional mean is $x/2$, and the second break's mean is $\ell/4$.

Common mistakes: treating a density height as probability, dividing by $P(Y = y)$, forgetting to normalize a slice, using the whole needle length in the crossing threshold, or integrating the stick's marginal from 0 instead of from y .

Check yourself: Why is a small square's probability approximately $f_{X,Y}(x, y)\delta^2$? What is a raw slice's area? Why is Buffon's threshold $(\ell/2) \sin \theta$? Why does the second stick break have a logarithmic marginal density?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 9: Multiple Continuous Random Variables](#), PDF pp. 1–2. The supplied definitions, integrals, and calculations are completed, and the diagrams redrawn. The first stick-break interval is written as $[0, \ell]$, consistent with the stated stick length and subsequent formulas. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 10: CONTINUOUS BAYES' RULE; DERIVED DISTRIBUTIONS

Bayes' rule updates the distribution of an unknown value after an observation. A transformation poses a different question: if a random variable is known, what is the distribution of a function of it? In both cases, track which variable is being described and which values it can take.

10.1 Review and the discrete Bayes rule

For discrete variables, probabilities are sums of masses. For continuous variables with densities, they are integrals of densities. The corresponding joint, marginal, and conditional formulas are

	Discrete	Continuous
Joint	$p_{X,Y}(x, y)$	$f_{X,Y}(x, y)$
Marginal of X	$\sum_y p_{X,Y}(x, y)$	$\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$
Conditional on $Y = y$	$\frac{p_{X,Y}(x, y)}{p_Y(y)}$	$\frac{f_{X,Y}(x, y)}{f_Y(y)}$

Use a conditional ratio only when its denominator is positive. Both descriptions have the CDF $F_X(x) = P(X \leq x)$, expectation $\mathbb{E}[X]$, and variance $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$, when these moments are finite.

The **PRIOR DISTRIBUTION** describes X before observing Y . The **POSTERIOR DISTRIBUTION** describes X given the observed value $Y = y$. To obtain it, write the joint law using the prior and the model for the observation, then normalize over x .

For discrete X, Y and $p_Y(y) > 0$,

$$\boxed{p_{X|Y}(x | y) = \frac{p_X(x)p_{Y|X}(y | x)}{p_Y(y)}}, \quad p_Y(y) = \sum_x p_X(x)p_{Y|X}(y | x).$$

The numerator is $p_{X,Y}(x, y)$ by the multiplication rule. The denominator adds this joint probability over every possible hidden value x , so the posterior masses sum to one.

Example: *Radar.* Let $X = 1$ mean that an airplane is present and $X = 0$ that it is absent. Let $Y = 1$ mean that the radar registers something and $Y = 0$ that it does not. For either observed value $y \in \{0, 1\}$,

$$P(X = 1 | Y = y) = \frac{p_X(1)p_{Y|X}(y | 1)}{p_X(0)p_{Y|X}(y | 0) + p_X(1)p_{Y|X}(y | 1)}.$$

The two denominator terms account for the same observation with an airplane absent or present.

10.2 Continuous and mixed Bayes rules

When both variables are continuous, the same calculation uses densities and replaces the sum over hidden values by an integral. For $f_Y(y) > 0$,

$$\boxed{f_{X|Y}(x | y) = \frac{f_X(x)f_{Y|X}(y | x)}{f_Y(y)}}, \quad f_Y(y) = \int_{-\infty}^{\infty} f_X(u)f_{Y|X}(y | u) du.$$

Here u is an integration variable. The numerator is the joint density $f_{X,Y}(x, y)$; integrating it over x gives the denominator. The resulting posterior density therefore integrates to one. The denominator is a density at y , rather than $P(Y = y)$.

Example: *A noisy continuous signal.* The unknown signal X has prior density f_X . The observation Y is a noisy version of it, with conditional density $f_{Y|X}(y | x)$ specifying the noise model. For a fixed observation y , multiply the prior by this conditional density and divide by its integral over x . This gives the distribution of the signal after the observation.

The observation and the hidden value need not have the same type. The type of X determines whether the posterior is a PMF or a PDF, and whether normalization sums or integrates over x .

Discrete X , continuous Y . Each possible x has mass $p_X(x)$, but the observation has a conditional density. For $f_Y(y) > 0$,

$$\boxed{p_{X|Y}(x | y) = \frac{p_X(x)f_{Y|X}(y | x)}{f_Y(y)}}, \quad f_Y(y) = \sum_u p_X(u)f_{Y|X}(y | u).$$

For a discrete signal observed with continuous noise, the posterior assigns a probability mass to each possible signal value. The denominator adds the contributions of all possible signals to the observed density at y .

Continuous X , discrete Y . Now the prior is a density, while the observation model gives a conditional probability mass. For $p_Y(y) > 0$,

$$\boxed{f_{X|Y}(x | y) = \frac{f_X(x)p_{Y|X}(y | x)}{p_Y(y)}}, \quad p_Y(y) = \int_{-\infty}^{\infty} f_X(u)p_{Y|X}(y | u) du.$$

A light beam's intensity X can be continuous, while an observed photon count Y is discrete. The model $p_{Y|X}(y | x)$ gives the probability of that count at intensity x . Multiplying by $f_X(x)$ and normalizing gives a posterior density for intensity. In each mixed case, summing or integrating the numerator over x produces exactly the stated denominator.

10.3 Derived distributions and discrete transformations

Definition 10.1 [DERIVED DISTRIBUTION]. A DERIVED DISTRIBUTION is the probability law of a function of one or more random variables whose law is known. For a discrete output, describe this law by a PMF; for an output with a density, describe it by a PDF.

For example, suppose (X, Y) has joint density one on the unit square and zero outside it. The ratio $g(X, Y) = Y/X$ is itself a random variable. Finding its PDF means deriving a distribution from the known joint law. Since $P(X = 0) = 0$, the value assigned to the ratio on that boundary does not affect its distribution.

If only an expectation is needed, a derived PDF is unnecessary:

$$\mathbb{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dy dx,$$

when the absolute-value integral is finite. This computes the expectation directly from the joint law.

For a discrete transformation $Z = g(X)$, collect every input value that produces the same output value z :

$$p_Z(z) = P(g(X) = z) = \sum_{x: g(x)=z} p_X(x).$$

The events $\{X = x\}$ in this sum are disjoint. If several inputs map to one output, their masses must be added.

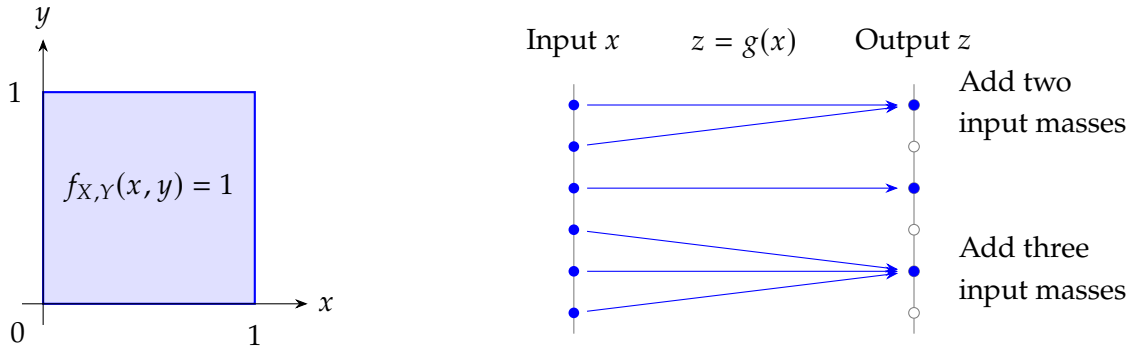


Figure 42: Left: a known joint law from which a function such as Y/X derives a new distribution. Right: a schematic discrete map; inputs with the same output contribute to the same mass. Empty output points have zero mass.

10.4 Continuous transformations: start with the CDF

For $Z = g(X)$, first find the input values that make $g(X) \leq z$. Their probability gives the CDF; differentiating gives a density.

$$F_Z(z) = P(g(X) \leq z), \quad f_Z(z) = \frac{d}{dz} F_Z(z).$$

If the output has a density, this procedure gives it, as in the examples below. Below the range of possible outputs, the CDF is zero; above that range, it is one.

Example: *Cubing a uniform variable.* Let X be uniform on $[0, 2]$ and let $Y = X^3$. Find the density of Y .

Cubing is increasing on $[0, 2]$, with output support $[0, 8]$. For $0 \leq y \leq 8$, take a cube root without reversing the inequality:

$$\begin{aligned} F_Y(y) &= P(X^3 \leq y) = P(X \leq y^{1/3}) \\ &= \int_0^{y^{1/3}} \frac{1}{2} dx = \frac{y^{1/3}}{2}. \end{aligned}$$

Complete the CDF outside the support and differentiate on its interior:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y^{1/3}/2, & 0 \leq y \leq 8, \\ 1, & y > 8. \end{cases} \quad f_Y(y) = \begin{cases} \frac{1}{6y^{2/3}}, & 0 < y < 8, \\ 0, & \text{otherwise.} \end{cases}$$

The density grows without bound near zero, but its area is $\int_0^8 (6y^{2/3})^{-1} dy = [y^{1/3}/2]_0^8 = 1$. The continuous CDF has no jump at zero, so there is no point mass there. Individual endpoint density values do not affect the law.

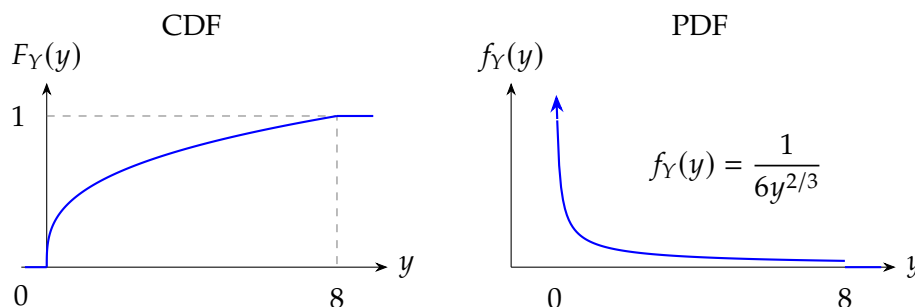


Figure 43: For $Y = X^3$, the CDF reaches one at 8. Its density decreases on $(0, 8)$ and grows without bound as $y \downarrow 0$. The upward arrow indicates the omitted continuation of the PDF.

10.5 Driving time: a decreasing transformation

Example: *Boston to New York.* Joan drives a distance of 200 miles at a random speed V that is uniform between 30 and 60 miles per hour. Find the distribution of the trip duration $T = 200/V$.

The speed density is $f_V(v) = 1/30$ for $30 < v < 60$, and zero elsewhere. A greater speed gives a shorter trip. Thus the possible times, in hours, run from

$$t_{\min} = \frac{200}{60} = \frac{10}{3} \quad \text{to} \quad t_{\max} = \frac{200}{30} = \frac{20}{3}.$$

For $10/3 \leq t \leq 20/3$, the event $T \leq t$ requires a speed *at least* $200/t$. Since V, t are positive,

$$\begin{aligned} F_T(t) &= P(200/V \leq t) = P(V \geq 200/t) \\ &= \int_{200/t}^{60} \frac{1}{30} dv = \frac{60 - 200/t}{30} = 2 - \frac{20}{3t}. \end{aligned}$$

The full CDF and its density are therefore

$$F_T(t) = \begin{cases} 0, & t < 10/3, \\ 2 - \frac{20}{3t}, & 10/3 \leq t \leq 20/3, \\ 1, & t > 20/3, \end{cases} \quad \boxed{f_T(t) = \begin{cases} \frac{20}{3t^2}, & 10/3 < t < 20/3, \\ 0, & \text{otherwise.} \end{cases}}$$

The CDF equals zero and one at the two support endpoints. The density is positive on the interior, and it integrates to one:

$$\int_{10/3}^{20/3} \frac{20}{3t^2} dt = \left[-\frac{20}{3t} \right]_{10/3}^{20/3} = 1.$$

Uniform speed does not give uniform time. The reciprocal map changes interval widths, yielding the decreasing density above.

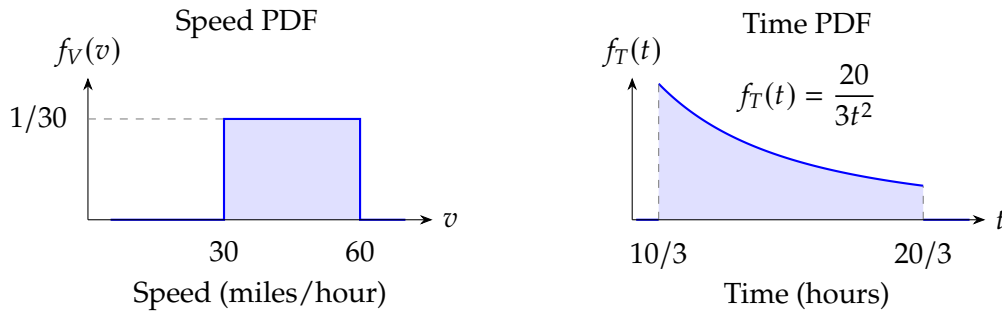


Figure 44: The speed PDF is flat, while the time PDF decreases across its support. The horizontal axes use different units; each density has total area one.

10.6 Affine transformations and normal variables

Scaling by a changes interval widths by $|a|$; adding b shifts their locations. The density must adjust its height to preserve probability.

Proposition 10.2 [DENSITY UNDER AN AFFINE MAP]. *Let X have a density and let $Y = aX + b$, where $a \neq 0$. Then*

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

Proof. Invert the inequality in the CDF, treating the sign of a explicitly. If $a > 0$,

$$F_Y(y) = P\left(X \leq \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right).$$

Differentiating gives $f_Y(y) = f_X((y-b)/a)/a$. If $a < 0$, the inequality reverses. Since X has no point masses,

$$F_Y(y) = P\left(X \geq \frac{y-b}{a}\right) = 1 - F_X\left(\frac{y-b}{a}\right).$$

Differentiating gives $f_Y(y) = -f_X((y-b)/a)/a$. In either case, the positive scaling factor is $1/|a|$. $\square$

For $Y = 2X + 5$, first stretch the horizontal axis by two and halve the density height; then shift the stretched density five units to the right. In particular,

$$f_Y(y) = \frac{1}{2} f_X\left(\frac{y-5}{2}\right).$$

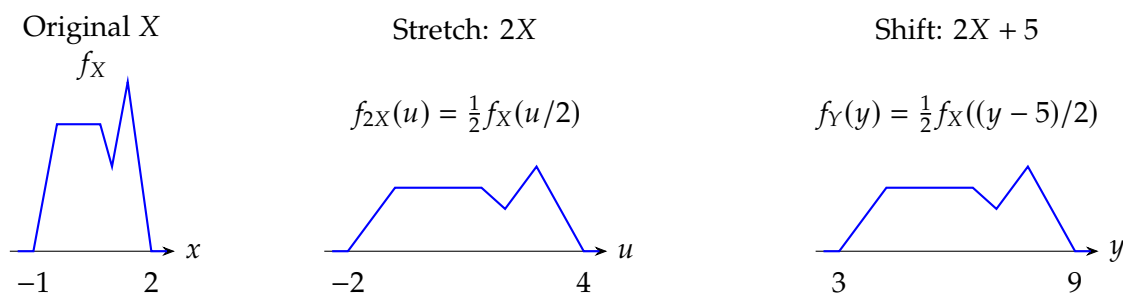


Figure 45: A schematic density with support $[-1, 2]$ is stretched to support $[-2, 4]$, then shifted to $[3, 9]$. The last two panels have half the original height. All panels use the same horizontal and vertical scales.

Example: *A transformed normal variable.* Let X be normal with mean μ and variance $\sigma^2 > 0$. For $a \neq 0$, substituting its density into the affine formula gives

$$\begin{aligned} f_Y(y) &= \frac{1}{|a|\sigma\sqrt{2\pi}} \exp\left(-\frac{((y-b)/a - \mu)^2}{2\sigma^2}\right) \\ &= \frac{1}{\sqrt{a^2\sigma^2}\sqrt{2\pi}} \exp\left(-\frac{(y - (a\mu + b))^2}{2a^2\sigma^2}\right). \end{aligned}$$

This is the normal density with mean $a\mu + b$ and variance $a^2\sigma^2$. Thus $Y \sim N(a\mu + b, a^2\sigma^2)$. If $a = 0$, then $Y = b$ with probability one and has no ordinary PDF.

Revision. Bayes' rule multiplies a prior law by the conditional observation law, then normalizes over the hidden value. Sum for discrete X ; integrate for continuous X . For a derived PMF, add the masses of all inputs that produce an output. For a continuous transformation, find the support and CDF before differentiating. An affine map has density $f_X((y-b)/a)/|a|$ and preserves the normal family for $a \neq 0$.

Common mistakes: using $P(Y = y)$ for a continuous observation, integrating over the observed variable instead of the hidden one, missing inputs that map to the same output, forgetting to reverse the inequality for $200/V$, or omitting the absolute value in $1/|a|$.

Check yourself: What determines whether a posterior is a PMF or a PDF? Why is the cubic density unbounded near zero without a point mass there? Why does $T \leq t$ require $V \geq 200/t$? How do width and height change when a density is stretched by two?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 10: Continuous Bayes' Rule; Derived Distributions](#), PDF pp. 1–2. The Bayes formulas, transformation examples, affine proof, and normal-density calculation are completed, and the diagrams redrawn. The unit-square ratio remains a motivating example of a derived distribution. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 11: DERIVED DISTRIBUTIONS; CONVOLUTION; COVARIANCE AND CORRELATION

Find a derived law by describing its CDF event. Independent sums use convolution. Covariance and correlation describe how variables vary together.

11.1 A ratio from the unit square

Example: Find the density of $Z = Y/X$. The joint density of (X, Y) is one on the unit square and zero elsewhere.

Ignoring the boundary $X = 0$, which has probability zero, $Z \leq z$ means $Y \leq zX$. Its probability is the area below $y = zx$ in the square. For $z < 0$, this area is zero.

For $0 \leq z \leq 1$, the line stays below the square's top edge:

$$F_Z(z) = \int_0^1 \int_0^{zx} 1 \, dy \, dx = \int_0^1 zx \, dx = \frac{z}{2}.$$

For $z \geq 1$, the line reaches the top at $x = 1/z$. To its right, every y in the square is allowed:

$$\begin{aligned} F_Z(z) &= \int_0^{1/z} zx \, dx + \int_{1/z}^1 1 \, dx \\ &= \frac{1}{2z} + 1 - \frac{1}{z} = 1 - \frac{1}{2z}. \end{aligned}$$

Complete the CDF and differentiate on its two positive intervals:

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ z/2, & 0 \leq z \leq 1, \\ 1 - 1/(2z), & z > 1, \end{cases} \quad \boxed{f_Z(z) = \begin{cases} 1/2, & 0 < z \leq 1, \\ 1/(2z^2), & z > 1, \\ 0, & \text{otherwise.} \end{cases}}$$

Both CDF formulas give $1/2$ at $z = 1$, and $F_Z(z) \rightarrow 1$ as $z \rightarrow \infty$. The density integrates to $\int_0^1 (1/2) \, dz + \int_1^\infty (1/(2z^2)) \, dz = 1$. Values assigned to the PDF at 0 and 1 do not change the law.

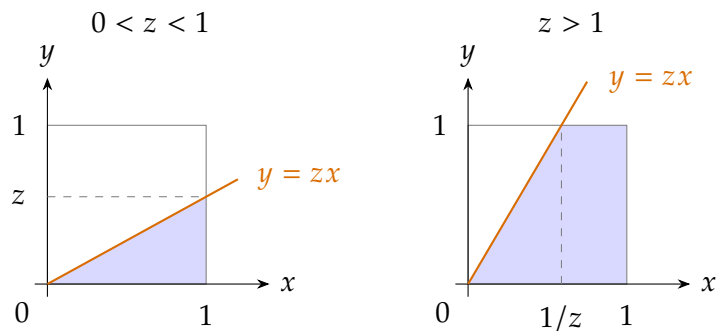


Figure 46: The shaded area is $P(Y/X \leq z)$. For $0 < z < 1$, it is a triangle. For $z > 1$, it includes a full strip to the right of $x = 1/z$. Slopes represent the two regimes schematically.

11.2 A density formula for a monotone transformation

A STRICTLY MONOTONE function is either strictly increasing or strictly decreasing, so each output has at most one input. For $Y = g(X)$, this lets us convert a density by accounting for the change in interval width.

For a short input interval of length δ near x , the output interval has length approximately $|g'(x)|\delta$. Both intervals describe the same event, hence have the same probability:

$$f_X(x)\delta \approx f_Y(g(x))|g'(x)|\delta.$$

An increasing map preserves the order of endpoints. A decreasing map reverses them, but its output interval still has positive width.

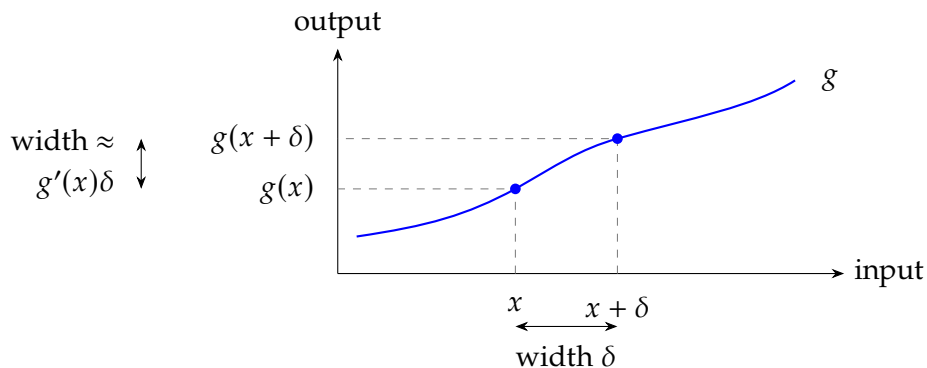


Figure 47: An increasing map sends an input interval of width δ to an output interval of width approximately $g'(x)\delta$. The corresponding probabilities agree. The curve and widths are schematic.

Proposition 11.1 [MONOTONE CHANGE OF VARIABLE]. *Let X have a density on an interval I . Suppose g is continuously differentiable and strictly monotone there, with $g'(x) \neq 0$ in its interior. Writing $h = g^{-1}$, the density of $Y = g(X)$, for y in the interior of $g(I)$, is*

$$f_Y(y) = f_X(h(y)) |h'(y)| = \frac{f_X(h(y))}{|g'(h(y))|}.$$

It is zero outside the output range. Endpoint values are immaterial.

Proof. Express the output CDF through the inverse map. If g is increasing, $F_Y(y) = F_X(h(y))$, and differentiation gives $f_X(h(y))h'(y)$. If g is decreasing, $F_Y(y) = 1 - F_X(h(y))$, because X has no point masses. Differentiation gives $-f_X(h(y))h'(y)$. Thus the positive factor in either case is $|h'(y)|$. Differentiating $g(h(y)) = y$ gives $g'(h(y))h'(y) = 1$, which yields the second formula. $\square$

The interval-width argument explains the factor: a wider output interval needs a lower density to retain the same probability. For $g(x) = ax + b$, this reduces to the factor $1/|a|$.

11.3 Discrete convolution: the distribution of a sum

Let $W = X + Y$, with X, Y independent and discrete. Each possible value x contributes to $W = w$ exactly when $Y = w - x$. The events for different x are disjoint, so

$$\begin{aligned} p_W(w) &= \sum_x P(X = x, Y = w - x) \\ &= \sum_x p_X(x) p_Y(w - x). \end{aligned}$$

The second line uses independence. The sum ranges over the possible values of X ; a term is zero when $w - x$ is not a possible value of Y .

Definition: The CONVOLUTION of two PMFs is the sum $\sum_x p_X(x) p_Y(w - x)$ as a function of w . It is the PMF of the sum of independent variables with those two PMFs.

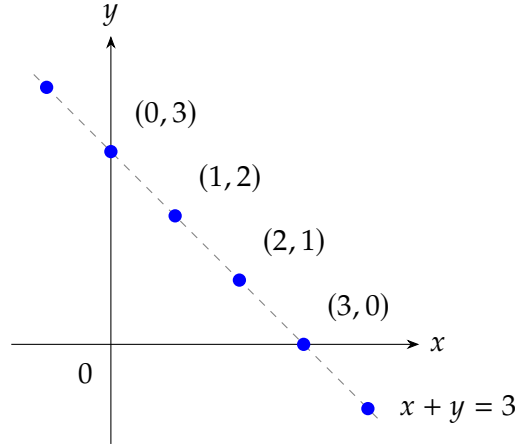


Figure 48: For integer values, the pairs that produce $W = 3$ lie on $x + y = 3$, including $(0, 3), (1, 2), (2, 1), (3, 0)$. Add the probability masses of all allowed pairs on this line.

To compute a convolution graphically, align the two PMFs on a common input axis, then carry out these steps:

1. Flip the PMF of Y , giving $p_Y(-x)$.
2. Shift it by w , giving $p_Y(w - x)$. A positive w shifts the flipped graph to the right.
3. Multiply the two heights at each x .
4. Add all products to obtain $p_W(w)$.

Changing w moves the flipped graph and produces another value of the output PMF. Independence supplies the product of masses; the constraint $x + y = w$ selects which masses to combine.

11.4 Continuous convolution

For independent continuous variables, condition on one summand. Given $X = x$, the sum $W = x + Y$ is a shifted version of Y . Independence makes the conditional law of Y its original law, so

$$f_{W|X}(w | x) = f_Y(w - x).$$

Use the multiplication rule to form a joint density, then integrate over all possible values of x :

$$f_{W,X}(w, x) = f_X(x)f_{W|X}(w | x) = f_X(x)f_Y(w - x).$$

$$f_W(w) = \int_{-\infty}^{\infty} f_X(x)f_Y(w - x) dx.$$

This integral is the continuous convolution. The flip, shift, and multiply steps are the same as for PMFs, with integration replacing addition.

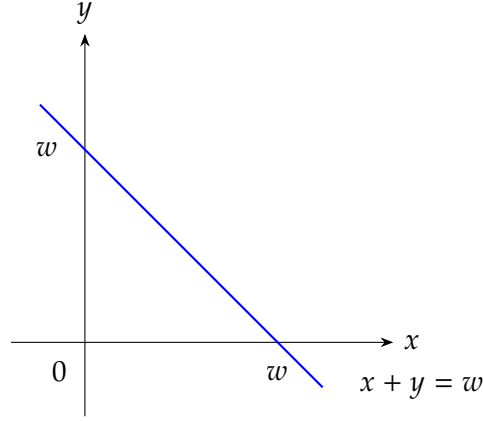


Figure 49: For a positive w , pairs producing that sum lie on $x + y = w$. The convolution density integrates $f_X(x)f_Y(w - x)$ over x . The line itself has probability zero for a joint density; its integral here produces a density in w , not a point probability.

The formula automatically respects support: only values of x for which both $f_X(x)$ and $f_Y(w - x)$ can be nonzero contribute. Its total integral is one, since

$$\begin{aligned} \int_{-\infty}^{\infty} f_W(w) dw &= \int_{-\infty}^{\infty} f_X(x) \left(\int_{-\infty}^{\infty} f_Y(w - x) dw \right) dx \\ &= \int_{-\infty}^{\infty} f_X(x) dx = 1. \end{aligned}$$

For each fixed x , the inner integral is a translation of the integral of f_Y , so it equals one.

11.5 The joint density of independent normal variables

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ be independent, with $\sigma_X, \sigma_Y > 0$. Multiplying their densities gives

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left(-\frac{(x - \mu_X)^2}{2\sigma_X^2} - \frac{(y - \mu_Y)^2}{2\sigma_Y^2}\right).$$

The density has the same height wherever

$$\frac{(x - \mu_X)^2}{2\sigma_X^2} + \frac{(y - \mu_Y)^2}{2\sigma_Y^2} = c$$

for a fixed $c > 0$. These **CONTOURS**, curves on which the density is constant, are ellipses centered at (μ_X, μ_Y) . Their horizontal and vertical half-widths are $\sqrt{2c} \sigma_X$ and $\sqrt{2c} \sigma_Y$. If $\sigma_X = \sigma_Y$, the ellipses are circles.

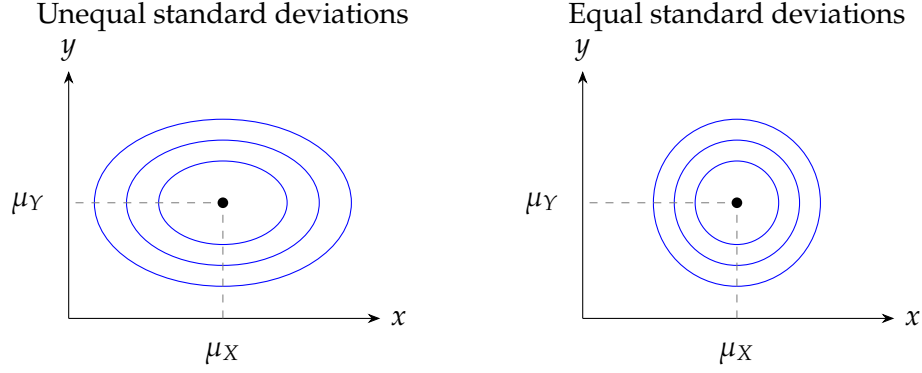


Figure 50: Equal-density contours of independent normal variables are centered at their means. Unequal standard deviations give ellipses; equal standard deviations give circles. Larger contours have a larger c and therefore a lower density. Scales are schematic.

11.6 The sum of independent normal variables

Proposition 11.2 [NORMAL CONVOLUTION]. *For the independent normal variables above,*

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

Proof. First take both means to be zero. Complete the square in the convolution integral, leaving a normal integral in x and an exponential factor in w . Write $A = \sigma_X^2$, $B = \sigma_Y^2$, and $S = A + B$. Then

$$f_W(w) = \frac{1}{2\pi\sqrt{AB}} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2A} - \frac{(w-x)^2}{2B}\right) dx,$$

$$\frac{x^2}{A} + \frac{(w-x)^2}{B} = \frac{S}{AB} \left(x - \frac{A}{S}w\right)^2 + \frac{w^2}{S}.$$

The second identity follows by expanding both sides: the coefficient of x^2 is $S/(AB)$, the coefficient of x is $-2w/B$, and the constant is w^2/B . Substitution gives

$$f_W(w) = \frac{e^{-w^2/(2S)}}{2\pi\sqrt{AB}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x - Aw/S)^2}{2AB/S}\right) dx.$$

The integrand is a normal density without its normalizing factor. Its integral is $\sqrt{2\pi AB/S}$, so

$$f_W(w) = \frac{1}{\sqrt{2\pi S}} \exp\left(-\frac{w^2}{2S}\right).$$

This is the density of $N(0, S)$.

For nonzero means, $X - \mu_X$ and $Y - \mu_Y$ remain independent and are zero-mean normal variables with variances A, B . Their sum is $N(0, S)$. Adding $\mu_X + \mu_Y$ shifts its mean and leaves its variance unchanged, proving the stated result. $\square$

Normality of the sum follows from the convolution calculation. Its mean and variance alone would not establish its distribution.

11.7 Covariance and the variance of a sum

Subtract each mean to record deviations. Their product is positive for matching signs and negative for opposite signs.

Definition 11.3 [COVARIANCE]. For variables with finite second moments, their COVARIANCE is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Covariance averages the products of the centered values. When both means are zero, it reduces to $\mathbb{E}[XY]$. Expanding the centered product gives a useful equivalent formula:

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \end{aligned}$$

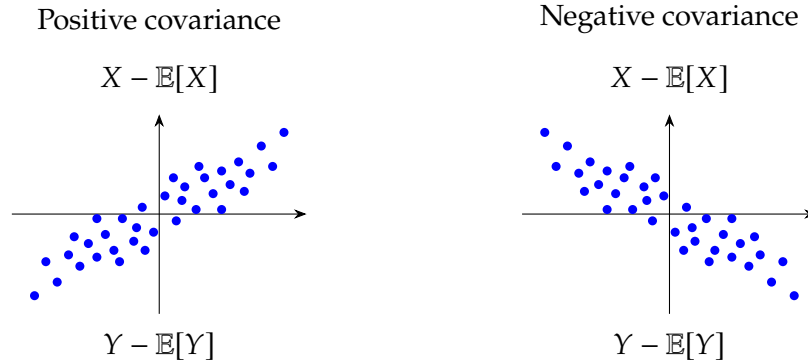


Figure 51: Schematic centered clouds show positive and negative covariance. Products of deviations tend to be positive in the left drawing and negative in the right. The dots illustrate the geometry and are not measured data.

To expand a sum's variance, put $D_i = X_i - \mathbb{E}[X_i]$. Linearity of expectation gives

$$\begin{aligned}\text{Var}\left(\sum_{i=1}^n X_i\right) &= \mathbb{E}\left[\left(\sum_{i=1}^n D_i\right)^2\right] \\ &= \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).\end{aligned}$$

The factor two appears because each unordered pair occurs twice in the square. With a sum over all ordered pairs $i \neq j$, the same formula uses a factor one instead.

If X, Y are independent, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, so $\text{Cov}(X, Y) = 0$. Consequently, pairwise independent summands have additive variances. Zero covariance does not imply independence: it constrains one expected product, rather than the whole joint law.

11.8 Correlation: covariance on a common scale

Covariance carries the product of the variables' units. Divide each centered variable by its standard deviation to compare their variation without units.

Definition 11.4 [CORRELATION COEFFICIENT]. For finite, positive variances, write $\sigma_X = \sqrt{\text{Var}(X)}$ and $\sigma_Y = \sqrt{\text{Var}(Y)}$. The CORRELATION COEFFICIENT is

$$\rho = \mathbb{E}\left[\frac{X - \mathbb{E}[X]}{\sigma_X} \frac{Y - \mathbb{E}[Y]}{\sigma_Y}\right] = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Let $U = (X - \mathbb{E}[X])/\sigma_X$ and $V = (Y - \mathbb{E}[Y])/\sigma_Y$. Both have mean zero and second moment one; $\mathbb{E}[UV] = \rho$. The nonnegativity of squared differences and sums gives

$$0 \leq \mathbb{E}[(U - V)^2] = 2 - 2\rho, \quad 0 \leq \mathbb{E}[(U + V)^2] = 2 + 2\rho.$$

The first inequality gives $\rho \leq 1$ and the second gives $\rho \geq -1$. Therefore

$$-1 \leq \rho \leq 1.$$

If $\rho = 1$, then $\mathbb{E}[(U - V)^2] = 0$, so $U = V$ with probability one. If $\rho = -1$, the same reasoning gives $U = -V$ with probability one. Rewriting these relations yields

$$X - \mathbb{E}[X] = \pm \frac{\sigma_X}{\sigma_Y} (Y - \mathbb{E}[Y]).$$

Conversely, suppose $X - \mathbb{E}[X] = c(Y - \mathbb{E}[Y])$ with probability one for a constant $c \neq 0$. Then $\sigma_X = |c|\sigma_Y$ and $\text{Cov}(X, Y) = c\sigma_Y^2$, so $\rho = c/|c|$. Thus $|\rho| = 1$ holds exactly when the centered variables are linearly related with probability one.

Variables with zero covariance, or equivalently zero correlation when both variances are positive, are **UNCORRELATED**. Independence implies $\rho = 0$, but the converse is false. If either variance is zero, the stated correlation coefficient is undefined because its denominator vanishes.

Revision. Find a derived distribution by describing its CDF event. A monotone map changes density by the inverse interval-width factor. Independent sums use convolution: sum or integrate $p_X(x)p_Y(w - x)$ or $f_X(x)f_Y(w - x)$. Independent normal summands give a normal sum, with means and variances added. Covariance averages centered products; correlation divides it by the two standard deviations and lies in $[-1, 1]$.

Common mistakes: missing the full strip when Y/X exceeds one, omitting the derivative's absolute value, using convolution without independence, counting covariance pairs twice after including a factor two, or treating zero correlation as independence.

Check yourself: Why does the ratio CDF change at $z = 1$? What does flipping and shifting do in a convolution? Which completed-square term determines the normal sum's variance? Why does $|\rho| = 1$ force a linear relation?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 11: Derived Distributions; Convolution; Covariance and Correlation](#), PDF pp. 1–2. The ratio, monotone-map, convolution, normal-sum, and correlation calculations are completed, and the diagrams redrawn. Decreasing maps reverse interval endpoints. The variance sum is written over $i < j$ with factor two, avoiding the ambiguity of a factor two beside an ordered $i \neq j$ sum. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 12: ITERATED EXPECTATIONS; SUM OF A RANDOM NUMBER OF RANDOM VARIABLES

Conditioning can turn a difficult calculation into several easier ones. Find a mean or variance after fixing another variable, then account for how that variable varies. This also handles sums whose number of terms is random.

12.1 Conditional expectation as a random variable

For a fixed value y , the conditional mean is a number computed from the conditional distribution:

$$m(y) = \mathbb{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y).$$

For a conditional density, replace the sum by $\int_{-\infty}^{\infty} x f_{X|Y}(x \mid y) dx$. These formulas apply where the conditional law is defined; assume $\mathbb{E}[|X|] < \infty$.

Definition: The **CONDITIONAL EXPECTATION AS A RANDOM VARIABLE**, $\mathbb{E}[X \mid Y]$, is $m(Y)$. Its value is $m(y)$ whenever $Y = y$. It varies because the value of Y varies.

Example: *Break a stick twice.* Start with a stick of length $\ell > 0$. Choose the first break point Y uniformly on $[0, \ell]$. Keep the piece from 0 to Y , then choose the second break point X uniformly on that piece.

Given $Y = y > 0$, X is uniform on $[0, y]$, so

$$\mathbb{E}[X \mid Y = y] = \int_0^y x \frac{1}{y} dx = \frac{y}{2}, \quad \boxed{\mathbb{E}[X \mid Y] = \frac{Y}{2}}.$$

The event $Y = 0$ has probability zero, so its conditional value does not affect any average.

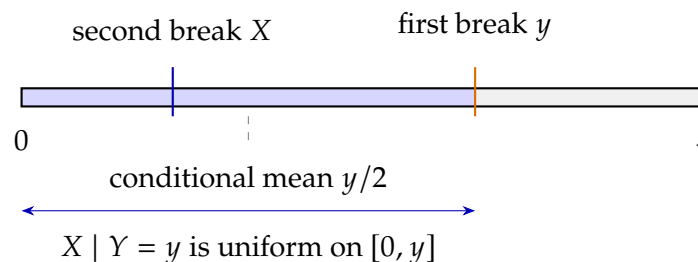


Figure 52: After fixing the first break at y , the second break lies uniformly on $[0, y]$. The conditional mean is its midpoint $y/2$. The positions shown are schematic.

12.2 The law of iterated expectations

Average the conditional means using the probabilities of the conditions. This recovers the original mean.

Theorem 12.1 [LAW OF ITERATED EXPECTATIONS]. *If $\mathbb{E}[|X|] < \infty$, then*

$$\boxed{\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]}.$$

For discrete Y , the left side is $\sum_y \mathbb{E}[X | Y = y] p_Y(y)$. For continuous Y with a density, it is $\int_{-\infty}^{\infty} \mathbb{E}[X | Y = y] f_Y(y) dy$.

Proof. Multiply each conditional mean by the probability of its condition. In the discrete case, the multiplication rule gives

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X | Y]] &= \sum_y \sum_x x p_{X|Y}(x | y) p_Y(y) \\ &= \sum_x x \sum_y p_{X,Y}(x, y) \\ &= \sum_x x p_X(x) = \mathbb{E}[X]. \end{aligned}$$

The finite absolute expectation justifies rearranging these sums. For a joint density, the same steps use integrals and $f_{X|Y}(x | y) f_Y(y) = f_{X,Y}(x, y)$:

$$\int \int x f_{X,Y}(x, y) dx dy = \int x f_X(x) dx = \mathbb{E}[X].$$

□

For the stick, averaging the random conditional mean gives

$$\boxed{\mathbb{E}[X] = \mathbb{E}[Y/2] = \frac{1}{2} \frac{\ell}{2} = \frac{\ell}{4}}.$$

The first averaging step fixes the retained length; the second averages over that length. No marginal density of X is needed.

12.3 Conditional variance and total variance

For a fixed condition, measure deviations from its own conditional mean, rather than from the overall mean.

Definition 12.2 [CONDITIONAL VARIANCE]. Assume $\mathbb{E}[X^2] < \infty$. The CONDITIONAL VARIANCE at $Y = y$ is

$$\text{Var}(X \mid Y = y) = \mathbb{E}[(X - \mathbb{E}[X \mid Y = y])^2 \mid Y = y].$$

The random variable $\text{Var}(X \mid Y)$ takes this value when $Y = y$. Equivalently,

$$\text{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - (\mathbb{E}[X \mid Y])^2.$$

Its expectation measures the average spread within conditions. The conditional means may also vary from one condition to another. Total variance combines these two sources of spread.

Theorem 12.3 [LAW OF TOTAL VARIANCE]. If $\mathbb{E}[X^2] < \infty$, then

$$\boxed{\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y])}.$$

Proof. Use the second-moment formula on each term. Write $M = \mathbb{E}[X \mid Y]$. Iterated expectations applied to X^2 and to X give

$$\begin{aligned}\mathbb{E}[\text{Var}(X \mid Y)] &= \mathbb{E}[\mathbb{E}[X^2 \mid Y]] - \mathbb{E}[M^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[M^2], \\ \text{Var}(M) &= \mathbb{E}[M^2] - (\mathbb{E}[M])^2 \\ &= \mathbb{E}[M^2] - (\mathbb{E}[X])^2.\end{aligned}$$

Adding cancels $\mathbb{E}[M^2]$ and leaves $\mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X)$. □

12.4 Variability within and between class sections

Example: Choose a student uniformly from two class sections. Section 1 has 10 students with mean score 90; section 2 has 20 students with mean score 60. Let X be the chosen student's score and Y the section number.

Let the scores be $x_1, \dots, x_{30}$, with the first ten in section 1. The given section means are

$$\frac{1}{10} \sum_{i=1}^{10} x_i = 90, \quad \frac{1}{20} \sum_{i=11}^{30} x_i = 60.$$

The section probabilities are $P(Y = 1) = 1/3$ and $P(Y = 2) = 2/3$. Thus

$$\mathbb{E}[X | Y] = \begin{cases} 90, & Y = 1, \\ 60, & Y = 2, \end{cases} \quad \mathbb{E}[X] = \frac{1}{3} \cdot 90 + \frac{2}{3} \cdot 60 = 70.$$

This equals the direct average $(90 \cdot 10 + 60 \cdot 20)/30$. The variability of the conditional means is

$$\text{Var}(\mathbb{E}[X | Y]) = \frac{1}{3}(90 - 70)^2 + \frac{2}{3}(60 - 70)^2 = 200.$$

Now suppose the within-section squared deviations satisfy

$$\frac{1}{10} \sum_{i=1}^{10} (x_i - 90)^2 = 10, \quad \frac{1}{20} \sum_{i=11}^{30} (x_i - 60)^2 = 20.$$

Because a student is sampled uniformly within a section, these are its conditional variances.

Section y	$P(Y = y)$	$\mathbb{E}[X Y = y]$	$\text{Var}(X Y = y)$
1	1/3	90	10
2	2/3	60	20

Average the conditional variances with the same section weights:

$$\mathbb{E}[\text{Var}(X | Y)] = \frac{1}{3} \cdot 10 + \frac{2}{3} \cdot 20 = \frac{50}{3}.$$

Therefore

$$\text{Var}(X) = \underbrace{\frac{50}{3}}_{\text{within sections}} + \underbrace{200}_{\text{between section means}} = \frac{650}{3}.$$

The average within-section variance omits the separation between the two means. Total variance restores that contribution.

12.5 A two-part density example

Example: Let X have density $1/3$ on $(0, 1)$ and $2/3$ on $(1, 2)$, and zero elsewhere. Set $Y = 1$ on the first interval and $Y = 2$ on the second. Compute the conditional moments and total variance.

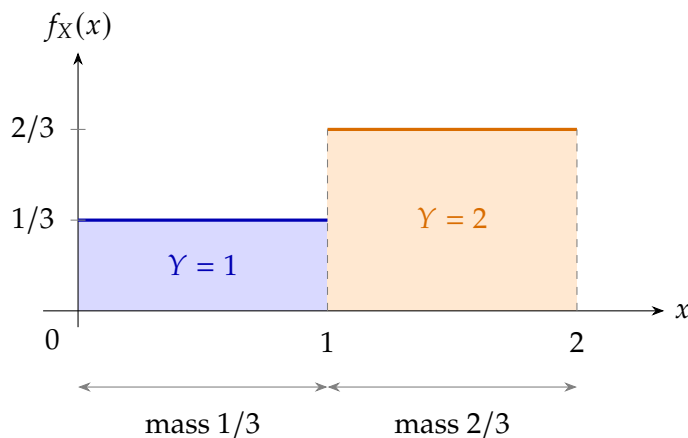


Figure 53: Each interval has width one. Its area is the probability of the corresponding value of Y : $1/3$ on the left and $2/3$ on the right. Conditioning divides the density by that area.

Integrating on the two intervals gives $P(Y = 1) = 1/3$ and $P(Y = 2) = 2/3$. After dividing by these probabilities, the conditional density is one on the corresponding interval. Thus the conditional laws are uniform, with

$$\begin{aligned}\mathbb{E}[X | Y = 1] &= \int_0^1 x \, dx = \frac{1}{2}, & \mathbb{E}[X | Y = 2] &= \int_1^2 x \, dx = \frac{3}{2}, \\ \text{Var}(X | Y = 1) &= \int_0^1 (x - \frac{1}{2})^2 \, dx = \frac{1}{12}, & \text{Var}(X | Y = 2) &= \int_1^2 (x - \frac{3}{2})^2 \, dx = \frac{1}{12}.\end{aligned}$$

The overall mean and the variance of the conditional means are

$$\begin{aligned}\mathbb{E}[X] &= \frac{1}{3} \frac{1}{2} + \frac{2}{3} \frac{3}{2} = \frac{7}{6}, \\ \text{Var}(\mathbb{E}[X | Y]) &= \frac{1}{3} \left(\frac{1}{2} - \frac{7}{6} \right)^2 + \frac{2}{3} \left(\frac{3}{2} - \frac{7}{6} \right)^2 = \frac{2}{9}.\end{aligned}$$

Both conditional variances equal $1/12$, so their average is $1/12$. Total variance gives

$$\boxed{\text{Var}(X) = \frac{1}{12} + \frac{2}{9} = \frac{11}{36}}.$$

As a direct check, $\mathbb{E}[X^2] = \frac{1}{3} \int_0^1 x^2 \, dx + \frac{2}{3} \int_1^2 x^2 \, dx = 5/3$. Subtracting $(7/6)^2$ again gives $11/36$.

12.6 The mean of a sum with a random number of terms

Example: Visit a random number N of stores, spending X_i in store i . Assume N is a nonnegative integer, the X_i are independent and identically distributed, and the whole spending sequence is independent of N . The total spending is

$$Y = X_1 + \cdots + X_N = \sum_{i=1}^N X_i.$$

When $N = 0$, the empty sum is zero.

Write $\mu = \mathbb{E}[X_1]$ and $\sigma^2 = \text{Var}(X_1)$ for the common mean and variance. Fixing $N = n$ converts the random-length sum into a sum of exactly n terms. Independence from N leaves the summands' joint law unchanged.

$$\begin{array}{ccc} \boxed{N = n} & \longrightarrow & \boxed{Y = X_1 + \cdots + X_n} \\ & & \mathbb{E}[Y \mid N = n] = n\mu \\ & & \text{Var}(Y \mid N = n) = n\sigma^2 \end{array}$$

Average over N to include variation in the count.

Figure 54: Condition on the number of visits first. For each nonnegative integer n , the total has mean $n\mu$ and variance $n\sigma^2$. Then average over the law of N . The row for n represents any allowed count.

Assume $\mathbb{E}[N] < \infty$ and $\mathbb{E}[|X_1|] < \infty$. Linearity gives

$$\begin{aligned} \mathbb{E}[Y \mid N = n] &= \mathbb{E}[X_1 + \cdots + X_n \mid N = n] \\ &= \mathbb{E}[X_1 + \cdots + X_n] = n\mu. \end{aligned}$$

As a random conditional mean, this is $\mathbb{E}[Y \mid N] = N\mu$. Iterated expectations now yield

$$\boxed{\mathbb{E}[Y] = \mathbb{E}[N\mu] = \mathbb{E}[N]\mu}.$$

Expected total spending equals the expected number of visits times the common expected spending per visit.

12.7 The variance of a sum with a random number of terms

Assume also $\mathbb{E}[N^2] < \infty$ and $\mathbb{E}[X_1^2] < \infty$. Given $N = n$, independence of the summands makes their variances add:

$$\text{Var}(Y \mid N = n) = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2.$$

Thus $\text{Var}(Y \mid N) = N\sigma^2$, and

$$\mathbb{E}[\text{Var}(Y \mid N)] = \mathbb{E}[N] \sigma^2.$$

The conditional mean $N\mu$ varies with the number of visits:

$$\text{Var}(\mathbb{E}[Y \mid N]) = \text{Var}(N\mu) = \mu^2 \text{Var}(N).$$

Apply total variance to combine these contributions:

$$\begin{aligned} \text{Var}(Y) &= \mathbb{E}[\text{Var}(Y \mid N)] + \text{Var}(\mathbb{E}[Y \mid N]) \\ &= \boxed{\mathbb{E}[N] \sigma^2 + \mu^2 \text{Var}(N)}. \end{aligned}$$

The first term is the average spending variability for a fixed number of visits. The second comes from changes in that number. If N is fixed, its variance is zero and the formula reduces to the variance of a fixed-length independent sum.

Revision. $\mathbb{E}[X \mid Y = y]$ is a number; $\mathbb{E}[X \mid Y]$ is a function of the random Y . Averaging conditional means gives $\mathbb{E}[X]$. Total variance is average conditional variance plus variance of the conditional means. For a count N independent of an i.i.d. sequence with mean μ and variance σ^2 , $\mathbb{E}[\sum_{i=1}^N X_i] = \mathbb{E}[N]\mu$ and $\text{Var}(\sum_{i=1}^N X_i) = \mathbb{E}[N]\sigma^2 + \mu^2 \text{Var}(N)$.

Common mistakes: giving all conditions equal weight, centering a conditional variance at the overall mean, confusing $\mathbb{E}[\text{Var}(X \mid Y)]$ with $\text{Var}(\mathbb{E}[X \mid Y])$, or dropping the random count's variance term. Independence from the count is needed to keep the summands' laws unchanged after conditioning.

Check yourself: Why is the stick's conditional mean $Y/2$ random? Which term accounts for the separation between section means? Why are both conditional densities in the two-part example equal to one? What are the two sources of random-sum variance?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 12: Iterated Expectations; Sum of a Random Number of Random Variables](#), PDF pp. 1–2. The conditional-expectation and total-variance arguments are completed, the section and density calculations filled in, and the diagrams redrawn. Moment assumptions and the zero-term sum are made explicit. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 13: BERNOULLI PROCESS

A Bernoulli process records a sequence of independent successes and failures. The same sequence can be described by its outcomes, its number of successes, or its waiting times between successes.

13.1 The Bernoulli process

Definition 13.1 [BERNOULLI PROCESS]. A BERNOULLI PROCESS with parameter $p \in [0, 1]$ is a sequence $X_1, X_2, \dots$ of independent Bernoulli random variables, each with

$$P(X_i = 1) = p, \quad P(X_i = 0) = 1 - p.$$

The index i is a trial number or a discrete time slot. A value one is a success, or an arrival; a value zero is a failure, or no arrival.

The success probability is the same at every trial. Independence means that knowing earlier outcomes does not change the law of later trials.

Lottery wins and losses, upward and downward movements of the Dow Jones, and arrivals or no arrivals at a bank or server can all be encoded by zeros and ones. A Bernoulli model for such a sequence assumes independent outcomes with a common success probability. For an arrival model, each slot records whether an arrival occurs.

13.2 Two views of a random process

A **RANDOM PROCESS** is a collection of random variables indexed by time. One view follows the individual variables. For every fixed time t , X_t has a Bernoulli distribution, so

$$\mathbb{E}[X_t] = p, \quad \text{Var}(X_t) = \mathbb{E}[X_t^2] - (\mathbb{E}[X_t])^2 = p - p^2 = p(1 - p).$$

Here $X_t^2 = X_t$, since both possible values are zero or one.

A second view treats an entire sequence as an outcome. The sample space is

$$\Omega = \{(x_1, x_2, \dots) : x_i \in \{0, 1\}\}.$$

An individual sequence is a **SAMPLE PATH**. The random variable X_t reads its t th coordinate. Independence specifies the probabilities of finite blocks: a particular block of length n containing s ones has probability $p^s(1 - p)^{n-s}$.

For $0 \leq p < 1$, the probability of an all-success infinite path is zero. Indeed, the event of success at every time is contained in the event of success in the first n trials, whose probability is p^n . Since this tends to zero,

$$P(X_t = 1 \text{ for every } t) = 0 \quad (p < 1).$$

If $p = 1$, every trial succeeds and this probability is one. An infinite path is therefore different from a finite block.

Bernoulli processes model memoryless arrivals in discrete time; Poisson processes use continuous time. Markov chains allow dependence across time. The calculations here concern the Bernoulli model.

13.3 The number of successes in a fixed interval

Let $S = X_1 + \cdots + X_n$ be the number of successes in n trials. To have $S = k$, choose which k slots succeed. Each choice has probability $p^k(1-p)^{n-k}$, and the choices are disjoint. Consequently,

$$P(S = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

This is the binomial distribution. Its PMF is zero outside the listed integers. For the endpoint parameters, $p = 0$ gives $S = 0$ and $p = 1$ gives $S = n$ with probability one.

Linearity of expectation and independence give the moments:

$$\begin{aligned} \mathbb{E}[S] &= \sum_{i=1}^n \mathbb{E}[X_i] = np, \\ \text{Var}(S) &= \sum_{i=1}^n \text{Var}(X_i) = np(1-p). \end{aligned}$$

The mean adds without needing independence. The variance adds here because the trials are independent.

Counting successes fixes the number of trials and lets the count vary. Waiting for a success reverses this: the target count is fixed, while the number of trials varies.

13.4 The waiting time to the first arrival

Assume $p > 0$. Let T_1 be the number of trials up to and including the first success. Its smallest possible value is one. The event $T_1 = t$ requires $t - 1$ failures followed by a success:

$$P(T_1 = t) = (1 - p)^{t-1}p, \quad t = 1, 2, \dots$$

Thus T_1 has a geometric distribution on the positive integers. The probability of waiting more than s trials is

$$P(T_1 > s) = (1 - p)^s, \quad s = 0, 1, 2, \dots$$

This tends to zero, so the first success occurs in finite time with probability one. If $p = 0$, no success ever occurs.

Proposition 13.2 [GEOMETRIC WAITING-TIME MOMENTS]. For $0 < p \leq 1$,

$$\mathbb{E}[T_1] = \frac{1}{p}, \quad \text{Var}(T_1) = \frac{1 - p}{p^2}.$$

Proof. Compute the first two moments by differentiating the geometric series. Put $r = 1 - p$. For $0 < r < 1$,

$$\sum_{t=0}^{\infty} r^t = \frac{1}{1 - r}, \quad \sum_{t=1}^{\infty} t r^{t-1} = \frac{1}{(1 - r)^2}, \quad \sum_{t=2}^{\infty} t(t - 1) r^{t-2} = \frac{2}{(1 - r)^3}.$$

Multiplying the second series by p gives $\mathbb{E}[T_1] = p/(1 - r)^2 = 1/p$. Multiplying the third by pr gives

$$\mathbb{E}[T_1(T_1 - 1)] = \frac{2pr}{(1 - r)^3} = \frac{2r}{p^2}.$$

Since $T_1^2 = T_1(T_1 - 1) + T_1$,

$$\text{Var}(T_1) = \frac{2r}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{r}{p^2}.$$

For $p = 1$, $T_1 = 1$ with probability one, giving mean one and variance zero as claimed. $\square$

13.5 Memorylessness and a string of lottery losses

A waiting time is **MEMORYLESS** when its remaining wait, conditional on having waited so far, has its original law. For geometric waiting, past failures do not alter the future trials.

Proposition 13.3 [MEMORYLESS PROPERTY]. For $0 < p < 1$ and nonnegative integers s, t ,

$$P(T_1 > s + t \mid T_1 > s) = P(T_1 > t) = (1 - p)^t.$$

Equivalently, conditional on $T_1 > s$, the remaining wait $T_1 - s$ has the same distribution as T_1 .

Proof. The event $T_1 > s + t$ implies $T_1 > s$. Divide their probabilities:

$$P(T_1 > s + t \mid T_1 > s) = \frac{(1 - p)^{s+t}}{(1 - p)^s} = (1 - p)^t.$$

The conditioning probability is positive for $p < 1$. □

Example: Buy one lottery ticket every day, with independent wins of probability $p > 0$. Find the length of the initial string of losing days before the first win.

If L is that length, then $L = T_1 - 1$. Unlike T_1 , it can equal zero: the first ticket may win. For $\ell = 0, 1, 2, \dots$,

$$P(L = \ell) = (1 - p)^\ell p.$$

The power counts the losing tickets; the last factor accounts for the win that ends the string. Counting the final winning day would instead give T_1 , supported on $1, 2, \dots$. For $p = 1$, the string has length zero with probability one.

13.6 Interarrival times and the time of the k th success

Let T_2 be the number of additional trials from the first success to the second, and define subsequent gaps $T_3, T_4, \dots$ in the same way. The event $T_1 = t$ only fixes the first t trials. Future trials remain independent with success probability p . Hence

$$P(T_2 = s \mid T_1 = t) = (1 - p)^{s-1}p, \quad s = 1, 2, \dots$$

This conditional law does not depend on t , so T_2 is independent of T_1 and has the same geometric law.

More generally, prescribing gaps $t_1, \dots, t_k$ prescribes $t_i - 1$ failures and one success in each block. Independence of the underlying trials gives

$$P(T_1 = t_1, \dots, T_k = t_k) = \prod_{i=1}^k (1 - p)^{t_i-1} p.$$

Thus the INTERARRIVAL TIMES $T_1, T_2, \dots$ are independent and identically distributed geometric variables.

Let Y_k be the trial number of the k th success. Then

$$Y_k = T_1 + \dots + T_k, \quad \boxed{\mathbb{E}[Y_k] = \frac{k}{p}, \quad \text{Var}(Y_k) = \frac{k(1-p)}{p^2}}.$$

The mean and variance add because the gaps are independent.

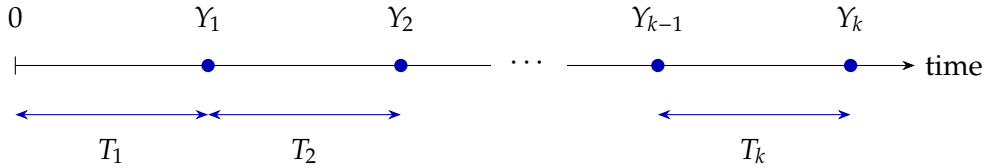


Figure 55: Arrival times Y_i accumulate the successive gaps T_i . Every gap includes the trial on which its ending success occurs. The spacing is schematic; the dots omit intermediate arrivals.

To have $Y_k = t$, the first $t - 1$ trials must contain exactly $k - 1$ successes, and trial t must succeed. Therefore

$$\boxed{P(Y_k = t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, \quad t = k, k+1, \dots}$$

The probability is zero for $t < k$. For $k = 1$, this reduces to the first-arrival PMF. If $p = 1$, the k th success occurs at k .

13.7 Splitting a Bernoulli process

Start with an arrival process of parameter p . Whenever an arrival occurs, route it to a kept stream with probability q and to a discarded stream with probability $1 - q$. The routing choices are independent of the arrivals and of each other.

Represent the choice in slot i by an independent Bernoulli(q) variable C_i , independent of all X_j . The two output indicators are

$$K_i = X_i C_i, \quad D_i = X_i (1 - C_i).$$

A kept arrival requires both an original arrival and the keep choice. A discarded arrival requires the other choice, so

$$P(K_i = 1) = pq, \quad P(D_i = 1) = p(1 - q).$$

For different slots, the pairs (X_i, C_i) are independent. Applying the routing rule to each pair preserves independence across slots. Each output is therefore a Bernoulli process, with parameters pq and $p(1 - q)$, respectively.

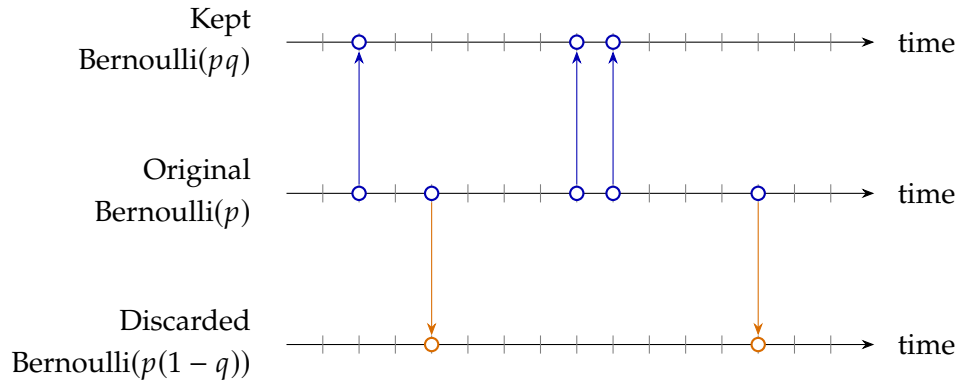


Figure 56: Each original arrival enters exactly one output stream, in the same slot. Arrows show the routing choices in a schematic path. Independent routing produces a Bernoulli law for each output stream.

The two output streams are generally dependent: they cannot both have an arrival in the same slot. Independence across time within each stream does not make the two streams independent of one another.

13.8 Merging independent Bernoulli processes

Let A_i and B_i be the indicators of two independent Bernoulli processes, with parameters p and q . Record one merged arrival if either process has an arrival in that slot. Even when both arrive, the merged slot records just one.

The merged indicator is $M_i = \max\{A_i, B_i\}$. It is zero exactly when both original indicators are zero. Independence gives

$$\begin{aligned} P(M_i = 1) &= 1 - P(A_i = 0, B_i = 0) \\ &= 1 - (1 - p)(1 - q) = \boxed{p + q - pq}. \end{aligned}$$

Equivalently, add the two arrival probabilities and subtract the overlap pq , which was counted twice.

The pairs (A_i, B_i) are independent across slots. The merged indicators depend on separate pairs, so they too are independent. Thus the merged process is Bernoulli with parameter $p + q - pq$.

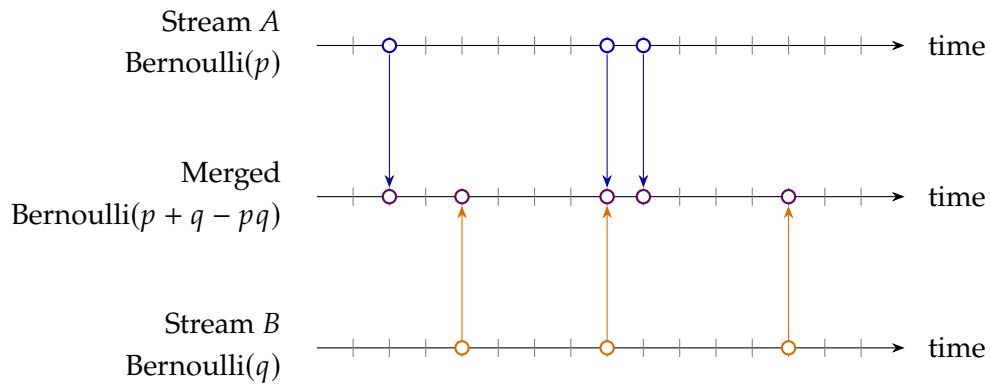


Figure 57: The merged stream contains the union of the original arrivals. At the shared slot, two incoming arrows meet one merged marker: a collision counts as one arrival. The paths are schematic.

The sum $A_i + B_i$ would count a collision as two. The indicator M_i instead records whether at least one arrival occurs, which explains the subtraction of pq .

Revision. A Bernoulli process has independent binary trials with a fixed success probability p . A count in n trials is binomial, with mean np and variance $np(1-p)$. Interarrival times are i.i.d. geometric variables on $1, 2, \dots$, with mean $1/p$ and variance $(1-p)/p^2$. The k th success time is their sum. Independent routing splits an arrival probability into pq and $p(1-q)$. Merging independent streams gives $p + q - pq$.

Common mistakes: assuming independence from binary outcomes alone, giving an infinite path the probability of a finite block, counting failures rather than trials in T_1 , using $\binom{t}{k}$ for the k th-arrival PMF, or counting a merged collision twice. Split outputs need not be independent.

Check yourself: What assumptions define a Bernoulli process? Why does the remaining geometric wait have the same law after failures? Why does $Y_k = t$ involve $t - 1$ trials followed by a success? Which independence assumptions justify the split and merged processes?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 13: Bernoulli Process](#), PDF pp. 1–2. The count, waiting-time, and arrival-time formulas are filled in; the sample-path, memoryless, splitting, and merging arguments are completed. The lottery convention and endpoint cases are explicit, and the arrival diagrams redrawn. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 14: POISSON PROCESS - I

Arrivals need not occur on a fixed grid of time slots. A Poisson process places them in continuous time, while retaining a constant arrival rate and independence between disjoint time intervals.

14.1 From Bernoulli trials to continuous arrivals

For a Bernoulli process with success probability p , the number of arrivals in n slots is binomial. The wait to the first arrival is geometric, and the trial number of the k th arrival has a PASCAL DISTRIBUTION, the law of a sum of k independent geometric waiting times. The relevant PMFs are

$$\begin{aligned}P(S_n = j) &= \binom{n}{j} p^j (1-p)^{n-j}, & j = 0, \dots, n, \\P(T_1 = m) &= (1-p)^{m-1} p, & m = 1, 2, \dots, \\P(Y_k = m) &= \binom{m-1}{k-1} p^k (1-p)^{m-k}, & m = k, k+1, \dots\end{aligned}$$

For $0 < p < 1$, geometric waiting is memoryless: $P(T_1 > s + u \mid T_1 > s) = (1-p)^u$ for nonnegative integers s, u . Past failures do not change the remaining wait.

To move to continuous time, divide time into slots of width δ . A rate λ suggests a success probability near $\lambda\delta$ in each short slot. As the slots become finer, the arrival rate stays fixed while their individual success probabilities shrink. This leads to Poisson counts and continuous waiting times.

14.2 Defining a Poisson process

Let N_t be the number of arrivals in $(0, t]$, with $N_0 = 0$. For $0 \leq a < b$, the count in $(a, b]$ is $N_b - N_a$. We describe the process using the laws of these interval counts.

Definition 14.1 [POISSON PROCESS]. A POISSON PROCESS of rate $\lambda > 0$ is an arrival process with the following properties.

1. TIME HOMOGENEITY: the probability of j arrivals in an interval depends only on its duration τ . Write it as $P(j, \tau)$.
2. INDEPENDENT INCREMENTS: counts in disjoint time intervals are independent.
3. For a short interval of duration δ ,

$$P(0, \delta) = 1 - \lambda\delta + o(\delta),$$

$$P(1, \delta) = \lambda\delta + o(\delta),$$

$$P(\text{at least two arrivals in } \delta) = o(\delta).$$

Here $o(\delta)$ denotes an error whose ratio to δ tends to zero as $\delta \downarrow 0$.

The ARRIVAL RATE λ has units of arrivals per unit time. Thus $\lambda\delta$ is a dimensionless probability approximation. Multiple arrivals in one short interval are possible, but their probability is negligible compared with its duration. The small-interval formulas are approximations, not exact formulas for a long interval.

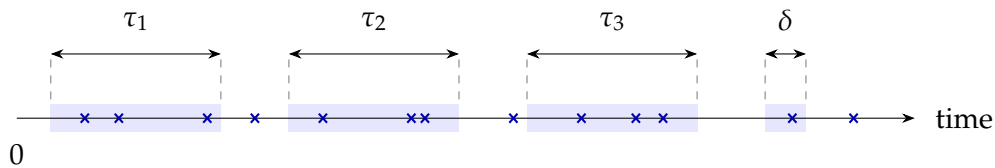


Figure 58: The highlighted intervals are disjoint, so their counts are independent. Each count law depends on the interval's duration, not its position. Crosses represent a schematic arrival path; the short interval has duration δ .

14.3 The distribution of an arrival count

Fix $t > 0$. Divide $(0, t]$ into n equal slots of width $\delta = t/n$. Let I_i indicate whether slot i contains at least one arrival, and put $B_n = I_1 + \cdots + I_n$. Homogeneity and independent increments make these indicators independent Bernoulli variables, with a common parameter

$$p_n = \lambda \frac{t}{n} + o(1/n), \quad np_n \longrightarrow \lambda t.$$

Consequently, B_n is binomial. It counts occupied slots, which agrees with N_t unless some slot has multiple arrivals. By the union bound,

$$P(B_n \neq N_t) \leq n P(\text{at least two arrivals in } t/n) = n o(1/n) \longrightarrow 0.$$

Thus the slot approximation becomes exact in probability.

For a fixed nonnegative integer j , set $\theta = \lambda t$. The binomial PMF can be written as

$$\begin{aligned} P(B_n = j) &= \frac{(np_n)^j}{j!} \left(\prod_{r=0}^{j-1} \left(1 - \frac{r}{n} \right) \right) (1 - p_n)^n (1 - p_n)^{-j} \\ &\longrightarrow \frac{\theta^j}{j!} e^{-\theta}. \end{aligned}$$

The finite product tends to one, $(1 - p_n)^{-j}$ tends to one, and $(1 - p_n)^n \rightarrow e^{-\theta}$ because $n \log(1 - p_n) = -np_n + o(1) \rightarrow -\theta$. For $j = 0$, the product is empty and equals one. Since the probabilities for B_n and N_t differ by at most $P(B_n \neq N_t)$, the same limit gives the exact count law:

$$P(j, t) = P(N_t = j) = e^{-\lambda t} \frac{(\lambda t)^j}{j!}, \quad j = 0, 1, 2, \dots$$

Hence N_t is Poisson with parameter λt . The same formula applies to any interval of duration t .

14.4 Count moments and the email example

The count's parameter combines rate and elapsed time. For $\theta = \lambda t$, the exponential series gives

$$\begin{aligned}\mathbb{E}[N_t] &= e^{-\theta} \sum_{j=1}^{\infty} j \frac{\theta^j}{j!} = \theta e^{-\theta} \sum_{r=0}^{\infty} \frac{\theta^r}{r!} = \theta, \\ \mathbb{E}[N_t(N_t - 1)] &= e^{-\theta} \sum_{j=2}^{\infty} j(j-1) \frac{\theta^j}{j!} = \theta^2.\end{aligned}$$

Use $N_t^2 = N_t(N_t - 1) + N_t$ to obtain

$$\boxed{\mathbb{E}[N_t] = \lambda t, \quad \text{Var}(N_t) = \theta^2 + \theta - \theta^2 = \lambda t}.$$

These moments follow directly from the exact Poisson PMF.

Example: Email arrives according to a Poisson process at a rate of 5 messages per hour. You check every thirty minutes. Find the probabilities of no new messages and one new message between consecutive checks.

Use hours for both rate and duration: $t = 1/2$ hour, so $\lambda t = 5/2$. The number of new messages N is $\text{Poisson}(5/2)$, and

$$\begin{aligned}P(N = 0) &= e^{-5/2} \frac{(5/2)^0}{0!} = \boxed{e^{-5/2} \approx 0.0821}, \\ P(N = 1) &= e^{-5/2} \frac{(5/2)^1}{1!} = \boxed{\frac{5}{2} e^{-5/2} \approx 0.2052}.\end{aligned}$$

The interval's starting time does not change these probabilities, and counts between different checks are independent.

14.5 The time of the k th arrival

Let Y_k be the elapsed time from zero to the k th arrival, where k is a positive integer. By time $y \geq 0$, that arrival has occurred exactly when at least k arrivals have occurred:

$$\{Y_k \leq y\} = \{N_y \geq k\}.$$

This converts a waiting-time question into a count question:

$$F_{Y_k}(y) = 1 - e^{-\lambda y} \sum_{j=0}^{k-1} \frac{(\lambda y)^j}{j!}, \quad y \geq 0.$$

The CDF is zero for $y < 0$. To differentiate, put $u = \lambda y$. The derivative of $e^{-u} \sum_{j=0}^{k-1} u^j / j!$ cancels all but its last term:

$$e^{-u} \left(\sum_{j=0}^{k-2} \frac{u^j}{j!} - \sum_{j=0}^{k-1} \frac{u^j}{j!} \right) = -e^{-u} \frac{u^{k-1}}{(k-1)!}.$$

For $k = 1$, the first sum is empty. Multiplying by the chain-rule factor λ gives the ERLANG DENSITY:

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \quad y \geq 0.$$

It is zero for $y < 0$; its value at the endpoint does not affect the law. The CDF tends to one, since an exponential dominates each of the finitely many polynomial terms in its tail.

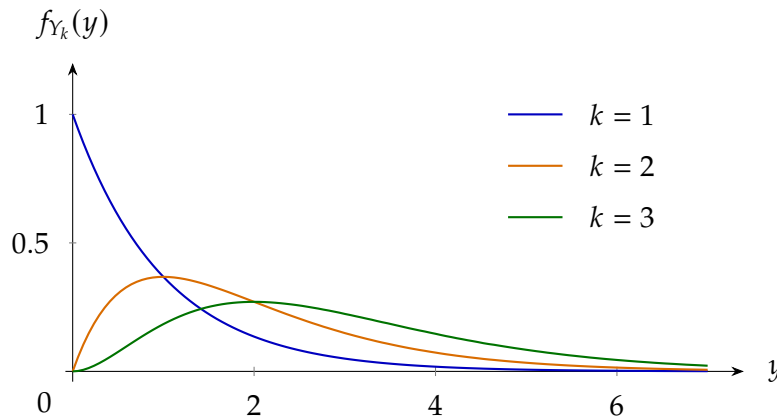


Figure 59: Erlang densities for $k = 1, 2, 3$ at rate $\lambda = 1$. The first-arrival density decreases from one. Later-arrival densities begin at zero and rise before decaying. These are formula plots, not measured arrival data.

14.6 Exponential waiting and memorylessness

For $k = 1$, the Erlang law becomes exponential:

$$f_{Y_1}(y) = \lambda e^{-\lambda y}, \quad y \geq 0.$$

The wait exceeds y exactly when there are no arrivals in $(0, y]$. Thus

$$P(Y_1 > y) = P(N_y = 0) = e^{-\lambda y}.$$

For $s, u \geq 0$, the survival probabilities give

$$\begin{aligned} P(Y_1 > s + u \mid Y_1 > s) &= \frac{e^{-\lambda(s+u)}}{e^{-\lambda s}} \\ &= e^{-\lambda u} = P(Y_1 > u). \end{aligned}$$

The remaining wait after no arrival in the first s units has the original exponential law. This is the continuous-time memoryless property.

More generally, observe the process at a fixed time a . The time W_a from a to the next arrival satisfies

$$\{W_a > u\} = \{N_{a+u} - N_a = 0\}.$$

Time homogeneity makes this event's probability $e^{-\lambda u}$, and independent increments make the future arrival behavior independent of the past arrival history. Previous arrivals do not change the waiting law. The intervals between successive arrivals likewise have the exponential waiting law of rate λ .

14.7 The Bernoulli–Poisson relation

The discrete approximation keeps the expected arrival count fixed. With $n = t/\delta$ slots and success probability $p = \lambda\delta$, it has

$$np = \lambda t.$$

This choice represents the first-order slot model; the exact occupied-slot probability has the smaller error described above. Taking $\delta \downarrow 0$ changes binomial counts into Poisson counts and discrete waits into continuous waits.

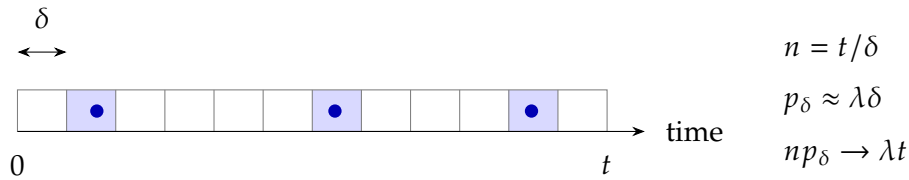


Figure 60: A fine time grid converts an arrival path into binary occupied-slot indicators. As the slot width shrinks, multiple arrivals in one slot become negligible. The displayed path and slot width are schematic.

Quantity	Poisson process	Bernoulli process
Arrival times	Continuous	Discrete
Rate or probability	λ per unit time	p per trial
Count in an interval	Poisson	Binomial
Interarrival time	Exponential	Geometric
Time to the k th arrival	Erlang	Pascal

The count laws have the same mean in the approximation. The units of the waiting variables differ: Bernoulli waits count trials, while Poisson waits measure elapsed time.

14.8 Sums and merged Poisson processes

First consider independent Poisson random variables A and B with parameters α and β . For $m = 0, 1, 2, \dots$, convolution and the binomial theorem give

$$\begin{aligned} P(A + B = m) &= \sum_{j=0}^m e^{-\alpha} \frac{\alpha^j}{j!} e^{-\beta} \frac{\beta^{m-j}}{(m-j)!} \\ &= \frac{e^{-(\alpha+\beta)}}{m!} \sum_{j=0}^m \binom{m}{j} \alpha^j \beta^{m-j} \\ &= \frac{e^{-(\alpha+\beta)} (\alpha + \beta)^m}{m!}. \end{aligned}$$

Hence

$$A + B \sim \text{Poisson}(\alpha + \beta).$$

Now merge two independent Poisson processes with rates λ_1 and λ_2 . Their count over any interval of duration t is the sum of independent Poisson variables with parameters $\lambda_1 t$ and $\lambda_2 t$. The merged count is therefore Poisson with parameter $(\lambda_1 + \lambda_2)t$. Counts over disjoint intervals remain independent, since they are functions of separate independent pairs of original counts. Thus the merged process is Poisson with rate

$$\lambda_1 + \lambda_2.$$

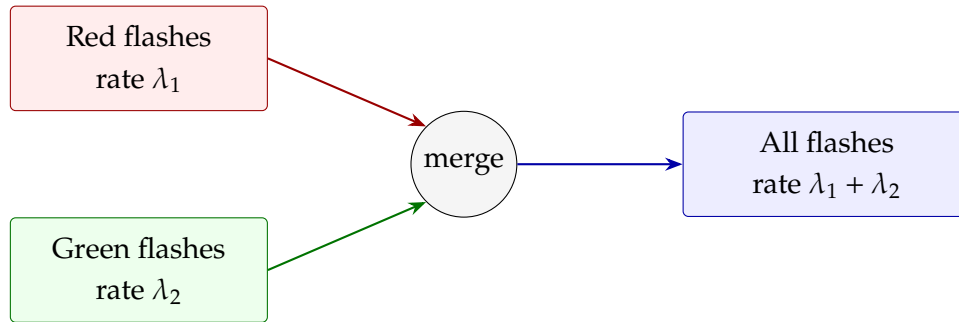


Figure 61: Independent red and green Poisson flashes form one Poisson stream when combined. The rates add. The arrows describe the model, not a physical arrangement of bulbs.

14.9 Which process supplies the next arrival?

Example: A red bulb and a green bulb flash according to independent Poisson processes with positive rates λ_1 and λ_2 . What is the probability that the next flash is red?

From a fixed observation time, let U and V be the remaining waits for a red and green flash. Their densities are exponential, and the waits are independent because the processes are independent. The next flash is red when $U < V$.

Condition on the red waiting time $U = u$. The green wait must exceed it. Integrating gives

$$\begin{aligned} P(U < V) &= \int_0^\infty f_U(u)P(V > u) du \\ &= \int_0^\infty \lambda_1 e^{-\lambda_1 u} e^{-\lambda_2 u} du \\ &= \boxed{\frac{\lambda_1}{\lambda_1 + \lambda_2}}. \end{aligned}$$

The green probability is $\lambda_2/(\lambda_1 + \lambda_2)$. Since independent continuous waits tie with probability zero, these two probabilities sum to one. The fraction of the total rate contributed by each process gives its probability of supplying the next arrival.

Revision. Poisson arrivals have a constant rate and independent counts in disjoint intervals. The count in duration t is $\text{Poisson}(\lambda t)$, with mean and variance λt . The first-arrival and interarrival laws are exponential; the k th arrival time is Erlang. Exponential waiting is memoryless. Independent Poisson processes merge by adding rates, and the next-arrival probability from stream i is its rate divided by the combined rate.

Common mistakes: mixing time units, treating $\lambda \delta$ as an exact long-interval probability, replacing elapsed time by a trial count, assuming overlapping interval counts are independent, or using a rate as a Poisson parameter without multiplying by duration.

Check yourself: Why does the occupied-slot count converge to the actual count? How does $Y_k \leq y$ become a statement about N_y ? Why does the exponential wait ignore past arrivals? How do independence and convolution justify merging?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 14: Poisson Process - I](#), PDF pp. 1–2. Small-interval approximations are expressed with explicit error terms. The Poisson limit, count moments, email example, Erlang density, memoryless property, and merging calculations are completed, and the diagrams redrawn. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 15: POISSON PROCESS - II

Poisson counts and exponential waiting times answer different questions about the same arrival process. We use both to study stopping rules, combined and routed arrivals, and the interval seen by an observer who arrives independently of the process.

15.1 The count and waiting-time laws

A Poisson process of rate $\lambda > 0$ has time-homogeneous counts and independent increments over disjoint intervals. For a short interval of duration δ ,

$$\begin{aligned}P(N_\delta = 0) &= 1 - \lambda\delta + o(\delta), \\P(N_\delta = 1) &= \lambda\delta + o(\delta), \\P(N_\delta \geq 2) &= o(\delta).\end{aligned}$$

The error divided by δ tends to zero. Over an interval of duration t , the exact law is

$$P(N_t = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots, \quad \mathbb{E}[N_t] = \text{Var}(N_t) = \lambda t.$$

If T is an interarrival time, then

$$f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0, \quad P(T > t) = e^{-\lambda t}, \quad \mathbb{E}[T] = \frac{1}{\lambda}.$$

The expectation follows by integrating the survival probability: $\mathbb{E}[T] = \int_0^\infty e^{-\lambda t} dt = 1/\lambda$. The time Y_k to the k th arrival has the Erlang density

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \quad y \geq 0.$$

Both densities are zero for negative arguments.

Exponential waiting is memoryless. For $s, u \geq 0$,

$$P(T > s + u \mid T > s) = e^{-\lambda u}.$$

At a fixed observation time, the wait to the next arrival is exponential with rate λ and independent of the past. This lets us restart a waiting-time calculation after learning that no arrival has yet occurred.

15.2 Poisson fishing: the stopping rule and probabilities

Example: Fish arrive as a Poisson process of rate $\lambda = 0.6$ per hour. Fish for two hours. If there has been no catch, keep fishing until the first catch.

Let T be the first-catch time, S the total fishing time, and K the total number of fish caught. The rule gives

$$S = \max\{2, T\}, \quad K = \max\{1, N_2\}.$$

Here N_2 counts catches in the first two hours. If it is positive, stop at hour 2 with all those catches. Otherwise, the extension ends with exactly one catch.

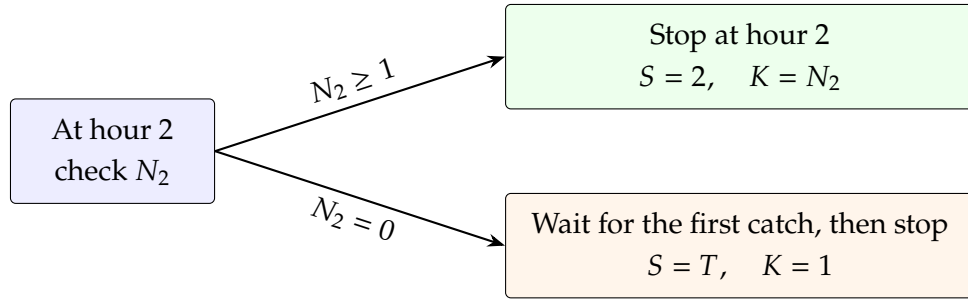


Figure 62: The decision at hour 2 determines whether there is an extension. During the extension, the first catch ends the trip.

(a) More than two hours. This occurs exactly when the first two hours have no catch:

$$P(S > 2) = P(N_2 = 0) = e^{-1.2} \approx 0.3012.$$

(b) More than two and less than five hours. The first catch must occur between hours 2 and 5:

$$P(2 < S < 5) = P(2 < T < 5) = e^{-1.2} - e^{-3} \approx 0.2514.$$

Arrival times have continuous laws, so the endpoints do not change this probability.

(c) At least two fish. The extension can produce only one fish. Thus

$$P(K \geq 2) = P(N_2 \geq 2) = 1 - e^{-1.2}(1 + 1.2) \approx 0.3374.$$

15.3 Poisson fishing: expected catches and time

(d) **Expected number of fish.** Write the catch count as

$$K = N_2 + \mathbf{1}_{\{N_2=0\}}.$$

The indicator adds the one catch obtained during an extension. Linearity of expectation gives

$$\mathbb{E}[K] = \mathbb{E}[N_2] + P(N_2 = 0) = 1.2 + e^{-1.2} \approx 1.5012.$$

(e) **Expected future time after four hours of fishing.** Still fishing at hour 4 means $S > 4$, equivalently $T > 4$. There has been no catch, and the remaining trip ends at the first one. Memorylessness gives

$$P(S - 4 > u \mid S > 4) = P(T > 4 + u \mid T > 4) = e^{-0.6u}.$$

Hence

$$\mathbb{E}[S - 4 \mid S > 4] = \frac{1}{0.6} = \frac{5}{3} \text{ hours}.$$

The four hours already spent do not lengthen the expected remaining exponential wait.

(f) **Expected total fishing time.** Every trip lasts at least two hours. An extension occurs with probability $e^{-1.2}$; conditional on an extension, its mean length is $1/0.6$ hour. Therefore

$$\begin{aligned} \mathbb{E}[S] &= 2 + P(T > 2)\mathbb{E}[T - 2 \mid T > 2] \\ &= 2 + \frac{e^{-1.2}}{0.6} \approx 2.5020 \text{ hours}. \end{aligned}$$

The unconditional extra time includes trips with zero extension. The conditional future time in part (e) concerns only trips that are still continuing.

15.4 Merging processes and identifying the next arrival

Let two independent Poisson processes have positive rates λ_1 and λ_2 . Combining all their arrivals gives a Poisson process with rate $\lambda_1 + \lambda_2$.

Over an interval of duration t , the combined count is the sum of independent Poisson counts with parameters $\lambda_1 t$ and $\lambda_2 t$. Their sum is Poisson with parameter $(\lambda_1 + \lambda_2)t$. Counts in disjoint intervals remain independent because the original interval counts are independent. These facts establish the merged process law.

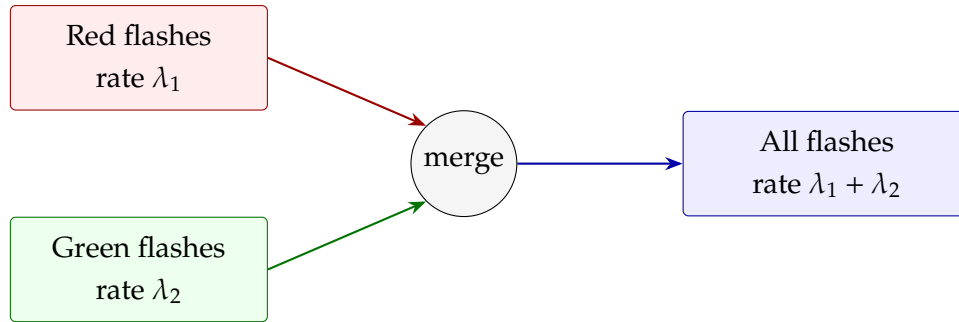


Figure 63: Independent streams combine by adding their rates. The merged stream keeps arrivals from both colors.

To identify the next arrival, let U and V be the independent exponential waits for the two streams from a fixed observation time. The first stream wins when $U < V$:

$$\begin{aligned} P(U < V) &= \int_0^\infty \lambda_1 e^{-\lambda_1 u} P(V > u) du \\ &= \int_0^\infty \lambda_1 e^{-(\lambda_1 + \lambda_2)u} du = \boxed{\frac{\lambda_1}{\lambda_1 + \lambda_2}}. \end{aligned}$$

The second stream supplies the next arrival with probability $\lambda_2/(\lambda_1 + \lambda_2)$. The waits tie with probability zero. Thus each stream's share of the combined rate is also its probability of supplying the next arrival.

15.5 The lifetime of the last of three bulbs

Example: Install three bulbs with independent exponential lifetimes of rate λ . Find the expected time until the last bulb dies.

Let the lifetimes be X_1, X_2, X_3 and put $L = \max\{X_1, X_2, X_3\}$. All three have failed by time t exactly when each lifetime is at most t . Independence gives

$$P(L \leq t) = (1 - e^{-\lambda t})^3, \quad t \geq 0.$$

Use the survival probability to compute the mean:

$$\begin{aligned} \mathbb{E}[L] &= \int_0^\infty [1 - (1 - e^{-\lambda t})^3] dt \\ &= \int_0^\infty (3e^{-\lambda t} - 3e^{-2\lambda t} + e^{-3\lambda t}) dt \\ &= \frac{3}{\lambda} - \frac{3}{2\lambda} + \frac{1}{3\lambda} = \boxed{\frac{11}{6\lambda}}. \end{aligned}$$

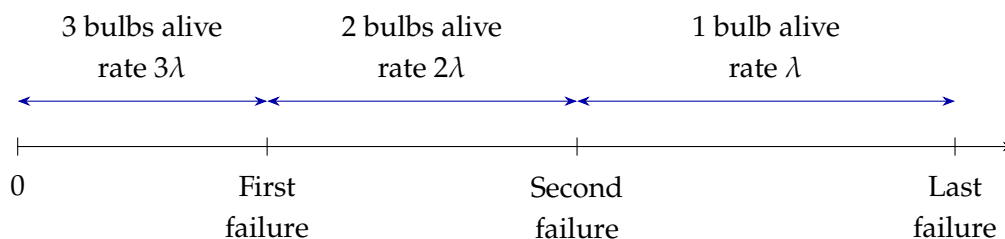


Figure 64: Each failure reduces the number of surviving bulbs by one. The indicated rates describe the next failure while that many bulbs remain alive; the drawn gaps are schematic.

The successive waits explain the same answer. With three bulbs alive, the probability of no failure for u time units is $(e^{-\lambda u})^3 = e^{-3\lambda u}$, so the first wait has mean $1/(3\lambda)$. After a failure, memorylessness leaves the surviving bulbs with independent exponential remaining lifetimes. The next two waits have means $1/(2\lambda)$ and $1/\lambda$. Adding their means gives

$$\mathbb{E}[L] = \frac{1}{3\lambda} + \frac{1}{2\lambda} + \frac{1}{\lambda} = \frac{11}{6\lambda}.$$

15.6 Splitting a Poisson process

Suppose email reaches a server as a Poisson process of rate λ . Each message independently goes to the USA with probability p and abroad with probability $1 - p$. Routing choices are also independent of the arrival process. This independent routing is called **SPLITTING**.

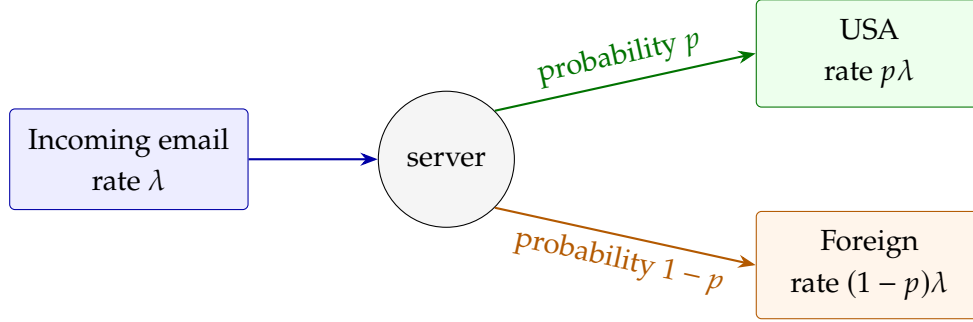


Figure 65: Each incoming message takes exactly one route. Independent routing produces output rates $p\lambda$ and $(1 - p)\lambda$.

To establish the output laws, fix an interval of duration t . Let A and B count its USA and foreign messages. The input count $N = A + B$ is $\text{Poisson}(\lambda t)$. Given $N = a + b$, exactly a of these messages choose the USA route with probability $\binom{a+b}{a} p^a (1-p)^b$. For $0 < p < 1$ and nonnegative integers a, b ,

$$\begin{aligned}
 P(A = a, B = b) &= e^{-\lambda t} \frac{(\lambda t)^{a+b}}{(a+b)!} \binom{a+b}{a} p^a (1-p)^b \\
 &= \left[e^{-p\lambda t} \frac{(p\lambda t)^a}{a!} \right] \left[e^{-(1-p)\lambda t} \frac{((1-p)\lambda t)^b}{b!} \right].
 \end{aligned}$$

Each bracket is a Poisson PMF. Summing over the other count shows that A is $\text{Poisson}(p\lambda t)$ and B is $\text{Poisson}((1-p)\lambda t)$. The product factorization also shows that these two counts are independent.

The laws depend only on interval duration. Counts and routing choices in disjoint intervals are independent, so both output streams have independent increments. Therefore the outputs are independent Poisson processes with rates

$$p\lambda \quad \text{and} \quad (1-p)\lambda.$$

At $p = 0$ or $p = 1$, one stream is empty and the other is the entire input. If the total count is fixed by conditioning, the two output counts instead satisfy the constraint $A + B = N$.

15.7 Random incidence in a Poisson process

An observer who arrives independently of the process sees the interval containing the observation time. This is called **RANDOM INCIDENCE**. Consider a homogeneous Poisson process that has been running indefinitely, so arrivals exist on both sides of an arbitrary fixed time a .

Let A be the time since the last arrival and B the time to the next one. The containing interval has length $L = A + B$.

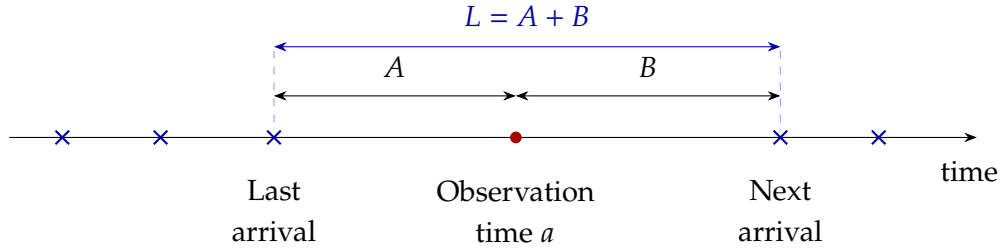


Figure 66: The observation time lies inside an interarrival interval. Its total length includes the elapsed part A and the remaining part B . Arrival positions are schematic.

For $x, y \geq 0$, the event $A > x$ means no arrival in the preceding x time units, and $B > y$ means no arrival in the following y time units. These intervals are disjoint. Hence

$$P(A > x) = e^{-\lambda x}, \quad P(B > y) = e^{-\lambda y},$$

$$P(A > x, B > y) = e^{-\lambda x} e^{-\lambda y}.$$

Thus A and B are independent exponential variables of rate λ . The same conclusion holds for a random observation time chosen independently of the arrivals.

Convolution gives the containing interval's density:

$$\begin{aligned} f_L(\ell) &= \int_0^\ell \lambda e^{-\lambda x} \lambda e^{-\lambda(\ell-x)} dx \\ &= \boxed{\lambda^2 \ell e^{-\lambda \ell}}, \quad \ell \geq 0. \end{aligned}$$

It is Erlang with shape 2, and $\mathbb{E}[L] = 2/\lambda$. An interval selected by observing a time has mean length twice the mean $1/\lambda$ of an ordinary interarrival interval. Longer intervals occupy more time and are more likely to contain the observer. The remaining wait B still has mean $1/\lambda$.

15.8 Renewal processes and the bus example

A RENEWAL PROCESS has successive arrivals with independent, identically distributed interarrival times. These times need not be exponential.

Suppose bus interarrival times are equally likely to be 5 or 10 minutes. Interpret a random observation time as a time sampled uniformly from a long period, independently of the bus arrivals. About half the intervals have each length, but the 10-minute intervals occupy twice as much time.

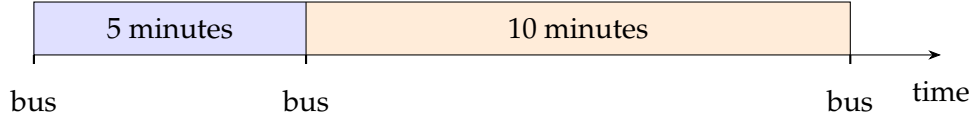


Figure 67: A 10-minute interval offers twice as much time in which an observer can arrive as a 5-minute interval. Widths are proportional to duration.

Let L be the containing interval's length. Weighting each length by the time it contributes gives

$$P(L = 5) = \frac{5(1/2)}{5(1/2) + 10(1/2)} = \boxed{\frac{1}{3}}, \quad P(L = 10) = \frac{2}{3}.$$

Within a selected interval of length ℓ , the observation point is uniform, so the mean remaining wait is $\ell/2$. For the time W to the next bus,

$$\boxed{\mathbb{E}[W] = \frac{1}{3} \frac{5}{2} + \frac{2}{3} \frac{10}{2} = \frac{25}{6} \text{ minutes} \approx 4.1667 \text{ minutes}.}$$

The observation-time probabilities are $1/3$ and $2/3$, even though the probabilities for a newly generated interval are $1/2$ and $1/2$.

Revision. Translate a stopping rule into events involving counts and arrival times. Memorylessness resets a remaining exponential wait. Independent Poisson streams merge by adding rates; independent routing splits a stream into Poisson outputs. A time observer favors longer intervals. For Poisson arrivals, the containing interval is Erlang(2), while its remaining part is exponential.

Common mistakes: counting extra catches after the fishing extension ends, confusing conditional and unconditional extra time, dropping the independence assumptions in splitting, or using interval frequencies as observation-time probabilities.

Check yourself: Why is $K = \max\{1, N_2\}$? Why do bulb failure rates decrease? How does the split-count joint PMF factor? Why is a 10-minute bus interval twice as likely to contain an observer?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 15: Poisson Process - II](#), PDF pp. 1–2. The fishing, bulb, splitting, and random-incidence calculations are completed, with explicit independence and observation-time assumptions. The diagrams are redrawn. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 16: MARKOV CHAINS - I

A useful state records everything needed to predict the next step. For a Markov chain, once the present state is known, earlier states add no information about the next transition. We begin with a queue, then compute transition probabilities and examine what a chain can do over a long time.

16.1 The checkout counter as a state model

Let X_n be the number of customers at a checkout counter at integer time n . Arrivals form a Bernoulli process with probability p per slot. A customer's service time is geometric with parameter q : in each slot of service, completion occurs with probability q .

Assume independent arrivals and service completions. For the finite queue with states $0, 1, \dots, 10$, use this slot rule: first attempt to complete service of a customer present at the start, then admit the arrival if space is available. An arrival finding ten customers is turned away.

For $1 \leq i \leq 9$, the possible changes and probabilities are

$$\begin{aligned} X_{n+1} = i + 1 & : u = p(1 - q), \\ X_{n+1} = i - 1 & : d = (1 - p)q, \\ X_{n+1} = i & : h = pq + (1 - p)(1 - q). \end{aligned}$$

An increase needs an arrival and no completion; a decrease needs a completion and no arrival. Both events or neither leave the count unchanged. Thus $u + d + h = 1$.

At the empty boundary, $p_{01} = p$ and $p_{00} = 1 - p$. At the full boundary, $p_{10,9} = d$ and $p_{10,10} = 1 - d$. There is no service completion when the queue is empty.

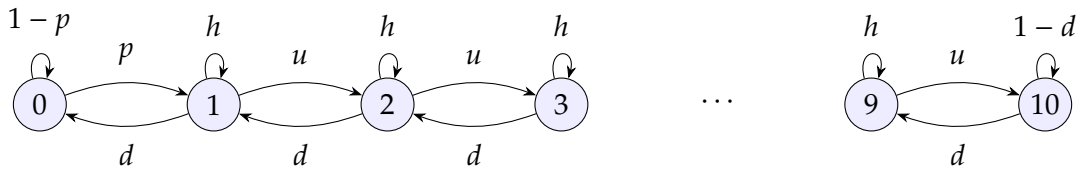


Figure 68: The finite queue moves up one, down one, or stays put. Interior rightward arrows have probability u , leftward arrows d , and loops h . The boundary labels implement the stated slot rule.

Geometric service is memoryless. For service time G , $P(G = s + 1 \mid G > s) = q$ after any s unsuccessful slots. Thus the number of customers is enough to determine the next-step law; elapsed service time need not be added to the state. The state includes the customer being served.

16.2 Finite-state Markov chains

Definition 16.1 [MARKOV CHAIN]. A FINITE-STATE MARKOV CHAIN is a sequence $X_0, X_1, \dots$ taking values in a finite state set, such as $\{1, \dots, m\}$, with the MARKOV PROPERTY

$$P(X_{n+1} = j \mid X_0 = i_0, \dots, X_n = i) = P(X_{n+1} = j \mid X_n = i) = p_{ij}.$$

The equality applies to histories with positive probability. Here the transition probabilities p_{ij} do not depend on n , so the chain is TIME HOMOGENEOUS.

The TRANSITION MATRIX is $P = (p_{ij})$. Its row i gives the distribution of the state after one step when the current state is i . Therefore

$$p_{ij} \geq 0, \quad \sum_{j=1}^m p_{ij} = 1.$$

Rows correspond to starting states and columns to destination states. A directed edge $i \rightarrow j$ appears in a state diagram when $p_{ij} > 0$; a self-loop represents staying in state i .

To specify a model, identify the states, the allowed transitions, their probabilities, and the initial state law. The starting state X_0 may be fixed or random. A complete description of a state must make the Markov property valid.

The property also determines path probabilities. For a fixed initial state i_0 and any path $i_0, i_1, \dots, i_n$, successive conditioning gives

$$P(X_1 = i_1, \dots, X_n = i_n \mid X_0 = i_0) = \prod_{r=1}^n p_{i_{r-1}, i_r}.$$

At every step the next factor depends only on the current state. This describes dependence through the state; it does not make successive states independent.

16.3 Transition probabilities over several steps

Define the n -STEP TRANSITION PROBABILITY by

$$r_{ij}(n) = P(X_n = j \mid X_0 = i).$$

At zero steps the starting state has not changed: $r_{ij}(0) = \mathbf{1}_{\{i=j\}}$. At one step, $r_{ij}(1) = p_{ij}$.

To arrive at j after n steps, first arrive at some state k after $n - 1$ steps, then make the last transition $k \rightarrow j$.

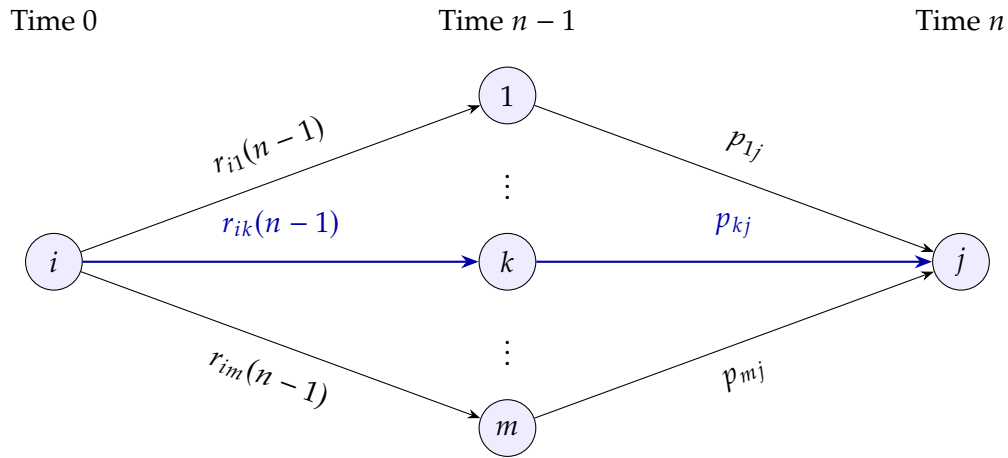


Figure 69: Partition paths by their state immediately before the last step. Each route through k contributes $r_{ik}(n-1)p_{kj}$. The middle column represents all states.

The law of total probability and the Markov property give

$$\begin{aligned} r_{ij}(n) &= \sum_{k=1}^m P(X_{n-1} = k \mid X_0 = i) P(X_n = j \mid X_{n-1} = k, X_0 = i) \\ &= \sum_{k=1}^m r_{ik}(n-1) p_{kj}. \end{aligned}$$

Terms with zero intermediate-state probability contribute zero. Writing $R(n) = (r_{ij}(n))$, this recursion is $R(n) = R(n-1)P$. Since $R(0) = I$, induction gives $R(n) = P^n$.

If the initial state is random, let $\alpha_i = P(X_0 = i)$. Conditioning on it yields

$$P(X_n = j) = \sum_{i=1}^m \alpha_i r_{ij}(n).$$

The state distribution is a weighted average of the conditional distributions for the possible starting states. For the row vector α , it is αP^n .

16.4 A two-state calculation

Consider the chain with transition matrix

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix}.$$

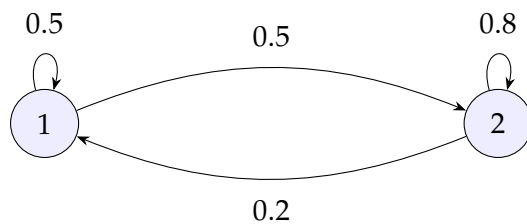


Figure 70: From either state, the two outgoing probabilities sum to one. Staying put is one of the possible transitions.

For example, a two-step return from state 1 can pass through either state:

$$r_{11}(2) = p_{11}p_{11} + p_{12}p_{21} = 0.5^2 + 0.5(0.2) = 0.35.$$

Applying the same calculation to each entry gives

$$R(0) = I, \quad R(1) = P, \quad R(2) = \begin{pmatrix} 0.35 & 0.65 \\ 0.26 & 0.74 \end{pmatrix}.$$

The requested values are

	$n = 0$	$n = 1$	$n = 2$	$n = 100$	$n = 101$
$r_{11}(n)$	1	0.5	0.35	$\approx 2/7$	$\approx 2/7$
$r_{12}(n)$	0	0.5	0.65	$\approx 5/7$	$\approx 5/7$
$r_{21}(n)$	0	0.2	0.26	$\approx 2/7$	$\approx 2/7$
$r_{22}(n)$	1	0.8	0.74	$\approx 5/7$	$\approx 5/7$

The large- n entries are approximations. An exact expression will show why both starting states produce the same limit.

16.5 The limit in the two-state example

Fix a starting state i and let $u_n = P(X_n = 1 \mid X_0 = i)$. The other state's probability is $1 - u_n$, so conditioning on the current state gives

$$u_{n+1} = 0.5u_n + 0.2(1 - u_n) = 0.2 + 0.3u_n.$$

The constant solution satisfies $u = 0.2 + 0.3u$, hence $u = 2/7$. Subtracting it from the recursion yields

$$u_{n+1} - \frac{2}{7} = 0.3 \left(u_n - \frac{2}{7} \right).$$

Iterating this equality gives

$$u_n = \frac{2}{7} + \left(u_0 - \frac{2}{7} \right) (0.3)^n.$$

Use $u_0 = 1$ for an initial state of 1 and $u_0 = 0$ for an initial state of 2. The exact matrix is

$$R(n) = \begin{pmatrix} \frac{2}{7} + \frac{5}{7}(0.3)^n & \frac{5}{7}(1 - (0.3)^n) \\ \frac{2}{7}(1 - (0.3)^n) & \frac{5}{7} + \frac{2}{7}(0.3)^n \end{pmatrix}.$$

This also gives the identity matrix when $n = 0$. Since $(0.3)^n \rightarrow 0$,

$$R(n) \rightarrow \begin{pmatrix} 2/7 & 5/7 \\ 2/7 & 5/7 \end{pmatrix}.$$

At $n = 100$, each displayed approximation has error at most $(5/7)(0.3)^{100} < 4 \times 10^{-53}$; at $n = 101$ it is smaller still. The limiting state probabilities do not depend on the starting state in this example. A random initial state gives the same limit by taking a weighted average of the two rows.

16.6 Transition probabilities can oscillate

Convergence is a question about a chain, not an automatic consequence of the Markov property. Consider

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}.$$

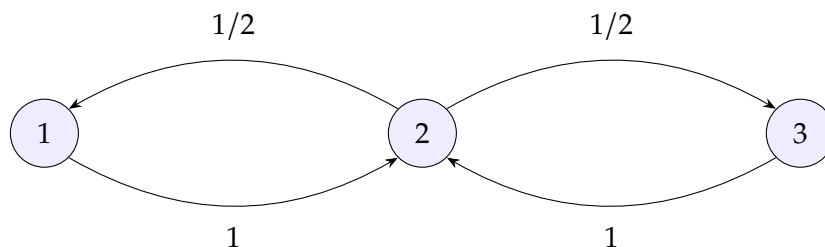


Figure 71: State 2 moves to an end state; either end state returns to 2. Starting at 2, the chain alternates between the center and the ends.

Starting at state 2, one step takes the chain to state 1 or 3. The next step returns to state 2 with certainty. Repeating this argument gives

$$r_{22}(n) = \begin{cases} 1, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}$$

The even and odd subsequences have different values, so $r_{22}(n)$ has no limit. The possibility of returning to a state does not by itself guarantee convergence of its occupancy probability.

16.7 Limits can depend on the initial state

Now consider the following transition structure. State 1 stays at 1. From state 2, the probabilities of moving to 1, staying at 2, and moving to 3 are 0.3, 0.4, and 0.3. States 3 and 4 can move only within their pair.

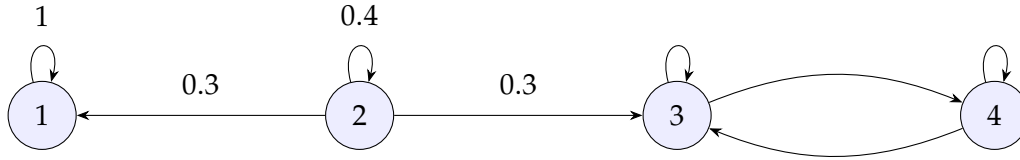


Figure 72: Once the chain enters state 1 or the pair $\{3, 4\}$, it cannot leave that set. Unlabeled arrows indicate positive-probability transitions; their values do not affect the probabilities of reaching state 1.

For every $n \geq 0$,

$$\boxed{r_{11}(n) = 1, \quad r_{31}(n) = 0}.$$

Starting at 1 means staying there. Starting at 3 gives no path to 1.

Starting at 2, arrival at state 1 by step n requires some number k of self-loops at 2, followed by the transition $2 \rightarrow 1$. For $n \geq 1$,

$$\begin{aligned} r_{21}(n) &= \sum_{k=0}^{n-1} (0.4)^k (0.3) \\ &= \frac{0.3}{1 - 0.4} (1 - (0.4)^n) = \boxed{\frac{1}{2} (1 - (0.4)^n)}. \end{aligned}$$

The final formula also gives $r_{21}(0) = 0$. It tends to $1/2$: eventually leaving 2 for 1 or for 3 is equally likely. Thus the limiting probability of being in state 1 is 1, 0, or $1/2$ for starting states 1, 3, or 2, respectively.

16.8 Recurrent states, transient states, and classes

A state j is **REACHABLE** from i if a path of positive-probability transitions leads from i to j . States **COMMUNICATE** if each is reachable from the other. A **COMMUNICATING CLASS** groups states that communicate. A class is **CLOSED** if no transition leads from it to a state outside it.

Definition 16.2 [RECURRENCE IN A FINITE CHAIN]. A state i is **RECURRENT** if every state reachable from i has a path back to i . Equivalently, its communicating class is closed. A state that fails this condition is **TRANSIENT**. A closed communicating class is a **RECURRENT CLASS**.

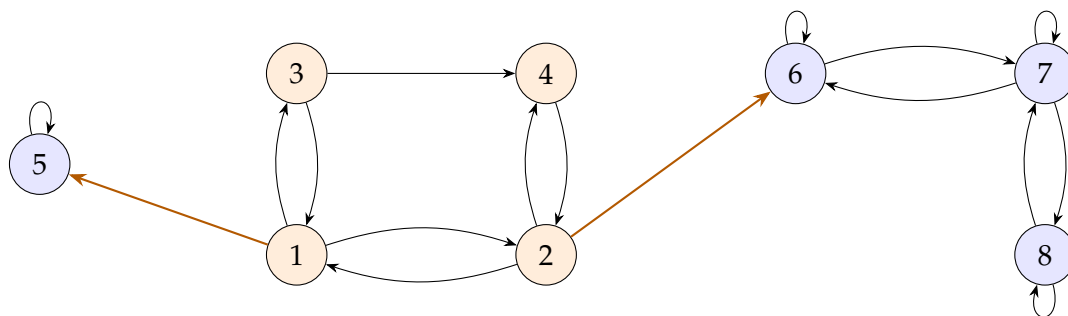


Figure 73: The arrows show possible transitions. Blue states form the recurrent classes $\{5\}$ and $\{6, 7, 8\}$. Orange states 1, 2, 3, 4 communicate but can escape to a recurrent class, so all four are transient.

States 1, 2, 3, 4 can reach one another, but $1 \rightarrow 5$ and $2 \rightarrow 6$ leave their class. There is no path back from 5 or from 6, 7, 8. Consequently, none of 1, 2, 3, 4 satisfies the return-path condition.

State 5 stays at 5, so $\{5\}$ is a recurrent class. The states 6, 7, 8 communicate within a closed set and form another recurrent class. An outside state may enter a recurrent class; closedness prohibits leaving it.

16.9 Why transient states disappear

The finite-state return-path condition has a probabilistic meaning. Starting from a recurrent state i , the chain returns to i with probability one. To see why, consider its closed communicating class. From each state there is a positive-probability path to i , and from i there is a positive-length return path. Finiteness lets us choose an integer M bounding all these path lengths and a number $\varepsilon > 0$ bounding their probabilities below.

From anywhere in the class, the chance of visiting i within the next M steps is at least ε . Repeated conditioning with the Markov property bounds the chance of avoiding i for k such blocks by $(1 - \varepsilon)^k$, which tends to zero. Applying the same reasoning after each return shows that i is visited infinitely often with probability one.

For a transient state i , choose a reachable state that cannot return to i . A shortest path to it does not revisit i . Following that path has some probability $\eta > 0$ and prevents any later visit to i . Thus at every visit there is at least probability η of escaping permanently. Conditional on a first visit,

$$P(\text{at least } k \text{ visits to } i) \leq (1 - \eta)^{k-1} \longrightarrow 0.$$

So the number of visits is finite almost surely. After the last visit the indicator $\mathbf{1}_{\{X_n=i\}}$ is zero; bounded convergence gives, for any initial distribution,

$$P(X_n = i) = \mathbb{E}[\mathbf{1}_{\{X_n=i\}}] \longrightarrow 0.$$

The probability of returning to a transient state can be positive. Transience means that a permanent escape is possible, and repeated visits eventually end almost surely.

Revision. A Markov state contains the information needed for the next step. A transition matrix has nonnegative entries and rows summing to one. Multi-step probabilities satisfy $R(n) = R(n-1)P = P^n$; a random initial law gives αP^n . Limits can fail to exist or depend on the starting state. In a finite chain, recurrent classes are closed communicating classes, while transient states are visited finitely often and their occupancy probabilities tend to zero.

Common mistakes: reversing the row and column roles, forgetting $R(0) = I$, assuming convergence from the Markov property alone, or treating any communicating class as closed.

Check yourself: Why does geometric service allow a queue-length state? Why sum over the state just before the last step? What causes the center state's oscillation? Which escape arrows make 1, 2, 3, 4 transient?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 16: Markov Processes - I](#), PDF pp. 1–2. The checkout slot order and capacity boundary are made explicit. The transition recursion, numerical examples, and finite-state recurrence arguments are completed; the state diagrams are redrawn. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 17: MARKOV CHAINS - II

The long-run behavior of a finite Markov chain depends on its recurrent classes and on whether transitions force a cyclic schedule. When the state probabilities converge, balance equations describe the limiting probability flow.

17.1 Review and the warmup chain

For a time-homogeneous Markov chain, p_{ij} is the probability of the next state being j given that the current state is i . The past adds no information once the current state is known. Multi-step probabilities satisfy

$$r_{ij}(n) = P(X_n = j \mid X_0 = i), \quad r_{ij}(n) = \sum_k r_{ik}(n-1)p_{kj}, \quad R(0) = I.$$

For a specified path, multiply the transition probabilities. For an endpoint event, sum over all paths of the required length.

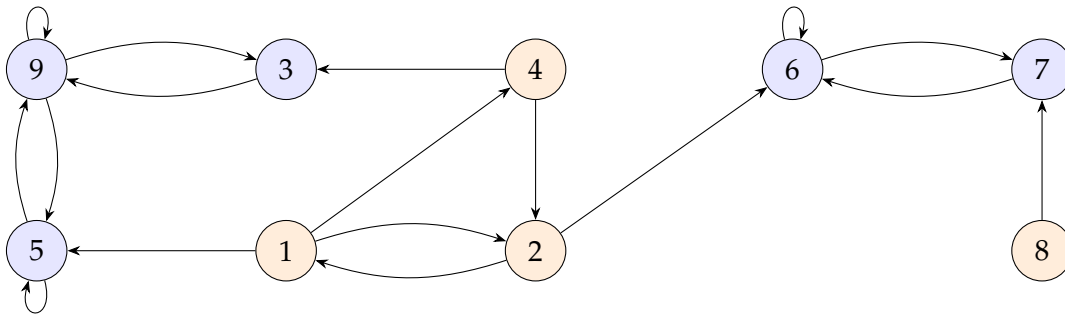


Figure 74: Arrows indicate positive-probability transitions. The blue states form the closed classes $\{3, 5, 9\}$ and $\{6, 7\}$. Orange states can leave permanently.

The event specifying the path $1 \rightarrow 2 \rightarrow 6 \rightarrow 7$ has probability

$$P(X_1 = 2, X_2 = 6, X_3 = 7 \mid X_0 = 1) = p_{12}p_{26}p_{67}.$$

There are exactly three four-step paths from 2 to 7:

$$2 \rightarrow 1 \rightarrow 2 \rightarrow 6 \rightarrow 7, \quad 2 \rightarrow 6 \rightarrow 6 \rightarrow 6 \rightarrow 7, \quad 2 \rightarrow 6 \rightarrow 7 \rightarrow 6 \rightarrow 7.$$

They are disjoint events, so

$$r_{27}(4) = p_{21}p_{12}p_{26}p_{67} + p_{26}p_{66}^2p_{67} + p_{26}p_{67}p_{76}p_{67}.$$

No numerical transition probabilities have been specified. These expressions retain the probabilities of the displayed edges.

Within a recurrent class every reachable state can return to the starting state. The sets $\{3, 5, 9\}$ and $\{6, 7\}$ are closed and communicating, so they are recurrent classes. States 1, 2, 4 communicate but can escape to either class; state 8 enters 7 and cannot return. Thus 1, 2, 4, 8 are transient, and their occupancy probabilities tend to zero.

17.2 Periodic recurrent classes

Definition 17.1 [PERIODICITY]. A recurrent class is **PERIODIC** if its states can be partitioned into $d > 1$ nonempty groups $G_0, \dots, G_{d-1}$ such that every transition from G_r leads to G_{r+1} , with indices taken modulo d . A class is **APERIODIC** if no such partition exists.

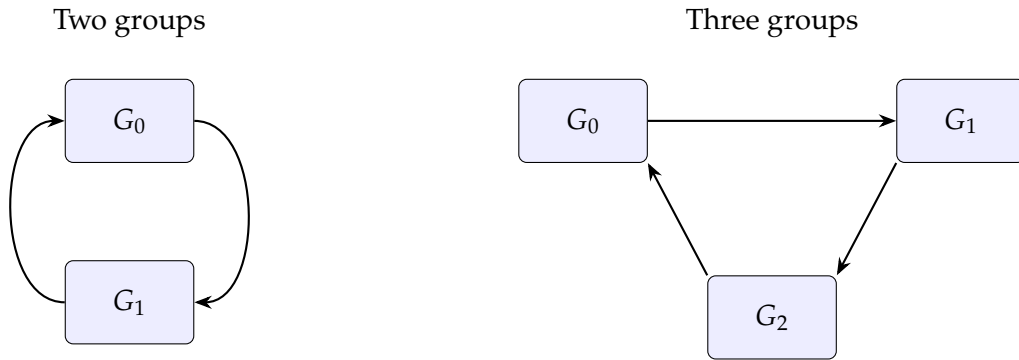


Figure 75: A cyclic partition forces the chain to advance one group at each step. The two diagrams illustrate two-group and three-group schedules.

Starting in G_0 , the chain is in $G_{n \bmod d}$ at time n . A return to the starting state is possible only after a number of steps divisible by d . This restriction can make state occupancy probabilities oscillate.

A positive-probability self-loop rules out a cyclic partition with $d > 1$: it would stay in the same group instead of moving to the next one. Thus a recurrent class containing a self-loop is aperiodic. The warmup's two recurrent classes are aperiodic, since the first contains loops at 5 and 9 and the second contains a loop at 6. The entire warmup chain still has two recurrent classes.

17.3 A periodic class with two return cycles

Consider the following nine-state chain. All displayed arrows have positive probability; at a state with more than one outgoing arrow, their probabilities sum to one.

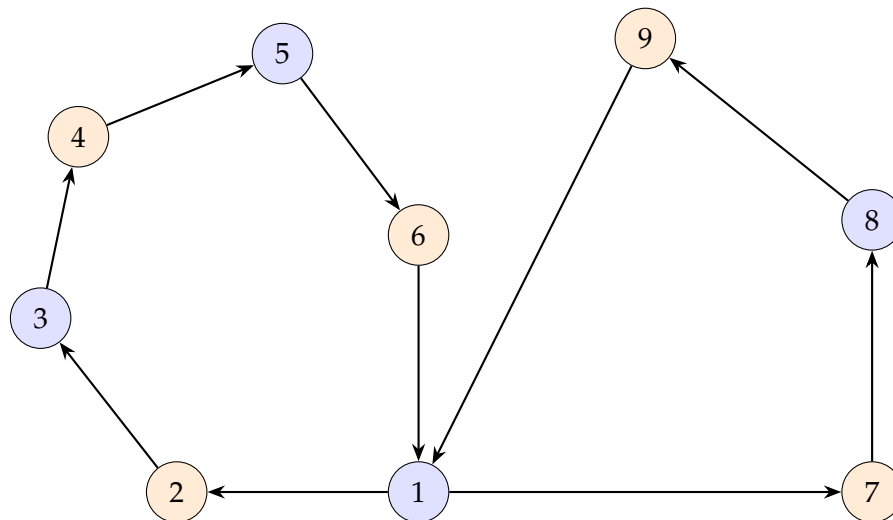


Figure 76: The blue group is $G_0 = \{1, 3, 5, 8\}$ and the orange group is $G_1 = \{2, 4, 6, 7, 9\}$. Every arrow changes group. The two directed cycles share state 1.

Every state can reach 1, and 1 can reach every state, so all nine states form one recurrent class. Its cycles are

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 1, \quad 1 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 1.$$

Both have even length. The displayed two-group partition forces every return time to be even. Four-step and six-step returns to 1 are both possible. Any cyclic grouping must therefore have a group count dividing both 4 and 6; the only possible count greater than one is 2.

This class has period 2, despite containing cycles of different lengths. Starting in the blue group, the chain is in the blue group with probability one at even times and probability zero at odd times. Consequently, the state probabilities cannot all converge to a common time-independent distribution.

17.4 Steady-state convergence and balance equations

Theorem 17.2 [STEADY-STATE CONVERGENCE]. *For a finite Markov chain with exactly one recurrent class, if that class is aperiodic, there are probabilities π_j such that, for every starting state i ,*

$$\lim_{n \rightarrow \infty} r_{ij}(n) = \pi_j.$$

The limit is independent of i . Transient states have $\pi_j = 0$; states in the recurrent class have $\pi_j > 0$.

These are the STEADY-STATE PROBABILITIES. The single-class condition makes the long-run destination common to all starting states. Aperiodicity removes the forced group schedule. Transient starting states are allowed.

To find the probabilities, take limits in the finite sum of the transition recursion:

$$\pi_j = \lim_{n \rightarrow \infty} \sum_k r_{ik}(n-1)p_{kj} = \sum_k \pi_k p_{kj}.$$

Taking limits in $\sum_j r_{ij}(n) = 1$ gives normalization. Thus the BALANCE EQUATIONS are

$$\boxed{\pi_j = \sum_k \pi_k p_{kj} \quad \text{for every } j, \quad \sum_j \pi_j = 1, \quad \pi_j \geq 0.}$$

For the row vector π , this is $\pi = \pi P$. A distribution satisfying it is STATIONARY: starting with $X_0 \sim \pi$ gives the same state distribution after every transition.

The convergence theorem supplies the conditions under which this stationary distribution is the common limit. Solving balance equations alone does not establish that arbitrary initial distributions converge. In the warmup, there are two recurrent classes. In the nine-state example, the single recurrent class is periodic. Neither satisfies both convergence conditions.

17.5 Solving the two-state balance equations

For the two-state chain

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix},$$

both states communicate and have self-loops. They form one aperiodic recurrent class, so the convergence theorem applies.

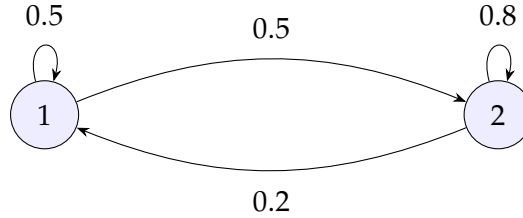


Figure 77: The self-loops permit the chain to stay in either state, preventing a forced alternation between the states.

The two balance equations and normalization are

$$\pi_1 = 0.5\pi_1 + 0.2\pi_2, \quad \pi_2 = 0.5\pi_1 + 0.8\pi_2, \quad \pi_1 + \pi_2 = 1.$$

The first two equations both reduce to $0.5\pi_1 = 0.2\pi_2$. Therefore

$$\pi_2 = \frac{5}{2}\pi_1, \quad \frac{7}{2}\pi_1 = 1, \quad \boxed{\pi_1 = \frac{2}{7}, \quad \pi_2 = \frac{5}{7}}.$$

Both starting states yield these limiting probabilities. The balance relation also says that the long-run flows between the two states agree:

$$\pi_1 p_{12} = \frac{2}{7} \frac{1}{2} = \frac{1}{7} = \frac{5}{7} \frac{1}{5} = \pi_2 p_{21}.$$

The chain spends more time in state 2, whose chance of leaving on a given step is smaller.

17.6 Visit and transition frequencies

Under the convergence conditions, π_j is the long-run fraction of time spent in state j . Among steps whose starting state is k , a fraction p_{kj} lead to j . Thus the long-run frequency of transitions $k \rightarrow j$ is

$$\pi_k p_{kj}.$$

This includes self-transitions, whose frequency is $\pi_j p_{jj}$.

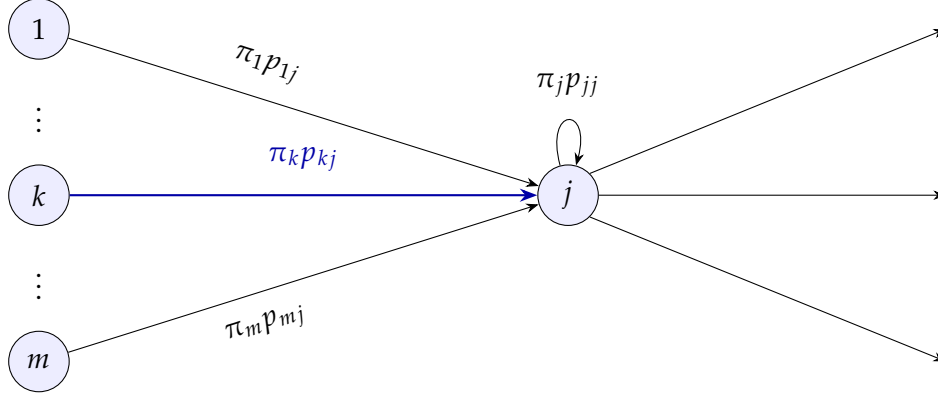


Figure 78: Each incoming transition contributes its frequency $\pi_k p_{kj}$. The self-loop contributes $\pi_j p_{jj}$. Their sum is the frequency π_j of visits to j .

For an expected-count calculation, define

$$C_{kj}(n) = \sum_{t=0}^{n-1} \mathbf{1}_{\{X_t=k, X_{t+1}=j\}}.$$

Linearity of expectation and one-step conditioning give

$$\frac{\mathbb{E}[C_{kj}(n)]}{n} = \frac{1}{n} \sum_{t=0}^{n-1} P(X_t = k) p_{kj} \longrightarrow \pi_k p_{kj},$$

because averaging a convergent sequence preserves its limit. Similarly, the expected frequency of visits to j tends to π_j .

Every visit at the end of a step has exactly one incoming transition. Adding the incoming frequencies gives

$$\pi_j = \sum_k \pi_k p_{kj}.$$

This is the probability-flow meaning of the balance equation. The initial endpoint contributes at most one visit, which has no effect after division by a long observation horizon.

17.7 Birth-death chains and flow across a cut

A BIRTH-DEATH CHAIN on $\{0, 1, \dots, m\}$ can move up one state, down one state, or stay put. Write p_i for the upward probability from i and q_i for the downward probability. Here these symbols are transition probabilities themselves.

For $1 \leq i \leq m-1$, the self-loop probability is $1 - p_i - q_i$. At 0 it is $1 - p_0$; at m it is $1 - q_m$. All probabilities must be nonnegative. Assume the adjacent upward and downward probabilities are positive so that the states communicate.



Figure 79: Only neighboring states exchange probability. The end states have one neighbor each; omitted states follow the same interior pattern.

In a stationary distribution, the total mass of $\{0, \dots, i\}$ does not change after a transition. The only transition leaving this set is $i \rightarrow i+1$, and the only transition entering it is $i+1 \rightarrow i$. Therefore their flows must agree:

$$\boxed{\pi_i p_i = \pi_{i+1} q_{i+1}, \quad 0 \leq i < m.}$$

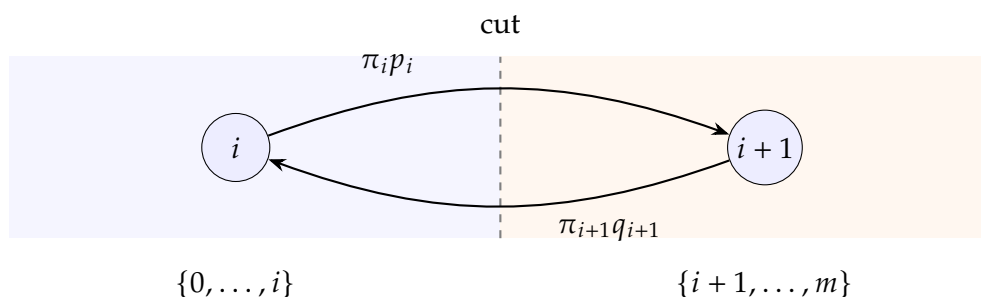


Figure 80: The cut separates $\{0, \dots, i\}$ from $\{i+1, \dots, m\}$. In stationarity the two crossing flows balance. All other transitions stay on their own side.

These adjacent balance relations imply the full state equations: for an interior state, incoming flow from its left neighbor replaces upward outgoing flow, and incoming flow from its right neighbor replaces downward outgoing flow. The self-loop accounts for the remaining probability.

17.8 Constant probabilities and the load factor

Suppose every allowed upward transition has probability $p > 0$ and every allowed downward transition has probability $q > 0$, with $p + q \leq 1$ for interior states. Define the **LOAD FACTOR**

$$\rho = \frac{p}{q}.$$

Adjacent balance gives

$$\pi_{i+1} = \frac{p}{q} \pi_i = \rho \pi_i, \quad \boxed{\pi_i = \pi_0 \rho^i, \quad 0 \leq i \leq m}.$$

The finite chain's normalization is exact:

$$\pi_0 = \frac{1}{\sum_{i=0}^m \rho^i}.$$

In particular, for $p < q$, so $0 < \rho < 1$,

$$\boxed{\pi_0 = \frac{1 - \rho}{1 - \rho^{m+1}}}.$$

For a large capacity with ρ^{m+1} negligible, this is approximately $1 - \rho$. The infinite-capacity geometric law is

$$\boxed{\pi_i = (1 - \rho) \rho^i, \quad i = 0, 1, 2, \dots}.$$

It requires $\rho < 1$ for the probabilities to sum to one.

In that geometric steady-state law, the expected state is

$$\begin{aligned} \mathbb{E}[X] &= (1 - \rho) \sum_{i=0}^{\infty} i \rho^i \\ &= (1 - \rho) \rho \frac{d}{d\rho} \left(\frac{1}{1 - \rho} \right) = \boxed{\frac{\rho}{1 - \rho}}. \end{aligned}$$

Here the geometric series can be differentiated for $0 < \rho < 1$. The mean grows as the load factor approaches one. For a finite capacity, the mean is instead computed from the normalized finite sum $\sum_{i=0}^m i \pi_i$; the infinite-capacity formula describes the limiting law.

Revision. Periodic classes force transitions through cyclic groups. A finite chain with one aperiodic recurrent class has initial-state-independent limiting probabilities. They satisfy $\pi = \pi P$ and sum to one. The transition frequency $k \rightarrow j$ is $\pi_k p_{kj}$. Birth-death balance equates flows across each adjacent cut; constant up and down probabilities give $\pi_i \propto \rho^i$.

Common mistakes: multiplying for an endpoint event without summing its paths, omitting self-loop flow, assuming that a stationary law guarantees convergence, or applying an infinite-capacity geometric law without $\rho < 1$.

Check yourself: Which four-step paths reach 7 from 2? Why does a self-loop rule out periodic grouping? Why must the two flows across a birth-death cut agree? Where does the factor $1 - \rho$ come from?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 17: Markov Processes - II](#), PDF pp. 1–2. The warmup paths, periodic grouping, balance calculations, and birth-death normalization are completed. Finite-capacity probabilities are distinguished from the infinite-capacity limit, and the diagrams redrawn. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 18: MARKOV CHAINS - III

Balance equations describe long-run occupancy. First-step conditioning answers a different question: which state will be reached, and how long will reaching it take?

18.1 Review of steady-state behavior

For a finite Markov chain with one aperiodic recurrent class, possibly together with transient states,

$$r_{ij}(n) = P(X_n = j \mid X_0 = i) \longrightarrow \pi_j.$$

The limit is independent of the initial state. The steady-state probabilities are the unique solution of

$$\pi_j = \sum_k \pi_k p_{kj}, \quad \sum_j \pi_j = 1, \quad \pi_j \geq 0.$$

These equations balance incoming probability at each state and normalize the total mass.

18.2 Joint events in the two-state chain

Consider the chain with transition matrix $P = \begin{pmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix}$ and steady-state probabilities $\pi_1 = 2/7$, $\pi_2 = 5/7$. Assume $X_0 = 1$ throughout this example.

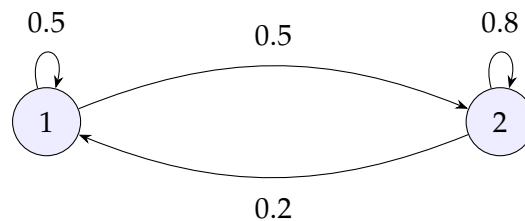


Figure 81: Condition on the earlier state when calculating a joint event. The gap between the two observation times determines the required multi-step transition.

Write $u_n = r_{11}(n)$. One-step conditioning gives $u_{n+1} = 0.5u_n + 0.2(1 - u_n) = 0.2 + 0.3u_n$, with $u_0 = 1$. Subtracting the fixed point $2/7$ and iterating gives

$$u_n - \frac{2}{7} = 0.3^n \left(1 - \frac{2}{7}\right), \quad u_n = \frac{2}{7} + \frac{5}{7}(0.3)^n.$$

The two joint probabilities are therefore

$$\begin{aligned}
P(X_1 = 1, X_{100} = 1 \mid X_0 = 1) &= p_{11}r_{11}(99) \\
&= \frac{1}{2} \left[\frac{2}{7} + \frac{5}{7}(0.3)^{99} \right] \approx \frac{1}{7}, \\
P(X_{100} = 1, X_{101} = 2 \mid X_0 = 1) &= r_{11}(100)p_{12} \\
&= \frac{1}{2} \left[\frac{2}{7} + \frac{5}{7}(0.3)^{100} \right] \approx \frac{1}{7}.
\end{aligned}$$

The first event uses 99 transitions after time 1. The second uses 100 transitions to reach state 1, then one transition to state 2. The Markov property supplies the conditional factor; it does not make successive states independent.

18.3 The phone company model

Calls arrive according to a Poisson process of rate $\lambda > 0$. Each accepted call occupies one of B available lines for an independent exponential duration with parameter $\mu > 0$, hence mean duration $1/\mu$. Durations are independent of the arrival process. An arriving call is blocked when all B lines are busy.

Let the state be the number of busy lines, in $\{0, 1, \dots, B\}$. Exponential memorylessness means that the remaining call durations depend only on how many calls are active, so this state description is Markov.

Over a short interval of length δ , an arrival has probability $\lambda\delta + o(\delta)$. In state i , each of the i active calls can end, so one departure has probability $i\mu\delta + o(\delta)$. Two or more events have probability $o(\delta)$.

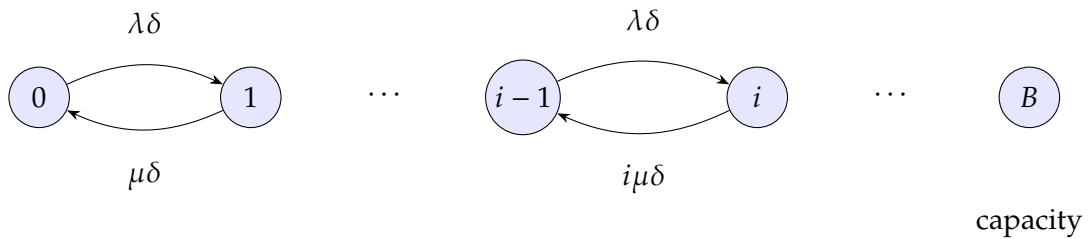


Figure 82: First-order transition probabilities over a short interval. Arrivals increase the busy-line count when capacity remains; departures decrease it. Self-loops are omitted for clarity.

For an interior state $0 < i < B$, the probability of no state change is $1 - (\lambda + i\mu)\delta + o(\delta)$. At 0 it is $1 - \lambda\delta + o(\delta)$; at B it is $1 - B\mu\delta + o(\delta)$. A blocked arrival leaves the busy-line count unchanged.

The state-dependent departure rate is the key difference from a birth-death chain with a constant downward rate: more busy lines create more opportunities for a call to end.

18.4 Steady-state occupancy and blocking

In stationarity, flows across the cut between $i-1$ and i balance. Using the short-interval probabilities, dividing by δ , and taking $\delta \rightarrow 0$ gives

$$\lambda \pi_{i-1} = i \mu \pi_i, \quad 1 \leq i \leq B.$$

Set $\rho = \lambda/\mu$. Repeated substitution yields

$$\pi_i = \frac{\rho}{i} \pi_{i-1} = \pi_0 \frac{\rho^i}{i!}.$$

Normalizing over the finite capacity gives

$$\pi_0 = \left(\sum_{k=0}^B \frac{\rho^k}{k!} \right)^{-1}, \quad \pi_i = \frac{\rho^i / i!}{\sum_{k=0}^B \rho^k / k!}.$$

The factorial comes from the successive departure rates $\mu, 2\mu, \dots, i\mu$. The sum stops at B because the system cannot hold more than B simultaneous calls.

A call is blocked exactly when it arrives to state B . To see why its blocking probability equals π_B , consider arrivals in a short interval while the system is stationary. Conditional on the state at its start, the Poisson arrival probability is still $\lambda\delta + o(\delta)$, independent of that state. Thus

$$P(\text{an arrival is blocked during the interval}) = \pi_B \lambda \delta + o(\delta),$$

$$P(\text{an arrival occurs during the interval}) = \lambda \delta + o(\delta).$$

An intervening departure or second arrival changes these expressions only by $o(\delta)$. Dividing and taking the short-interval limit gives

$$P(\text{a call is blocked}) = \pi_B = \frac{\rho^B / B!}{\sum_{k=0}^B \rho^k / k!}.$$

This is a steady-state probability. The argument uses the Poisson arrival model, rather than assuming that arbitrary arrival processes observe time averages.

18.5 Absorption probabilities

An **ABSORBING STATE** has $p_{ii} = 1$: once reached, it is never left. For the following chain, let a_i be the probability of eventual absorption in state 4 given $X_0 = i$.

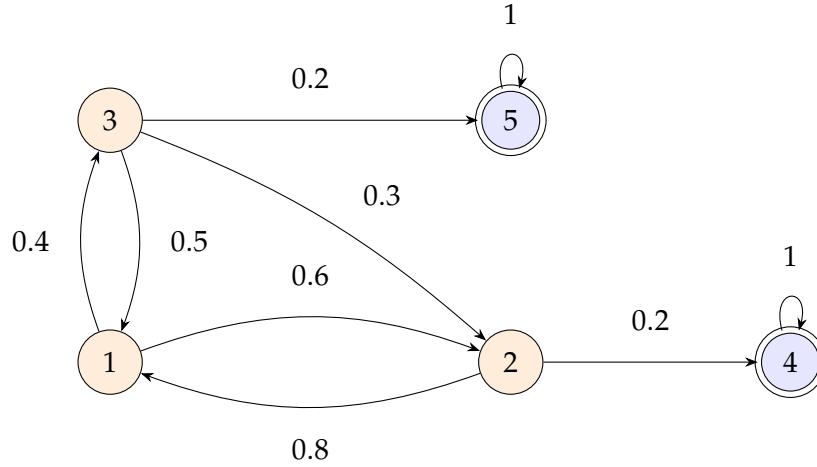


Figure 83: States 4 and 5 are absorbing. The transient states 1, 2, 3 can lead to either outcome. Labels give one-step transition probabilities.

Impose $a_4 = 1$ and $a_5 = 0$, since the desired outcome is absorption in 4. For a transient state, condition on the next state:

$$a_i = \sum_j p_{ij} a_j.$$

The future absorption probability after moving to j is a_j . Hence the unknowns satisfy

$$a_1 = \frac{2}{5}a_3 + \frac{3}{5}a_2, \quad a_2 = \frac{4}{5}a_1 + \frac{1}{5}, \quad a_3 = \frac{1}{2}a_1 + \frac{3}{10}a_2.$$

Substitute the second equation into the third:

$$a_3 = \frac{37}{50}a_1 + \frac{3}{50}.$$

Then the first equation becomes

$$a_1 = \frac{97}{125}a_1 + \frac{18}{125}, \quad 28a_1 = 18.$$

Therefore the unique solution is

$$\boxed{a_1 = \frac{9}{14}, \quad a_2 = \frac{5}{7}, \quad a_3 = \frac{15}{28}, \quad a_4 = 1, \quad a_5 = 0.}$$

18.6 Expected time to absorption

Now use the following four-state chain, whose only absorbing state is 4. Let μ_i be the expected number of transitions until reaching 4, given $X_0 = i$. Here μ_i denotes a mean transition count; it is separate from the call-duration parameter μ .

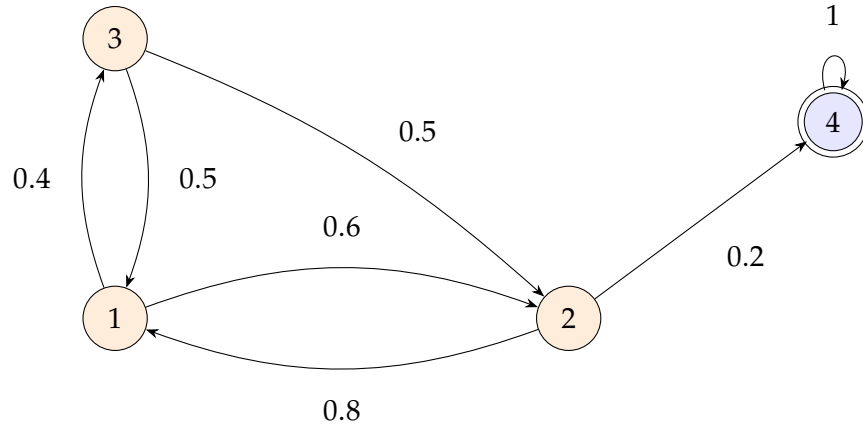


Figure 84: State 3 moves to 1 or 2 with equal probability. Every transient state can reach the absorbing state 4.

Starting at the absorbing state requires no transition, so $\mu_4 = 0$. From any other state, count the next transition, then condition on the remaining time:

$$\mu_i = 1 + \sum_j p_{ij} \mu_j, \quad i \neq 4.$$

For this chain,

$$\mu_1 = 1 + \frac{2}{5}\mu_3 + \frac{3}{5}\mu_2, \quad \mu_2 = 1 + \frac{4}{5}\mu_1, \quad \mu_3 = 1 + \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2.$$

The last two equations give

$$\mu_3 = \frac{3}{2} + \frac{9}{10}\mu_1.$$

Substitute into the first equation:

$$\mu_1 = \frac{11}{5} + \frac{21}{25}\mu_1, \quad \frac{4}{25}\mu_1 = \frac{11}{5}.$$

Thus the unique solution is

$$\mu_1 = \frac{55}{4}, \quad \mu_2 = 12, \quad \mu_3 = \frac{111}{8}, \quad \mu_4 = 0.$$

The 1 in the recursion counts a transition even when it reaches the absorbing state immediately. It is absent from the boundary equation $\mu_4 = 0$.

18.7 Mean first passage and recurrence times

Consider a finite chain with one recurrent class and fix a recurrent state s . Aperiodicity is not needed for the hitting-time calculations here.

The MEAN FIRST PASSAGE TIME from i to s is

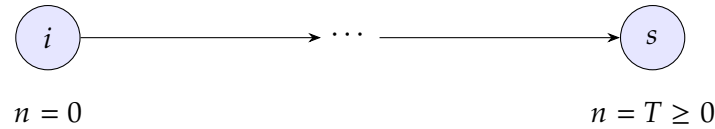
$$t_i = \mathbb{E}[\min\{n \geq 0 : X_n = s\} \mid X_0 = i].$$

Time zero is allowed, so $t_s = 0$. First-step conditioning gives the unique solution of

$$t_s = 0, \quad t_i = 1 + \sum_j p_{ij} t_j \quad (i \neq s).$$

To calculate a first passage time, stop counting upon the first visit to s ; its later transitions do not affect the answer.

First passage: start at i



Recurrence: start at s

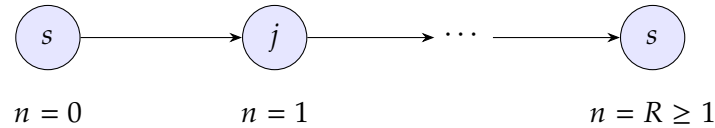


Figure 85: First passage permits a hit at time zero. Recurrence starts in s and requires a later visit. If the first transition is a self-loop, the recurrence time is already one. The paths are schematic.

The MEAN RECURRENCE TIME of s is instead

$$t_s^* = \mathbb{E}[\min\{n \geq 1 : X_n = s\} \mid X_0 = s].$$

After one transition from s to j , the expected remaining first passage time is t_j . Therefore

$$t_s^* = 1 + \sum_j p_{sj} t_j.$$

The sum includes $j = s$, whose contribution is $p_{ss} t_s = 0$. A self-loop returns in one transition, while a move elsewhere requires that first transition plus the time to reach s from the new state. The difference between $n \geq 0$ and $n \geq 1$ prevents confusing $t_s = 0$ with the positive recurrence time.

Revision. Balance equations determine steady-state occupancy. In the phone model, $\lambda\pi_{i-1} = i\mu\pi_i$ gives $\pi_i \propto (\lambda/\mu)^i / i!$, and the blocking probability is π_B . Absorption probabilities satisfy $a_i = \sum_j p_{ij}a_j$ with outcome-specific boundary values. Mean hitting times satisfy $t_i = 1 + \sum_j p_{ij}t_j$ away from the target, with zero time at the target.

Common mistakes: treating consecutive states as independent, using 100 instead of 99 transitions between times 1 and 100, forgetting the factor i in the total departure rate, or counting a transition when the process already starts at its target.

Check yourself: Why does occupancy contain an $i!$? Why does a Poisson arrival see blocking probability π_B ? Which boundary values describe absorption in 4 rather than 5? Why can $t_s = 0$ while $t_s^* > 0$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 18: Markov Processes - III](#), PDF pp. 1–2. Joint probabilities, phone-model blocking, and both absorption systems are completed; first-step recursions are explained and diagrams redrawn. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).